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COVER STORY

What the FTC got wrong in the Google antitrust investigation

From 2011-2012, the FTC investigated the possibility of pursuing antitrust action against Google.

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The FTC decided to close the investigation and not much was publicly known about what happened until Politico released 312 pages of internal FTC memos that from the investigation a decade later. As someone who works in tech, on reading the memos, the most striking thing is how one side, the side that argued to close the investigation, repeatedly displays a lack of basic understanding of tech industry and the memos from directors and other higher-ups don't acknowledge that this at all.

If you don't generally follow what regulators and legislators are saying about tech, seeing the internal c(or any other industry) when these decisions are, apparently, being made with little to no understanding of the industries 1 .

Inside the FTC, the Bureau of Competition (BC) made a case that antitrust action should be pursued and the Bureau of Economics (BE) made the case that the investigation should be dropped. The BC case is moderately strong. Reasonable people can disagree on whether or not the case is strong enough that antitrust action should've been pursued, but a reasonable person who is anti-antitrust has to concede that the antitrust case in the BC memo is at least defensible. The case against in the BE is not defensible. There are major errors in core parts of the BE memo. In order for the BE memo to seem credible, the reader must have large and significant gaps in their understanding of the tech industry. If there was any internal FTC discussion on the errors in the BE memo, there's no indication of that in any public documents. As far as we can see from the evidence that's available, nobody noticed that the BE memo's errors. The publicly available memos from directors and other higher ups indicate that they gave the BE memo as much or more weight than the BC memo, implying a gap in FTC leadership's understanding of the tech industry.

id="brief-summary">Brief summary

Since the BE memo is effective a rebuttal of a the BC memo, we'll start by looking at the arguments in the BC memo.

The bullet points below summarize the Executive Summary from the BC memo, which roughly summarizes the case made by the BC memo:

Google is dominant search engine and seller of search ads

This memo addresses 4 of 5 areas with anticompetitive conduct; mobile is in a supplemental memo

Google has monopoly power in the U. S. in Horizontal Search; Search Advertising; and Syndicated Search and Search Advertising

On the question of whether Google has unlawfully preferred its own content while demoting rivals, we do not recommend the FTC proceed; it's a close call and case law is not favorable to anticompetitive product design and Google's efficiency justifications are strong and there's some benefit to users

On whether Google has unlawfully scraped content from vertical rivals to improve their own vertical products, recommending condemning as a conditional refusal to deal under Section 2

Prior voluntary dealing was mutually beneficial

Threats to remove rival content from general search designed to coerce rivals into allowing Google to user their content for Google's vertical product

Natural and probable effect is to diminish incentives of vertical website R&D

On anticompetitive contractual restrictions on automated cross-management of ad campaigns, restrictions should be condemned under Section 2

They limit ability of advertisers to make use of their own data, reducing innovation and increasing transaction costs for advertisers and third-party businesses

Also degrade the quality of Google's rivals in search and search advertising

Google's efficiency justifications appears to be pretextual

On anticompetitive exclusionary agreements with websites for syndicated search and search ads, Google should be condemned under Section 2

Only modest anticompetitive effects on publishers, but deny scale to competitors, competitively significant to main rival (Bing) as well as significant barrier to entry in longer term

Google's efficiency justifications are, on balance, non-persuasive

Possible remedies

Could be required to provide an opt-out for snippets (reviews, ratings) from Google's vertical properties while retaining snippets in web search and/or Universal Search on main search results page

Could be required to limit use of content indexed from web search results

Campaign management restrictions

Could be required to remove problematic contractual restrictions from license agreements

Exclusionary syndication agreements

Could be enjoined from entering into exclusive search agreements with search syndication partners and required to loosen restrictions surrounding syndication partners' use of rival search ads

There are a number of risks to case, not named in summary except that Google can argue that Microsoft's most efficient distribution channel is bing.com and that any scale MS might gain will be immaterial to Bing's competitive position

[BC] Staff concludes Google's conduct has resulted and will result in real harm to consumers and to innovation in online search and ads.

In their supplemental memo on mobile, BC staff claim that Google dominates mobile search via exclusivity agreements and that mobile search was rapidly growing at the time. BC

staff claimed that, according to Google internal documents, mobile search went from 9.5% to 17.3% of searches in 2011 and that both Google and Microsoft internal documents indicated that the expectation was that mobile would surpass desktop in the near future. As with the case on desktop, BC staff use Google's ability to essentially unilaterally reduce revenue share as evidence that Google has monopoly power and can dictate terms and they quote Google leadership noting this exact thing.

BC staff acknowledge that many of Google's actions have been beneficial to consumers, but balance this against the harms of anticompetitive tactics, saying

the evidence paints a complex portrait of a company working toward an overall goal of maintaining its market share by providing the best user experience, while simultaneously engaging in tactics that resulted in harm to many vertical competitors, and likely helped to entrench Google's monopoly power over search and search advertising

BE staff strongly disagreed with BC staff. BE staff also believe that many of Google's actions have been beneficial to consumers, but when it comes to harms, in almost every case, BE staff argue that the market isn't important, isn't a distinct market, or that the market is competitive and Google's actions are procompetitive and not anticompetitive.

id="common-errors">Common errors

At least in the documents provided by Politico, BE staff generally declined to engage with BC staff's arguments and numbers directly. For example, in addition to arguing that Google's agreements and exclusivity (insofar as agreements are exclusive) are procompetitive and foreclosing the possibility of such agreements might have significant negative impacts on the market, they argue that mobile is a small and unimportant market. The BE memo argues that mobile is only 8% of the market and, while it's growing rapidly, is unimportant, as it's only a "small percentage of overall queries and an even smaller percentage of search ad revenues". They also claim that there is robust competition in mobile because, in addition to Apple, there's also BlackBerry and Windows Mobile. Between when the FTC investigation started and when the memo was written, BlackBerry's marketshare dropped from 14% to 6%, which was part of a long-term decline that showed no signs of changing. Windows Mobile's drop was less precipitous, from 6% to 4%, but in a market with such strong network effects, it's curious that BE staff would argue that these platforms with low and declining marketshare would provide robust competition going forward.

When the authors of the BE memo make a prediction, they seem to have a facility for predicting the opposite of what will happen. To do this, the authors of the BE memo took positions that were opposed to the general consensus at the time. Another example of this is when they imply that there is robust competition in the search market, which is implied

to be expected to continue without antitrust action. Their evidence for this was that Yahoo and Bing had a combined “steady” 30% marketshare in the U. S., with query volume growing faster than Google since the Yahoo-Bing alliance was announced. The BE memo authors even go even further and claim that Microsoft’s query volume is growing faster than Google’s and that Microsoft + Yahoo combined have higher marketshare than Google as measured by search MAU .

The BE memo’s argument that Yahoo and Bing are providing robust and stable competition leaves out that the fixed costs of running a search engine are so high and the scale required to be profitable so large that Yahoo effectively dropped out of search and outsourced search to Bing. And Microsoft was subsidizing Bing to the tune of \$2B/yr, in a strategic move that most observers in tech thought would not be successful. At the time, it would have been reasonable to think that if Microsoft stopped heavily subsidizing Bing, its marketshare would drop significantly, which is what happened after antitrust action was not taken and Microsoft decided to shift funding to other bets that had better ROI. Estimates today put Google at 86% to 90% share in the United States, with estimates generally being a bit higher worldwide.

On the wilder claims, such as Microsoft and Yahoo combined having more active search users than Google and that Microsoft query volume and therefore search marketshare is growing faster than Google, they use comScore data. There are a couple of curious things about this.

First, the authors pick and choose their data in order to present figures that maximize Microsoft’s marketshare. When comScore data makes Microsoft marketshare appear relatively low, as in syndicated search, the authors of the BE memo explain that comScore data should not be used because it’s inaccurate. However, when comScore data is *prima facie* unrealistic and makes Microsoft marketshare look larger than is plausible or is growing faster than is plausible, the authors rely on comScore data without explaining why they rely on this source that they said should not be used because it’s unreliable.

Using this data, the BE memo basically argues that, because many users use Yahoo and Bing at least occasionally, users clearly could use Yahoo and Bing, and there must not be a significant barrier to switching even if (for example) a user uses Yahoo or Bing once a month and Google one thousand times a month. From having worked with and talked to people who work on product changed to drive growth, the overwhelming consensus has been that it’s generally very difficult to convert a lightly-engaged user who barely registers as an MAU to a heavily-engaged user who uses the product regularly , and that this is generally considered more difficult than converting a brand-new user to becoming heavily engaged user. Like Boies’s argument about rangeCheck, it’s easy to see how this line of reasoning would sound plausible to a lay person who knows nothing

about tech, but the argument reads like something you’d expect to see from a lay person.

Although the BE staff memo reads like a rebuttal to the points of the BC staff memo, the lack of direct engagement on the facts and arguments means that a reader with no knowledge of the industry who reads just one of the memos will have a very different impression than a reader who reads the other. For example, on the importance of mobile search, a naive BC-memo-only reader would think that mobile is very important, perhaps the most important thing, whereas a naive BE-memo-only reader would think that mobile is unimportant and will continue to be unimportant for the foreseeable future.

Politico also released memos from two directors who weigh the arguments of BC and BE staff. Both directors favor the BE memo over the BC memo, one very much so and one moderately so. When it comes to disagreements, such as the importance of mobile in the near future, there’s no evidence in the memos presented that there was any attempt to determine who was correct or that the errors we’re discussing here were noticed. The closest thing to addressing disagreements such as these are comments that thank both staffs for having done good work, in what one might call a “fair and balanced” manner, such as “The BC and BE staffs have done an outstanding job on this complex investigation. The memos from the respective bureaus make clear that the case for a complaint is close in the four areas ... “. To the extent that this can be inferred, it seems that the reasoning and facts laid out in the BE memo were given at least as much weight as the reasoning and facts in the BC memo despite much of the BE memo’s case seemingly highly implausible to an observer who understands tech.

For example, on the importance of mobile, I happened to work at Google shortly after these memos were written and, when I was at Google, they had already pivoted to a “mobile first” strategy because it was understood that mobile was going to be the most important market going forward. This was also understood at other large tech companies at the time and had been understood going back further than the dates of these memos. Many consumers didn’t understand this and redesigns that degraded the desktop experience in order to unify desktop and mobile experiences were a common cause of complaints at the time. But if you looked at the data on this or talked to people at big companies, it was clear that, from a business standpoint, it made sense to focus on mobile and deal with whatever fallout might happen in desktop if that allowed for greater velocity in mobile development.

Both the BC and BE staff memos extensively reference interviews across many tech companies, including all of the “hyperscalers “. It’s curious that someone could have access to all of these internal documents from these companies as well as interviews and then make the argument that mobile was, at the time, not very important. And it’s strange that, at least to the extent that we can know what happened from these memos, directors took both sets of arguments

at face value and then decided that the BE staff case was as convincing or more convincing than the BC staff case.

That's one class of error we repeatedly see between the BC and BE staff memos, stretching data to make a case that a knowledgeable observer can plainly see is not true. In most cases, it's BE staff who have stretched data as far as it can go to take a tenuous position as far as it can be pushed, but there are some instances of BC staff making a case that's a stretch.

Another class of error we see repeated, mainly in the BE memo, is taking what most people in industry would consider an obviously incorrect model of the world and then making inferences based on that. An example of this is the discussion on whether or not vertical competitors such as Yelp and TripAdvisor were or would be significantly disadvantaged by actions BC staff allege are anticompetitive. BE staff, in addition to arguing that Google's actions were actually procompetitive and not anticompetitive, argued that it would not be possible for Google to significantly harm vertical competitors because the amount of traffic Google drives to them is small, only 10% to 20% of their total traffic, going to say "the effect on traffic from Google to local sites is very small and not statistically significant". Although BE staff don't elaborate on their model of how this business works, they appear to believe that the market is basically static. If Google removes Yelp from its listings (which they threatened to do if they weren't allowed to integrate Yelp's data into their own vertical product) or downranks Yelp to preference Google's own results, this will, at most, reduce Yelp's traffic by 10% to 20% in the long run because only 10% to 20% of traffic comes from Google.

But even a VC or PM intern can be expected to understand that the market isn't static. What one would expect if Google can persistently take a significant fraction of search traffic away from Yelp and direct it to Google's local offerings instead is that, in the long run, Yelp will end up with very few users and become a shell of what it once was. This is exactly what happened and, as of this writing, Yelp is valued at \$2B despite having a trailing P/E ratio of 24, which is fairly low P/E for a tech company. But the P/E ratio is unsurprisingly low because it's not generally believed that Yelp can turn this around due to Google's dominant position in search as well as maps making it very difficult for Yelp to gain or retain users. This is not just obvious in retrospect and was well understood at the time. In fact, I talked to a former colleague at Google who was working on one of a number of local features that leveraged the position that Google had and that Yelp could never reasonably attain; the expected outcome of these features was to cripple Yelp's business. Not only was it understood that this was going to happen, it was also understood that Yelp was not likely to be able to counter this due to Google's ability to leverage its market power from search and maps. It's curious that, at the time, someone would've seriously argued that cutting off Yelp's source of new users while simultaneously presenting virtually all of Yelp's then-current users with an alternative that's bundled into an app or website they already use would

not significantly impact Yelp's business, but the BE memo makes that case. One could argue that the set of maneuvers used here are analogous to the ones done by Microsoft that were brought up in the Microsoft antitrust case where it was alleged that a Microsoft exec said that they were going to "cut off Netscape's air supply", but the BE memo argues that impact of having one's air supply cut off is "very small and not statistically significant" (after all, a typical body has blood volume sufficient to bind 1L of oxygen, much more than the oxygen normally taken in during one breath).

Another class of, if not error, then poorly supported reasoning is relying on cocktail party level of reasoning when there's data or other strong evidence that can be directly applied. This happens throughout the BE memo even though, at other times, when the BC memo has some moderately plausible reasoning, the BE memo's counter is that we should not accept such reasoning and need to look at the data and not just reason about things in the abstract. The BE memo heavily leans on the concept that we must rely on data over reasoning and calls arguments from the BC memo that aren't rooted in rigorous data anecdotal, "beyond speculation", etc., but BE memo only does this in cases where knowledge or reasoning might lead one to conclude that there was some kind of barrier to competition. When the data indicates that Google's behavior creates some kind of barrier in the market, the authors of BE memo ignore all relevant data and instead rely on reasoning over data even when the reasoning is weak and has the character of the Boies argument we referenced earlier. One could argue that the standard of evidence for pursuing an antitrust case should be stronger the standard of evidence for not pursuing one, but if the asymmetry observed here were for that reason, the BE memo could have listed areas where the evidence wasn't strong enough without making its own weak assertions in the face of stronger evidence. An example of this is the discussion of the impact of mobile defaults.

The BE memo argues that defaults are essentially worthless and have little to no impact, saying multiple times that users can switch with just "a few taps", adding that this takes "a few seconds" and that, therefore, "[t]hese are trivial switching costs". The most obvious and direct argument piece of evidence on the impact of defaults is the amount of money Google pays to retain its default status. In a 2023 antitrust action, it was revealed that Google paid Apple \$26.3B to retain its default status in 2021. As of this writing, Apple's P/E ratio is 29.53. If we think of this payment as, at the margin, pure profit and having default status is as worthless as indicated by the BE memo, a naive estimate of how much this is worth to Apple is that it can account for something like \$776B of Apple's \$2.9T market cap. Or, looking at this from Google's standpoint, Google's P/E ratio is 27.49, so Google is willing to give up \$722B of its \$2.17T market cap. Google is willing to pay this to be the default search for something like 25% to 30% of phones in the world. This calculation is too simplistic, but there's no reasonable adjustment that could give anyone the impression that the value of being the default is as trivial as claimed by the BE memo. For reference, a \$776B tech company would be 7th

most valuable publicly traded U. S. tech company and the 8th most valuable publicly traded U. S. company (behind Meta/Facebook and Berkshire Hathaway, but ahead of Eli Lilly). Another reference is that YouTube's ad revenue in 2021 was \$28.8B. It would be difficult to argue that spending one YouTube worth of revenue, in profit, in order to retain default status makes sense if, in practice, user switching costs are trivial and defaults don't matter. If we look for publicly available numbers close to 2012 instead of 2021, in 2013, TechCrunch reported a rumor that Google was paying Apple \$1B/yr for search status and a lawsuit then revealed that Google paid Apple \$1B for default search status in 2014. This is not longer after these memos are written and \$1B/yr is still a non-trivial amount of money and it belies the BE memo's claim that mobile is unimportant and that defaults don't matter because user switching costs are trivial.

It's curious that, given the heavy emphasis in the BE memo on not trusting plausible reasoning and having to rely on empirical data, that BE staff appeared to make no attempt to find out how much Google was paying for its default status (a memo by a director who agrees with BE staff suggests that someone ought to check on this number, but there's no evidence that this was done and the FTC investigation was dropped shortly afterwards). Given the number of internal documents the FTC was able to obtain, it seems unlikely that the FTC would not have been able to obtain this number from either Apple or Google. But, even if it were the case that the number were unobtainable, it's *prima facie* implausible that defaults don't matter and switching costs are low in practice. If FTC staff interviewed product-oriented engineers and PMs or looked at the history of products in tech, so in order to make this case, BE staff had to ignore or avoid finding out how much Google was paying for default status, not talk to product-focused engineers, PM, or leadership, and also avoid learning about the tech industry.

One could make the case that, while defaults are powerful, companies have been able to overcome being non-default, which could lead to a debate on exactly how powerful defaults are. For example, one might argue about the impact of defaults when Google Chrome became the dominant browser and debate how much of it was due to Chrome simply being a better browser than IE, Opera, and Firefox, how much was due to blunders by Microsoft that Google is unlikely to repeat in search, how much was due to things like tricking people into making Chrome default via a bundle deal with badware installers and how much was due to pressuring people into setting Chrome as default via google.com. That's an interesting discussion where a reasonable person with an understanding of the industry could take either side of the debate, unlike the claim that defaults basically don't matter at all and user switching costs are trivial in practice, which is not plausible even without access to the data on how much Google pays Apple and others to retain default status. And as of the 2020 DoJ case against Google, roughly half of Google searches occur via a default search that Google pays for.

Another repeated error, closely related to the one above, is bringing up marketing statements, press releases, or other statements that are generally understood to be exaggerations, and relying on these as if they're meaningful statements of fact. For example, the BE memo states:

Microsoft's public statements are not consistent with statements made to antitrust regulators. Microsoft CEO Steve Ballmer stated in a press release announcing the search agreement with Yahoo: "This agreement with Yahoo! will provide the scale we need to deliver even more rapid advances in relevancy and usefulness. Microsoft and Yahoo! know there's so much more that search could be. This agreement gives us the scale and resources to create the future of search"

This is the kind of marketing pabulum that generally accompanies an acquisition or partnership. Because this kind of meaningless statement is common across many industries, one would expect regulators, even ones with no understanding of tech, to recognize this as marketing and not give it as much or more weight as serious evidence.

id="a-few-interesting-tidbits">A few interesting tidbits

Now that we've covered the main classes of errors observed in the memos, we'll look at a tidbits from the memos.

Between the approval of the compulsory process on June 3rd 2011 and the publication of the BC memo dated August 8th 2012, staff received 9.5M pages of documents across 2M docs and said they reviewed "many thousands of these documents", so staff were only able to review a small fraction of the documents.

Prior to the FTC investigation, there were a number of lawsuits related to the same issues, and all were dismissed, some with arguments that would, if they were taken as broad precedent, make it difficult for any litigation to succeed. In *SearchKing v. Google*, plaintiffs alleged that Google unfairly demoted their results but it was ruled that Google's rankings are constitutionally protected opinion and even malicious manipulation of rankings would not expose Google to liability. In *Kinderstart v. Google*, part of the ruling was that Google search is not an essential facility for vertical providers (such as Yelp, eBay, and Expedia). Since the memos are ultimately about legal proceedings, there is, of course, extensive discussion of *Verizon v. Trinko* and *Aspen Skiing Co. v. Aspen Highlands Skiing Corp* and the implications thereof.

As of the writing of the BC memo, 96% of Google's \$38B in revenue was from ads, mostly from search ads. The BC memo makes the case that other forms of advertising, other than social media ads, only have limited potential for growth. That's certainly wrong in retrospect. For example, video ads are a significant market. YouTube's ad revenue was \$28.8B in 2021 (a bit more than what Google pays to Apple to retain default search status), Twitch supposedly generated another \$2B-\$3B in video revenue, and a fair

amount of video ad revenue goes directly from sponsors to streamers without passing through YouTube and Twitch, e.g., the #137th largest streamer on Twitch was offered \$10M/yr stream online gambling for 30 minutes a day, and he claims that the #42 largest streamer, who he personally knows, was paid \$10M/mo from online gambling sponsorships. And this isn't just apparent in retrospect — even at the time, there were strong signs that video would become a major advertising market. It happens that those same signs also showed that Google was likely to dominate the market for video ads, but it's still the case that the specific argument here was overstated.

In general, the BC memo seems to overstate the expected primacy of search ads as well as how distinct a market search ads are, claiming that other online ad spend is not a substitute in any way and, if anything, is a complement. Although one might be able to reasonably argue that search ads are a somewhat distinct market and the elasticity of substitution is low once you start moving a significant amount of your ad spend away from search, the degree to which the BC memo makes this claim is a stretch. Search ads and other ad budgets being complements and not substitutes is a very different position than I've heard from talking to people about how ad spend is allocated in practice. Perhaps one can argue that it makes sense to try to make a strong case here in light of *Person V. Google*, where Judge Fogel of the Northern District of California criticized the plaintiff's market definition, finding no basis for distinguishing "search advertising market" from the larger market for internet advertising, which likely foreshadows an objection that would be raised in any future litigation. However, as someone who's just trying to understand the facts of the matter at hand and the veracity of the arguments, the argument here seems dubious.

For Google's integrated products like local search and product search (formerly Froogle), the BC memo claims that if Google treated its own properties like other websites, the products wouldn't be ranked and Google artificially placed their own vertical competitors above organic offerings. The web spam team declined to include Froogle results because the results are exactly the kind of thing that Google removes from the index because it's spammy, saying "[o]ur algorithms specifically look for pages like these to either demote or remove from the index". Bill Brouger, product manager for web search said "Generally we like to have the destination pages in the index, not the aggregated pages. So if our local pages are lists of links to other pages, it's more important that we have the other pages in the index". After the web spam team was overruled and the results were inserted, the ads team complained that the less clicked (and implied to be lower quality) results would lead to a loss of \$154M/yr. The response to this essentially contained the same content as the BC memo's argument on the importance of scale and why Google's actions to deprive competitors of scale are costly:

We face strong competition and must move quickly. Turning down onebox would hamper progress as follows -

Ranking: Losing click data harms ranking; [t]riggering Losing CTR and google.com query distribution data triggering accuracy; [c]omprehensiveness: Losing traffic harms merchant growth and therefore comprehensiveness; [m]erchant cooperation: Losing traffic reduces effort merchants put into offer data, tax, & shipping; PR: Turning off onebox reduces Google's credibility in commerce; [u]ser awareness: Losing shopping-related UI on google.com reduces awareness of Google's shopping features

Normally, CTR is used as a strong signal to rank results, but this would've resulted in a low ranking for Google's own vertical properties, so "Google used occurrence of competing vertical websites to automatically boost the ranking of its own vertical properties above that of competitors" — if a comparison shopping site was relevant, Google would insert Google Product search above any rival, and if a local search site like Yelp or CitySearch was relevant, Google automatically returned Google Local at top of SERP.

Additionally, in order to see content for Google local results, Google took Yelp content and integrated it into Google Places. When Yelp observed this was happening, they objected to this and Google threatened to ban Yelp from traditional Google search results and further threatened to ban any vertical provider that didn't allow its content to be used in Google Places. Marissa Mayer testified that it was, from a technical standpoint, extraordinarily difficult to remove Yelp from Google Places without also removing Yelp from traditional organic search results. But when Yelp sent a cease and desist letter, Google was able to remove Yelp results immediately, seemingly indicating that it was less difficult than claimed. Google then claimed that it was technically infeasible to remove Yelp from Google Places without removing Yelp from the "local merge" interface on SERP. BC staff believe this claim is false as well, and Marissa Mayer later admitted in a hearing that this claim was false and that Google was concerned about the consequences of allowing sites to opt out of Google Places while staying in "local merge". There was also a very similar story with Amazon results and product search. As noted above, the BE memo's counterargument to all of this is that Google traffic is "very small and not statistically significant"

The BC memo claims that the activities above both reduced incentives of companies Yelp, City Search, Amazon, etc., to invest in the area and also reduced the incentives for new companies to form in this area. This seems true. In addition to the evidence presented in the BC memo (which goes beyond what was summarized above), if you just talked to founders looking for an idea or VCs around the time of the FTC investigation, there had already been a real movement away from founding and funding companies like Yelp because it was understood that Google could seriously cripple any similar company in this space by cutting off its air supply.

We'll defer to the appendix BC memo discussion on the AdWords API restrictions that specifically disallow programmatic porting of campaigns to other platforms, such

as Bing. But one interesting bit there is that Google was apparently aware of the legal sensitivity of this matter, so meeting notes and internal documentation on the topic are unusually incomplete. On one meeting, apparently the most informative written record BC staff were able to find consists of a message from Director of PM Richard Holden to SVP of ads Susan Wojcicki which reads, “We didn’t take notes for obvious reasons hence why I’m not elaborating too much here in email but happy to brief you more verbally”.

We’ll also defer a detailed discussion of the BC memo comments on Google’s exclusive and restrictive syndication agreements to the appendix, except for a couple of funny bits. One is that Google claims they were unaware of the terms and conditions in their standard online service agreements. In particular, the terms and conditions contained a “preferred placement” clause, which a number of parties believe is a de facto exclusivity agreement. When FTC staff questioned Google’s VP of search services about this term, the VP claimed they were not aware of this term. Afterwards, Google sent a letter to Barbara Blank of the FTC explaining that they were removing the preferred placement clause in the standard online agreement.

Another funny bit involves Google’s market power and how it allowed them to collect an increasingly large share of revenue for themselves and decrease the revenue share their partner received. Only a small number of Google’s customers who were impacted by this found this concerning. Those that did find it concerning were some of the largest and most sophisticated customers (such as Amazon and IAC); their concern was that Google’s restrictive and exclusive provisions would increase Google’s dominance over Bing/Microsoft and allow them to dictate worse terms to customers. Even as Google was executing a systematic strategy to reduce revenue share to customers, which could only be possible due to their dominance of the market, most customers appeared to either not understand the long-term implications of Google’s market power in this area or the importance of the internet.

For example, Best Buy didn’t find this concerning because Best Buy viewed their website and the web as a way for customers to find presale information before entering a store and Walmart didn’t find this concerning because they viewed the web as an extension to brick and mortar retail. It seems that the same lack of understanding of the importance of the internet which led Walmart and Best Buy to express their lack of concern over Google’s dominance here also led to these retailers, which previously had a much stronger position than Amazon, falling greatly behind in both online and overall profit. Walmart later realized its error here and acquired Jet.com for \$3.3B in 2016 and also seriously (relative to other retailers) funded programmers to do serious tech work inside Walmart. Since Walmart started taking the internet seriously, it’s made a substantial comeback online and has averaged a 30% CAGR in online net sales since 2018, but taking two decades to mount a serious response to Amazon’s online presence has put Walmart solidly behind Amazon in online retail despite nearly

a decade of serious investment and Best Buy has still not been able to mount an effective response to Amazon after three decades.

The BE memo uses the lack of concern on the part of most customers as evidence that the exclusive and restrictive conditions Google dictated here were not a problem but, in retrospect, it’s clear that it was only a lack of understanding of the implications of online business that led customers to be unconcerned here. And when the BE memo refers to the customers who understood the implications here as sophisticated, that’s relative to people in lines of business where leadership tended to not understand the internet. While these customers are sophisticated by comparison to a retailer that took two decades to mount a serious response to the threat Amazon poses to their business, if you just talked to people in the tech industry at the time, you wouldn’t need to find a particularly sophisticated individual to find someone who understood what was going on. It was generally understood that retail revenue and even moreso, retail profit was going to move online, and you’d have to find someone who was extremely unusually out of the loop to find someone who didn’t at least roughly understand the implications here.

There’s a lengthy discussion on search and scale in both the BC and BE memos. On this topic, the BE memo seems wrong and the implications of the BC memo are, if not subtle, at least not obvious. Let’s start with the BE memo because that one’s simpler to discuss, although we’ll very briefly discuss the argument in the BC memo in order to frame the discussion in the BE memo. A rough sketch of the argument in the BC memo is that there are multiple markets (search, ads) where scale has a significant impact on product quality. Google’s own documents acknowledge this “virtuous cycle” where having more users lets you serve better ads, which gives you better revenue for ads and, likewise in search, having more scale gives you more data which can be used to improve results, which leads to user growth. And for search in particular, the BC memo claims that click data from users is of high importance and that more data allows for better results.

The BE memo claims that this is not really the case. On the importance of click data, the BE memo raises two large objections. First, that this is “contrary to the history of the general search market” and second, that “it is also contrary to the evidence that factors such as the quality of the web crawler and web index; quality of the search algorithm; and the type of content included in the search results [are as important or more important].

Of the first argument, the BE memo elaborates with a case that’s roughly “Google used to be smaller than it is today, and the click data at the time was sufficient, therefore being as large as Google used to be means that you have sufficient click data”. Independent of knowledge of the tech industry, this seems like a strange line of reasoning. “We now produce a product that’s 1/3 as good as our competitor for the same price, but that should be fine because our competitor

previously produced a product that's 1/3 as good as their current product when the market was less mature and no one was producing a better product" is generally not going to be a winning move. That's especially true in markets where there's a virtuous cycle between market share and product quality, like in search.

The second argument also seems like a strange argument to make even without knowledge of the tech industry in that it's a classic fallacious argument. It's analogous to saying something like "the BC memo claims that it's important for cars to have a right front tire, but that's contrary to evidence that it's at least as important for a car to have a left front tire and a right rear tire". The argument is even less plausible if you understand tech, especially search. Calling out the quality of the search algorithm as distinct doesn't feel quite right because scale and click data directly feed into algorithm development (and this is discussed at some length in the BE memo — the authors of the BC memo surely had access to the same information and, from their writing, seem to have had access to the argument). And as someone who's worked on search indexing, as much as I'd like to be agree with the BE memo and say that indexing is as important or more important than ranking, I have to admit that indexing is an easier and less important problem than ranking and likewise for crawling vs. ranking. This was generally understood at the time so, given the number of interviews FTC staff did, the authors of the BE memo should've known this as well. Moreover, given the "history of the general search market" which the BE memo refers to, even without talking to engineers, this should've been apparent.

For example, Cuil was famous for building a larger index than Google. While that's not a trivial endeavor, at the time, quite a few people had the expertise to build an index that rivaled Google's index in raw size or whatever other indexing metric you prefer, if given enough funding for a serious infra startup. Cuil and other index-focused attempts failed because having a large index without good search ranking is worth little. While it's technically true that having good ranking with a poor index is also worth little, this is not something we've really seen in practice because ranking is the much harder problem and a company that's competent to build a good search ranker will, as a matter of course, have a good enough index and good enough crawling.

As for the case in the BC memo, I don't know what the implications should be. The BC memo correctly points out that increased scale greatly improves search quality, that the extra data Bing got from the Yahoo greatly increased search quality and increased CTR, that further increased scale should be expected to continue to provide high return, that the costs of creating a competitor to Google are high (Bing was said to be losing \$2B/yr at the time and was said to be spending \$4.5B/yr "developing its algorithms and building the physical capacity necessary to operate Bing"), and that Google undertook actions that might be deemed anticompetitive which disadvantaged Bing's compared to the counterfactual world where Google did not take those

actions, and they make a similar case for ads. However, despite the strength of the stated BC memo case and the incorrectness of the stated BE memo case, the BE memo's case is correct in spirit, in that there are actions Microsoft could've taken but did not in order to compete much more effectively in search and one could argue that the FTC shouldn't be in the business of rescuing a company from competing ineffectively.

Personally, I don't think it's too interesting to discuss the position of the BC memo vs. the BE memo at length because the positions the BE memo takes seem extremely weak. It's not fair to call it a straw man because it's a real position, and one that carried the day at the FTC, but the decision to take action or not seemed more about philosophy than the arguments in the memos. But we can discuss what else might've been done.

id="what-might-ve-happened">What might've happened

What happened after the FTC declined to pursue antitrust action was that Microsoft effectively defunded Bing as a serious bet, taking resources that could've gone to continuing to fund a very expensive fight against Google, and moving them to other bets that it deemed to be higher ROI. The big bets Microsoft pursued were Azure, Office, and HoloLens (and arguably Xbox). Hololens was a pie-in-the-sky bet, but Azure and Office were lines of business where Microsoft could, instead of fighting an uphill battle where their competitor can use its dominance in related markets to push around competitors, Microsoft could fight downhill battles where they can use their dominance in related markets to push around competitors, resulting in a much higher return per dollar invested. As someone who worked on Bing and thought that BIng had the potential to seriously compete with Google given sustained, unprofitable, heavy investment, I find that disappointing but also likely the correct business decision. If you look at any particular sub-market, like Teams vs. Slack, the Microsoft product doesn't need to be nearly as good as the competing product to take over the market, which is the opposite of the case in search, where Google's ability to push competitors around means that Bing would have to be much better than Google to attain marketshare parity.

Based on their public statements, Biden's DoJ Antitrust AAG appointee, Jonathan Kanter, would argue for pursuing antitrust action under the circumstances, as would Biden's FTC commissioner and chair appointee Lina Khan. Prior to her appointment as FTC commissioner and chair, Khan was probably best known for writing Amazon's Antitrust Paradox, which has been influential as well as controversial. Obama appointees, who more frequently agreed with the kind of reasoning from the BE memo, would have argued against antitrust action and the investigation under discussion was stopped on their watch. More broadly, they argued against the philosophy driving Kanter and Khan. Obama's FTC Commissioner appointee, GMU economist and legal scholar Josh Wright actually wrote a rebuttal titled "Requiem for a Paradox: The Dubious Rise and Inevitable

Fall of Hipster Antitrust”, a scathing critique of Khan’s position.

If, in 2012, the FTC and DoJ were run by Biden appointees instead of Obama appointees, what difference would that have made? We can only speculate, but one possibility would be that they would’ve taken action and then lost, as happened with the recent cases against Meta and Microsoft which seem like they would not have been undertaken under an Obama FTC and DoJ. Under Biden appointees, there’s been much more vigorous use of the laws that are on the books, the Sherman Act, the Clayton Act, the FTC Act, the Robinson–Patman Act, as well as “smaller” antitrust laws, but the opinion of the courts hasn’t changed under Biden and this has led to a number of unsuccessful antitrust cases in tech. Both the BE and BC memos dedicate significant space to whether or not a particular line of reasoning will hold up in court. Biden’s appointees are much less concerned with this than previous appointees and multiple people in the DoJ and the FTC are on the record saying things like “it is our duty to enforce the law”, meaning that when they see violations of the antitrust laws that were put into place by elected officials, it’s their job to pursue these violations even if courts may not agree with the law.

Another possibility is that there would’ve been some action, but the action would’ve been in line with most corporate penalties we see. Something like a small fine that costs the company an insignificant fraction of marginal profit they made from their actions, or some kind of consent decree (basically a cease and desist), where the company will be required to stop doing specific actions while keeping their marketshare, keeping the main thing they wanted to gain, a massive advantage in a market dominated by network effects. Perhaps there will be a few more meetings where “[w]e didn’t take notes for obvious reasons” to work around the new limitations and business as usual will continue. Given the specific allegations in the FTC memos and the attitudes of the courts at the time, my guess is that something like this second set of possibilities would’ve been the most likely outcome had the FTC proceeded with their antitrust investigation instead of dropping it, some kind of nominal victory that makes little to no difference in practice. Given how long it takes for these cases to play out, it’s overwhelmingly likely that Microsoft would’ve already scaled back its investment in Bing and moved Bing from a subsidized bet it was trying to grow to a profitable business it wanted to keep by the time any decision was made. There are a number of cases that were brought by other countries which had remedies that were in line with what we might’ve expected if the FTC investigation continued. On Google using market power in mobile to push software Google wants to nearly all Android phones, an EU and was nominally successful but made little to no difference in practice. Cristina Caffara of the Centre for Economic Policy Research characterized this as

Europe has failed to drive change on the ground. Why? Because we told them, don’t do it again, bad dog, don’t do it again. But in fact, they all went and said ‘ok, ok’, and then

went out, ran back from the back door and did it again, because they’re smarter than the regulator, right? And that’s what happens.

So, on the tying case, in Android, the issue was, don’t tie again so they say, “ok, we don’t tie”. Now we got a new system. If you want Google Play Store, you pay \$100. But if you want to put search in every entry point, you get a discount of \$100 ... the remedy failed, and everyone else says, “oh, that’s a nice way to think about it, very clever”

Another pair of related cases are Yandex’s Russian case on mobile search defaults and a later EU consent decree. In 2015, Yandex brought a suit about mobile default status on Android in Russia, which was settled by adding a “choice screen” which has users pick their search engine without preferencing a default. This immediately caused Yandex to start gaining marketshare on Google and Yandex eventually surpassed Google in marketshare in Russia according to statcounter . In 2018, the EU required a similar choice screen in Europe, which didn’t make much of a difference , except maybe sort of in the Czech republic. There are a number of differences between the situation in Russia and in the EU. One, arguably the most important, is that when Yandex brought the case against Google in Russia, Yandex was still fairly competitive, with marketshare in the high 30% range. At the time of the EU decision in 2018, Bing was the #2 search engine in Europe, with about 3.6% marketshare. Giving consumers a choice when one search engine completely dominates the market can be expected to have fairly little impact. One argument the BE memo heavily relies on is the idea that, if we intervene in any way, that could have bad effects down the line, so we should be very careful and probably not do anything, just in case. But in these winner-take-most markets with such strong network effects, there’s a relatively small window in which you can cheaply intervene. Perhaps, and this is highly speculative, if the FTC required a choice screen in 2012, Bing would’ve continued to invest enough to at least maintain its marketshare against Google.

For verticals, in shopping, the EU required some changes to how Google presents results in 2017. This appears to have had little to no impact, being both perhaps 5-10 years too late and also a trivial change that wouldn’t have made much difference even if enacted a decade earlier. The 2017 ruling came out of a case that started in 2010, and in the 7 years it took to take action, Google managed to outcompete its vertical competitors, making them barely relevant at best.

Another place we could look is at the Microsoft antitrust trial. That’s a long story, at least as long as this document, but to very briefly summarize, in 1990, the FTC started an investigation over Microsoft’s allegedly anticompetitive conduct. A vote to continue the investigation ended up in a 2-2 tie, causing the investigation to be closed. The DoJ then did its own investigation, which led to a consent decree that was generally considered to not be too effective. There was then a 1998 suit by the DoJ about Microsoft’s use of monopoly power in the browser market, which initially led to a

decision to break Microsoft up. But, on appeal, the breakup was overturned, which led to a settlement in 2002. A major component of the 1998 case was about browser bundling and Microsoft's attack on Netscape. By the time the case was settled, in 2002, Netscape was effectively dead. The parts of the settlements having to do with interoperability were widely regarded as ineffective at the time, not only because Netscape was dead, but because they weren't going to be generally useful. A number of economists took the same position as the BE memo, that no intervention should've happened at the time and that any intervention is dangerous and could lead to a fettering of innovation. Nobel Prize winning economist Milton Friedman wrote a Cato Policy Forum essay titled "The Business Community's Suicidal Impulse", predicting that tech companies calling for antitrust action against Microsoft were committing suicide, and that a critical threshold had been passed and that this would lead to the bureaucratization of Silicon Valley

When I started in this business, as a believer in competition, I was a great supporter of antitrust laws; I thought enforcing them was one of the few desirable things that the government could do to promote more competition. But as I watched what actually happened, I saw that, instead of promoting competition, antitrust laws tended to do exactly the opposite, because they tended, like so many government activities, to be taken over by the people they were supposed to regulate and control. And so over time I have gradually come to the conclusion that antitrust laws do far more harm than good and that we would be better off if we didn't have them at all, if we could get rid of them. But we do have them.

Under the circumstances, given that we do have antitrust laws, is it really in the self-interest of Silicon Valley to set the government on Microsoft? ... you will rue the day when you called in the government. From now on the computer industry, which has been very fortunate in that it has been relatively free of government intrusion, will experience a continuous increase in government regulation. Antitrust very quickly becomes regulation. Here again is a case that seems to me to illustrate the suicidal impulse of the business community.

In retrospect, we can see that this wasn't correct and, if anything, was the opposite of correct. On the idea that even attempting antitrust action against Microsoft would lead to an inevitable increase in government intervention, we saw the opposite, a two-decade long period of relatively light regulation and antitrust activity. And in terms of the impacts on innovation, although the case against Microsoft was too little and too late to save Netscape, Google's success appears to be causally linked to the antitrust trial. At one point, in the early days of Google, when Google had no market power and Microsoft effectively controlled how people access the internet, Microsoft internally discussed proposals aimed at killing Google. One proposal involved redirecting users who tried to navigate to Google to Bing (at the time, called MSN Search, and of course this was before Chrome existed and IE dominated the browser market). Another idea was to put up a big scary warning that warned users that Google

was dangerous, much like the malware warnings browsers have today. Gene Burrus, a lawyer for Microsoft at the time, stated that Microsoft chose not to attempt to stop users from navigating to google.com due to concerns about further antitrust action after they'd been through nearly a decade of serious antitrust scrutiny. People at both Google and Microsoft who were interviewed about this both believe that Microsoft would've killed Google had they done this so, in retrospect, we can see that Milton Friedman was wrong about the impacts of the Microsoft antitrust investigations and that one can make the case that it's only because of the antitrust investigations that web 1.0 companies like Google and Facebook were able to survive, let alone flourish.

Another possibility is that a significant antitrust action would've been undertaken, been successful, and been successful quickly enough to matter. It's possible that, by itself, a remedy wouldn't have changed the equation for Bing vs. Google, but if a reasonable remedy was found and enacted, it still could've been in time to keep Yelp and other vertical sites as serious concerns and maybe even spur more vertical startups. And in the hypothetical universe where people with the same philosophy as Biden's appointees were running the FTC and the DoJ, we might've also seen antitrust action against Microsoft in markets where they can leverage their dominance in adjacent markets, making Bing a more appealing area for continued heavy investment. Perhaps that would've resulted in Bing being competitive with Google and the aforementioned concerns that "sophisticated customers" like Amazon and IAC had may not have come to pass. With antitrust against Microsoft and other large companies that can use their dominance to push competitors around, perhaps Slack would still be an independent product and we'd see more startups in enterprise tools (a number of commenters believe that Slack was basically forced into being acquired because it's too difficult to compete with Teams given Microsoft's dominance in related markets). And Slack continuing to exist and innovate is small potatoes — the larger hypothetical impact would be all of the new startups and products that would be created that no one even bothers to attempt because they're concerned that a behemoth with an integrated bundle like Microsoft would crush their standalone product. If you add up all of these, if not best-case, at least very-good-case outcomes for antitrust advocates, one could argue that consumers and businesses would be better off. But, realistically, it's hard to see how this very-good-case set of outcomes could have come to pass.

Coming back to the FTC memo, if we think about what it would take to put together a set of antitrust actions that actually fosters real competition, that seems extraordinarily difficult. A number of the more straightforward and plausible sounding solutions are off the table for political reasons, due to legal precedent, or due to arguments like the Boies argument we referenced or some of the arguments in the BE memo that are clearly incorrect, but appear to be convincing to very important people.

For the solutions that seem to be on the table, weighing the harms caused by them is non-trivial. For example, let's say the FTC mandated a mobile and desktop choice screen in 2012. This would've killed Mozilla in fairly short order unless Mozilla completely changed its business model because Mozilla basically relies on payments from Google for default status to survive. We've seen with Opera that even when you have a superior browser that introduces features that other browsers later copy, which has better performance than other browsers, etc., you can't really compete with free browsers when you have a paid browser. So then we would've quickly been down to IE/Edge and Chrome. And in terms of browser engines, just Chrome after not too long as Edge is now running Chrome under the hood. Maybe we can come up with another remedy that allows for browser competition as well, but the BE memo isn't wrong to note that antitrust remedies can cause other harms.

Another example which highlights the difficulty of crafting a politically suitable remedy are the restrictions the Bundeskartellamt imposed against Facebook, which have to do with user privacy and use of data (for personalization, ranking, general ML training, etc.), which is considered an antitrust issue in Germany. Michal Gal, Professor and Director of the Forum on Law and Markets at the University of Haifa pointed out that, of course Facebook, in response to the rulings, is careful to only limit its use of data if Facebook detects that you're German. If the concern is that ML models are trained on user data, this doesn't do much to impair Facebook's capability. Hypothetically, if Germany had a tech scene that was competitive with American tech and German companies were concerned about a similar ruling being leveled against them, this would be disadvantageous to nascent German companies that initially focus on the German market before expanding internationally. For Germany, this is only a theoretical concern as, other than SAP, no German company has even approached the size and scope of large American tech companies. But when looking at American remedies and American regulation, this isn't a theoretical concern, and some lawmakers will want to weigh the protection of American consumers against the drag imposed on American firms when compared to Korean, Chinese, and other foreign firms that can grow in local markets with fewer privacy concerns before expanding to international markets. This concern, if taken seriously, could be used to argue against nearly any pro-antitrust action argument.

id="what-can-we-do-going-forward">What can we do going forward?

This document is already long enough, so we'll defer a detailed discussion of policy specifics for another time, but in terms of high-level actions, one thing that seems like it would be helpful is to have tech people intimately involved in crafting remedies and regulation as well as during investigations². From the directors memos on the 2011-2021 FTC investigation that are publicly available, it would appear this was not done because the arguments from the BE memos that wouldn't pass the sniff test for a tech person appear

to have been taken seriously. Another example is the one EU remedy that Cristina Caffara noted was immediately worked around by Google, in a way that many people in tech would find to be a delightful "hack".

There's a long history of this kind of "hacking the system" being lauded in tech going back to before anyone called it "tech" and it was just physics and electrical engineering. To pick a more recent example, one of the reasons Sam Altman become President of Y Combinator, which eventually led to him becoming CEO of Open AI was that Paul Graham admired his ability to hack systems; in his 2010 essay on founders, under the section titled "Naughtiness", Paul wrote:

Though the most successful founders are usually good people, they tend to have a piratical gleam in their eye. They're not Goody Two-Shoes type good. Morally, they care about getting the big questions right, but not about observing proprieties. That's why I'd use the word naughty rather than evil. They delight in breaking rules, but not rules that matter. This quality may be redundant though; it may be implied by imagination.

Sam Altman of Loopt is one of the most successful alumni, so we asked him what question we could put on the Y Combinator application that would help us discover more people like him. He said to ask about a time when they'd hacked something to their advantage—hacked in the sense of beating the system, not breaking into computers. It has become one of the questions we pay most attention to when judging applications.

Or, to pick one of countless examples from Google, in order to reduce travel costs at Google, Google engineers implemented a system where they computed some kind of baseline "expected cost for flights, and then gave people a credit for taking flights that came in under the baseline costs that could be used to upgrade future flights and travel accommodations. This was a nice experience for employees compared to what stodgier companies were doing in terms of expense limits and Google engineers were proud of creating a system that made things better for everyone, which was one kind of hacking the system. The next level of hacking the system was when some employees optimized their flights and even set up trips to locations that were highly optimizable (many engineers would consider this a fun challenge, a variant of classic dynamic programming problems that are given in interviews, etc.), allowing them to upgrade to first class flights and the nicest hotels.

When I've talked about this with people in management in traditional industries, they've frequently been horrified and can't believe that these employees weren't censured or even fired for cheating the system. But when I was at Google, people generally found this to be admirable, as it exemplified the hacker spirit.

We can see, from the history of antitrust in tech going back at least two decades, that courts, regulators, and legislators

have not been prepared for the vigor, speed, and delight with which tech companies hack the system.

And there's precedent for bringing in tech folks to work on the other side of the table. For example, this was done in the big Microsoft antitrust case. But there are incentive issues that make this difficult at every level that stem from, among other things, the sheer amount of money that tech companies are willing to pay out. If I think about tech folks I know who are very good at the kind of hacking the system described here, the ones who want to be employed at big companies frequently make seven figures (or more) annually, a sum not likely to be rivaled by an individual consulting contract with the DoJ or FTC. If we look at the example of Microsoft again, the tech group that was involved was managed by Ron Schnell, who was taking a break from working after his third exit, but people like that are relatively few and far between. Of course there are people who don't want to work at big companies for a variety of reasons, often moral reasons or a dislike of big company corporate politics, but most people I know who fit that description haven't spent enough time at big companies to really understand the mechanics of how big companies operate and are the wrong people for this job even if they're great engineers and great hackers.

At an antitrust conference a while back, a speaker noted that the mixing and collaboration between the legal and economics communities was a great boon for antitrust work. Notably absent from the speech as well as the conference were practitioners from industry. The conference had the feel of an academic conference, so you might see CS academics at the conference some day, but even if that were to happen, many of the policy-level discussions are ones that are outside the area of interest of CS academics. For example, one of the arguments from the BE memo that we noted as implausible was the way they used MAU to basically argue that switching costs were low. That's something outside the area of research of almost every CS academic, so even if the conference were to expand and bring in folks who work closely with tech, the natural attendees would still not be the right people to weigh in on the topic when it comes to the plausibility of nitty gritty details.

Besides the aforementioned impact on policy discussions, the lack of collaboration with tech folks also meant that, when people spoke about the motives of actors, they would often make assumptions that were unwarranted. On one specific example of what someone might call a hack of the system, the speaker described an exec's reaction (high-fives, etc.), and inferred a contempt for lawmakers and the law that was not in evidence. It's possible the exec in question does, in fact, have a contempt and disdain for lawmakers and the law, but that celebration is exactly what you might've seen after someone at Google figured out how to get upgraded to first class "for free" on almost all their flights by hacking the system at Google, which wouldn't indicate contempt or disdain at all.

Coming back to the incentive problem, it goes beyond getting people who understand tech on the other side of the table in antitrust discussions. If you ask Capitol Hill staffers who were around at the time, the general belief is that the primary factor that scuttled the FTC investigation was Google's lobbying, and of course Google and other large tech companies spend more on lobbying than entities that are interested in increased antitrust scrutiny.

And in the civil service, if we look at the lead of the BC investigation and the first author on the BC memo, they're now Director and Associate General Counsel of Competition and Regulatory Affairs at Facebook. I don't know them, so I can't speak to their motivations, but if I were offered as much money as I expect they make to work on antitrust and other regulatory issues at Facebook, I'd probably take the offer. Even putting aside the pay, if I was a strong believer in the goals of increased antitrust enforcement, that would still be a very compelling offer. Working for the FTC, maybe you lead another investigation where you write a memo that's much stronger than the opposition memo, which doesn't matter when a big tech company pours more lobbying money into D. C. and the investigation is closed. Or maybe your investigation leads to an outcome like the EU investigation that led to a "choice screen" that was too little and far too late. Or maybe it leads to something like the Android Play Store untying case where, seven years after the investigation was started, an enterprising Google employee figures out a "hack" that makes the consent decree useless in about five minutes. At least inside Facebook, you can nudge the company towards what you think is right and have some impact on how Facebook treats consumers and competitors.

Looking at it from the standpoint of people in tech (as opposed to people working in antitrust), in my extended social circles, it's common to hear people say "I'd never work at company X for moral reasons". That's a fine position to take but, almost everyone I know who does this ends up working at a much smaller company that has almost no impact on the world. If you want to take a moral stand, you're more likely to make a difference by working from the inside or finding a smaller direct competitor and helping it become more successful.

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id="appendix-non-statements">Appendix: non-statements

This is analogous to the "non-goals" section of a technical design doc, but weaker, in that a non-goal in a design doc is often a positive statement that implies something that couldn't be implied from reading the doc, whereas the non-goal statements themselves don't add any informatio

Antitrust action against Google should have been pursued in 2012

Not that anyone should care what my opinion is, but if you'd asked me at the time if antitrust action should be pursued, I would've said "probably not". The case for antitrust action seems stronger now and the case against seems weaker, but you could still mount a fairly strong argument against antitrust action today.

Even if you believe that, *ceteris paribus*, antitrust action would've been good for consumers and the "very good case" outcome in "what might've happened" would occur if antitrust action were pursued, it's still not obvious that Google and other tech companies are the right target as opposed to (just for example) Visa and Mastercard's dominance of payments, hospital mergers leading to increased concentration that's had negative impacts on both consumers and workers, Ticketmaster's dominance, etc.. Or perhaps you think the government should focus on areas where regulation specifically protects firms, such as in shipping (which is except from the Sherman Act) or car dealerships (which have special protections in the law in many U. S. states that prevent direct sales and compel car companies to abide by their demands in certain ways), etc.

Weaker or stronger antitrust measures should be taken today

I don't think I've spent enough time reading up on the legal, political, historical, and philosophical background to have an opinion on what should be done, but I know enough about tech to point out a few errors that I've seen and to call out common themes in these errors.

id="bc-staff-memo">BC Staff Memo

By "Barbara R. Blank, Gustav P. Chiarello, Melissa Westman-Cherry, Matthew Accornero, Jennifer Nagle, Anticompetitive Practices Division; James Rhilinger, Healthcare Division; James Frost, Office of Policy and Coordination; Priya B. Viswanath, Office of the Director; Stuart Hirschfeld, Danica Noble, Northwest Region; Thomas Dahdouh, Western Region-San Francisco, Attorneys; Daniel Gross, Robert Hilliard, Catherine McNally, Cristobal Ramon, Sarah Sajewski, Brian Stone, Honors Paralegals; Stephanie Langley, Investigator"

Dated August 8, 2012

id="executive-summary">Executive Summary

Google is dominant search engine and seller of search ads

This memo addresses 4 of 5 areas with anticompetitive conduct; mobile is in a supplemental memo

Google has monopoly power in the U. S. in Horizontal Search; Search Advertising; and Syndicated Search and Search Advertising

On the question of whether Google has unlawfully preferred its own content while demoting rivals, we do not recommend the FTC proceed; it's a close call and case

law is not favorable to anticompetitive product design and Google's efficiency justifications are strong and at there's some benefit to users

On whether Google has unlawfully scraped content from vertical rivals to improve their own vertical products, recommending condemning as a conditional refusal to deal under Section 2

Prior voluntary dealing was mutually beneficial

Threats to remove rival content from general search designed to coerce rivals into allowing Google to use their content for Google's vertical product

Natural and probable effect is to diminish incentives of vertical website R&D

On anticompetitive contractual restrictions on automated cross-management of ad campaigns, restrictions should be condemned under Section 2

They limit ability of advertisers to make use of their own data, reducing innovation and increasing transaction costs for advertisers and third-party businesses

Also degrade the quality of Google's rivals in search and search advertising

Google's efficiency justifications appears to be pretextual

On anticompetitive exclusionary agreements with websites for syndicated search and search ads, Google should be condemned under Section 2

Only modest anticompetitive effects on publishers, but deny scale to competitors, competitively significant to main rival (Bing) as well as significant barrier to entry in longer term

Google's efficiency justifications are, on balance, non-persuasive

Possible remedies

Could be required to provide an opt-out for snippets (reviews, ratings) from Google's vertical properties while retaining snippets in web search and/or Universal Search on main search results page

Could be required to limit use of content indexed from web search results

Campaign management restrictions

Could be required to remove problematic contractual restrictions from license agreements

Exclusionary syndication agreements

Could be enjoined from entering into exclusive search agreements with search syndication partners and required

to loosen restrictions surrounding syndication partners' use of rival search ads

There are a number of risks to case, not named in summary except that Google can argue that Microsoft's most efficient distribution channel is bing.com and that any scale MS might gain will be immaterial to Bing's competitive position

Staff concludes Google's conduct has resulted and will result in real harm to consumer, innovation in online search and ads.

id="i-history-of-the-investigation-and-related-proceedings">I. HISTORY OF THE INVESTIGATION AND RELATED PROCEEDINGS

id="a-ftc-investigation">A. FTC INVESTIGATION

Compulsory process approved on June 03 2011

Received over 2M docs (9.5M pages) "and have reviewed many thousands of those documents"

Reviewed documents procured to DoJ in Google-Yahoo (2008) and ITA (2010) investigations and documents produced in response to European Commission and U. S. State investigations

Interviewed dozens of parties including vertical competitors in travel, local, finance, and retail; U.. advertisers and ad agencies; Google U. S. syndication and distribution partners; mobile device manufacturers and wireless carriers

17 investigational hearings of Google execs & employees

id="b-european-commission-investigation">B. EUROPEAN COMMISSION INVESTIGATION

Parallel investigation since November 2010

May 21, 2012: Commissioner Joaquin Almunia issued letter signaling EC's possible intent to issue Statement of Objections for abuse of dominance in violation of Article 102 of EC Treaty

"favourable treatment of its own vertical search services as compared to those of its competitors in its natural search results"

"practice of copying third party content" to supplement own vertical content

"exclusivity agreements with publishers for the provision of search advertising intermediation services"

"restrictions with regard to the portability and cross-platform management of online advertising campaigns"

offered opportunity to resolve concerns prior to issuance of SO by producing description of solutions

Google denied infringement of EU law, but proposed several commitments to address stated concerns

FTC staff coordinated with EC staff

id="c-multi-state-investigation">C. MULTI-STATE INVESTIGATION

Texas investigating since June 2010, leader of multi-state working group

FTC working closely with states

id="d-private-litigation">D. PRIVATE LITIGATION

Several private lawsuits related to issues in our investigation; all dismissed

Two categories, manipulation of search rankings and increases in minimum prices for AdWords search ads

Kinderstart.com LLC v. Google, Inc.,¹¹ and SearchKing, Inc. v. Google Tech., Inc., plaintiffs alleged that Google unfairly demoted their results

SearchKing court ruled that Google's rankings are constitutionally protected opinion; even malicious manipulation of rankings would not expose Google to tort liability

Kinderstart court rejected Google search being an essential facility for vertical websites

In AdsWords cases, plaintiffs argue that Google increased minimum bids for keywords they'd purchases, making those keywords effectively unavailable, depriving plaintiff website of traffic

TradeComet.com, LLC v. Google, Inc. dismissed for improper venue and Google, Inc. v. myTriggers.com, Inc. dismissed for failing to describe harm to competition has a whole

both dismissed with little discussion of merits

Person V. Google, Inc.: Judge Fogel of the Northern District of California criticized plaintiff's market definition, finding no basis for distinguishing "search advertising market" from larger market for internet advertising

id="ii-statement-of-facts">II. STATEMENT OF FACTS

id="a-the-parties">A. THE PARTIES

id="1-google">1. Google

Products include "horizontal" search engine and integrated "vertical" websites that focus on specific areas (product or shopping comparisons, maps, finance, books, video), search advertising via AdWords, search and search advertising syndication through AdSense, computer and software applications such as Google Toolbar, Gmail, Chrome, also

have Android for mobile and applications for mobile devices and recently acquired Motorola Mobility

32k people, \$38B annual revenue

id="2-general-search-competitors">2. General search competitors

id="a-microsoft">a. Microsoft

MSN search released in 1998, rebranded Bing in 2009. Filed complaints against Google in 2011 with FTC and EC

id="b-yahoo">b. Yahoo

Partnership with Bing since 2010; Bing provides search results and parties jointly operate a search ad network

id="3-major-vertical-competition">3. Major Vertical Competition

In general, these companies complain that Google's practice of preferencing its own vertical results has negatively impacted ability to compete for users and advertisers

Product search directly competes with Google Product Search

eBay

product search competes with Google Product Search

NexTag

shopping comparison website that competes with Google Product Search

UK product comparison website that competes with Google Product Search

Complaint to EC, among others, prompted EC to open its investigation into Google's web search practices

First vertical website to publicly accuse Google of preferencing its own vertical content over competitors on Google's search page

competes against Google's fledgling Google Flight Search

TripAdvisor

TripAdvisor competes with Google Local (formerly Google Places)

has complained that Google has appropriated / scraped its user-generated reviews, placing them on Google's own local property

has complained that Google has appropriated / scraped its user-generated reviews, placing them on Google's own local property

Competes with Google's recently introduced Google Plus

has complained that Google's preferencing of Google Plus results over Facebook results is negatively impacting ability to compete for users

id="b-industry-background">B. INDUSTRY BACKGROUND

id="1-general-search">1. General Search

[nice description of search engines for lay people omitted]

id="2-online-advertising">2. Online Advertising

Google's core business is ads; 96% of its nearly \$38B in revenue was from ad sales

[lots of explanations of ad industry for lay people, mostly omitted]

Reasons advertisers have shifted business to web include high degree of tracking possible and quantifiable, superior, ROI

Search ads make up most of online ad spend, primarily because advertisers believe search ads provided best precision in IDing customers, measurability, and the highest ROI

Online advertising continues to evolve, with new offerings that aren't traditional display or search ads, such as contextual ads, re-targeted behavioral ads, and social media ads

these new ad products don't account for a significant portion of online ads today and, with the exception of social media ads, appear to have only limited potential for growth [Surely video is pretty big now, especially if you include "sponsorships" and not just ads inserted by the platform?]

id="3-syndicated-search-and-search-advertising">3. Syndicated Search and Search Advertising

Search engines "syndicate" search and/or search ads

E.g., if you go "AOL or Ask.com", you can do a search which is powered by a search Provider, like Google

Publisher gets to keep user on own platform, search provider gets search volume and can monetize traffic

End-user doesn't pay; publisher pays Google either on cost-per-user-query basis or by accepting search ads and spitting revenues from search ads run on publisher's site. Revenue sharing agreement often called "traffic acquisition cost" (TAC)

Publishers can get search ads without offering search (AdSense) and vice versa

id="4-mobile-search">4. Mobile Search

Focus of search has been moving from desktop to "rapid emerging — and lucrative — frontier of mobile"

Android at forefront; has surpassed iPhone in U. S. market share

Mobile creates opportunities for location-based search ads; even more precise intent targeting than desktop search ads

Google and others have signed distribution agreements with device makers and wireless carriers, so user-purchased devices usually come pre-installed with search and other apps

id="c-the-significance-of-scale-in-internet-search">C. THE SIGNIFICANCE OF SCALE IN INTERNET SEARCH

Scale (user queries and ad volume) important to competitive dynamics

id="1-search-query-volume">1. Search Query Volume

Microsoft claims it needs higher query volume to improve Bing

Logs of queries can be used to improve tail queries

Suggestions, instant search, spelling correction

Trend identification, fresh news stories

Click data important for evaluating search quality

Udi Manber (former Google chief of search quality) testimony: "The ranking itself is affected by the click data. If we discover that, for a particular query, hypothetically, 80 percent of people click on Result No. 2 and only 10 percent click on Result No. 1, after a while we figure out, well, probably Result 2 is the one people want. So we'll switch it."

Testimony from Eric Schmidt and Sergey Brin confirms click data important and provides feedback on quality of search results

Scale / volume allows more experiments

Larry and Sergei's annual letter in 2005 notes importance of experiments, running multiple simultaneous experiments

More scale allows for more experiments as well as for experiments to complete more quickly

Susan Athey (Microsoft chief economist) says Microsoft search quality team is greatly hampered by insufficient search volume to run experiments

2009 comment from Udi Manber: "The bottom line is this. If Microsoft had the same traffic we have their quality will improve *significantly*, and if we had the same traffic they have, ours will drop significantly. That's a fact"

id="2-advertising-volume">2. Advertising Volume

Microsoft claims they need more ad volume to improve relevance and quality of ads

More ads means more choices over what ads to serve to use, better matched ads / higher conversion rates

Also means more queries

Also has similar feedback loop to search

Increase volume of advertisers increases competitiveness for ad properties, gives more revenue to search engine

Allows search engine to amortize costs, re-invest in R&D, provide better advertiser coverage, revenue through revenue-sharing agreements to syndication partners (website publishers). Greater revenue to partners attracts more publishers and more advertisers

id="3-scale-curve">3. Scale Curve

Google acknowledges the importance of scale (outside of the scope of this particular discussion)

Google documents replete with references to "virtuous cycle" among users, advertisers, and publishers

Testimony from Google execs confirms this

But Google argues scale no longer matters at Google's scale or Microsoft's scale, that additional scale at Microsoft's scale would not "significantly improve" Microsoft search quality

Susan Athey argues that relative scale, Bing being 1/5th the size of Google, matters, not absolute size

Microsoft claims that 5% to 10% increase in query volume would be "very meaningful", notes that gaining access to Yahoo queries and ad volume in 2010 was significant for search quality and monetization

Claim that Yahoo query data increased click through rate for "auto suggest" from 44% to 61% [the timeframe here is July 2010 to September 2011 — too bad they didn't provide an A/B test here, since this more than 1 year timeframe allows for many other changes to impact the suggest feature as well; did that ship a major change here without A/B testing it? That seems odd]

Microsoft also claims search quality improvements due to experiment volume enabled by extra query volume

id="d-google-s-suspect-conduct">D. GOOGLE'S SUSPECT CONDUCT

Five main areas of staff investigation of alleged anticompetitive conduct:

id="1-google-s-preferencing-of-google-vertical-properties-within-its-search-engine-results-page-serp">1. Google's Preferencing of Google Vertical Properties Within Its Search Engine Results Page ("SERP")

Allegation is that Google's conduct is anticompetitive because "it forecloses alternative search platforms that might

operate to constraint Google's dominance in search and search advertising"

"Although it is a close call, we do not recommend that the Commission issue a complaint against Google for this conduct."

id="a-overview-of-changes-to-google-s-serp">a. Overview of Changes to Google's SERP

Google makes changes to UI and algorithms, sometimes without user testing

sometimes with testing with launch review process, typically including:

"the sandbox", internal testing by engineers

"SxS", side-by-side testing by external raters who compare existing results to proposed results

Testing on a small percent of live traffic

"launch report" for Launch Committee

Google claims to have run 8000 SxS tests and 2500 "live" click tests in 2010, with 500 changes launched

"Google's stated goal is to make its ranking algorithms better in order to provide the user with the best experience possible."

id="b-google-s-development-and-introduction-of-vertical-properties">b. Google's Development and Introduction of Vertical Properties

Google vertical properties launched in stages, initially around 2001

Google News, Froogle (shopping), Image Search, and Groups

Google has separate indexes for each vertical

Around 2005 ,Google realized that vertical search engines, i.e., aggregators in some categories were a "threat" to dominance in web search, feared that these could cause shift in some searches away from Google

From GOOG-Texas-1325832-33 (2010): "Vertical search is of tremendous strategic importance to Google. Otherwise the risk is that Google is the go-to place for finding information only in the cases where there is sufficiently low monetization potential that no niche vertical search competitor has filled the space with a better alternative."

2008 presentation titled "Online Advertising Challenges: Rise of the Aggregators":

"Issue 1. Consumers migrating to MoneySupermarket. Driver: General search engines not solving consumer queries as well as specialized vertical search Consequence: Increas-

ing proportion of visitors going directly to MoneySupermarket. Google Implication: Loss of query volumes."

Issue 2: "MoneySupermarket has better advertiser proposition. Driver: MoneySupermarket offers cheaper, lower risk (CPA-based) leads to advertisers. Google Implication: Advertiser pull: Direct advertisers switch spend to MoneySupermarket/other channels"

In response to this threat, Google invested in existing verticals (shopping, local) and invested in new verticals (mortgages, offers, hotel search, flight search)

id="c-the-evolution-of-display-of-google-s-vertical-properties-on-the-serp">c. The Evolution of Display of Google's Vertical Properties on the SERP

Google initially had tabs that let users search within verticals

In 2003, Marissa Mayer started developing "Universal Search" (launched in 2007), to put this content directly on Google's SERP. Mayer wrote:

"Universal Search is an effort to redesign the user interface of the main Google.com results page SO that Google deliver[s] the most relevant information to the user on Google.com no matter what corpus that information comes from. This design is motivated by the fact that very few users are motivated to click on our tabs, SO they often miss relevant results in the other corpora."

Prior to Universal Search launch, Google used "OneBoxes", which put vertical content above Google's SERP

After launching Universal Search, vertical results could go anywhere

id="d-google-s-preferential-display-of-google-vertical-properties-on-the-serp">d. Google's Preferential Display of Google Vertical Properties on the SERP

Google used control over Google SERP both to improve UX for searches and to maximize benefit to its own vertical properties

Google wanted to maximize percentage of queries that had Universal Search results and drive traffic to Google properties

In 2008, goal to "[i]ncrease google.com product search inclusion to the level of google.com searches with 'product intent', while preserving clickthrough rate." (GOOG-Texas-0227159-66)

Q1 2008, goal of triggering Product Universal on 6% of English searches

Q2 2008, goal changed to top OneBox coverage of 50% with 10% CTR and "[i]ncrease coverage on head queries. For example, we should be triggering on at least 5 of the top

10 most popular queries on amazon.com at any given time, rather than only one.”

“Larry thought product should get more exposure”, GOOG-ITA-04-0004120-46 (2009)

Mandate from exec meeting to push product-related queries as quickly as possible

Launch Report for one algorithm change: “To increase triggering on head queries, Google also implemented a change to trigger the Product Universal on google.com queries if they appeared often in the product vertical. “Using Exact Corpusboost to Trigger Product Onebox” compares queries on www.google.com with queries on Google Shopping, triggers the Product OneBox if the same query is often searched in Google Shopping, and automatically places the universal in position 4, regardless of the quality of the universal results or user “bias” for top placement of the box.’

“presentation stating that Google could take a number of steps to be “#1” in verticals, including “[e]ither [getting] high traffic from google.com, or [developing] a separate strong brand,” and asking: “How do we link from Search to ensure strong traffic without harming user experience or AdWords proposition for advertisers?””, GOOGFOX-000082469 (2009)

Jon Hanke, head of Google Local, to Marissa Mayer: “long term, I think we need to commit to a more aggressive path w/ google where we can show non-webpage results on google outside of the universal ‘box’ most of us on geo think that we won’t win unless we can inject a lot more of local directly into google results.”

“Google’s key strengths are: Google.com real estate for the 70MM of product queries/day in US/UK/DE alone”

“I think the mandate has to come down that we want to win [in local] and we are willing to take some hits [i.e., trigger incorrectly sometimes]. I think a philosophical decision needs to get made that results that are not web search results and that displace web pages are “OK” on google.com and nothing to be ashamed of. That would open the door to place page or local entities as ranked results outside of some ‘local universal’ container. Arguably for many queries all of the top 10 results should be local entities from our index with refinement options. The current mentality is that the google results page needs to be primarily about web pages, possibly with some other annotations if they are really, really good. That’s the big weakness that Bing is shooting at w/ the ‘decision engine’ pitch - not a sea of pointers to possible answers, but real answers right on the page. “

In spring 2008, Google estimated top placement of Product Universal would lead to loss of \$154M/yr on product queries. Ads team requested reduction in triggering frequency and Product Universal team objected, “We face strong competition and must move quickly.

Turning down onebox would hamper progress as follows - Ranking: Losing click data harms ranking; [t]riggering

Losing CTR and google.com query distribution data triggering accuracy; [c]omprehensiveness: Losing traffic harms merchant growth and therefore comprehensiveness; [m]erchant cooperation: Losing traffic reduces effort merchants put into offer data, tax, & shipping; PR: Turning off onebox reduces Google’s credibility in commerce; [u]ser awareness: Losing shopping-related UI on google.com reduces awareness of Google’s shopping features.“

“Google embellished its Universal Search results with photos and other eye-catching interfaces, recognizing that these design choices would help steer users to Google’s vertical properties”

“Third party studies show the substantial difference in traffic with prominent, graphical user interfaces”; “These ‘rich’ user interfaces are not available to competing vertical websites”

Google search results near or at top of SERP, pushing other results down, resulting in reduced CTR to “natural search results”

Google did this without comparing quality of Google’s vertical content to competitors or evaluating whether users prefer Google’s vertical content to displaced results

click-through from eBay indicates that (Jan-Apr 2012), Google Product Search appeared in top 5 position 64% of time when displayed and Google Product Search had lower CTR than web search in same position regardless of position [below is rank: natural result CTR / Google Shopping CTR / eBay CTR]

- 1: 38% / 21% / 31%
- 2: 21% / 14% / 20%
- 3: 16% / 12% / 18%
- 4: 13% / 9% / 11%
- 5: 10% / 8% / 10%
- 6: 8% / 6% / 9%
- 7: 7% / 5% / 9%
- 8: 6% / 2% / 7%
- 9: 6% / 3% / 6%
- 10 5% / 2% / 6%
- 11: 5% / 2% / 5%
- 12: 3% / 1% / 4%

Although Google tracks CTR and relies on CTR to improve web results, it hasn’t relied on CTR to rank Universal Search results against other web search results

Marissa Mayer said Google didn't use CTR "because it would take too long to move up on the SERP on the basis of user click-through rate"

Instead, "Google used occurrence of competing vertical websites to automatically boost the ranking of its own vertical properties above that of competitors"

If comparison shopping site was relevant, Google would insert Google Product search above any rival

If local search like Yelp or CitySearch was relevant, Google automatically returned Google Local at top of SERP

Google launched commission-based verticals, mortgage, flights, offers, in ad space reserved exclusively for its own properties

In 2012, Google announced that google product search would transition to paid and Google would stop including product listings for merchants who don't pay to be listed

Google's dedicated ads don't competition with other ads via AdWords and automatically get the most effective ad spots, usually above natural search results

As with Google's Universal results, its own ads have a rich user interface not available to competitors which results in higher CTR

id="e-google-s-demotion-of-competing-vertical-websites">e. Google's Demotion of Competing Vertical Websites

"While Google embarked on a multi-year strategy of developing and showcasing its own vertical properties, Google simultaneously adopted a strategy of demoting, or refusing to display, links to certain vertical websites in highly commercial categories"

"Google has identified comparison shopping websites as undesirable to users, and has developed several algorithms to demote these websites on its SERP. Through an algorithm launched in 2007, Google demoted all comparison shopping websites beyond the first two on its SERP"

"Google's own vertical properties (inserted into Google's SERP via Universal Search) have not been subject to the same demotion algorithms, even though they might otherwise meet the criteria for demotion."

Google has acknowledged that its own vertical sites meet the exact criteria for demotion

Additionally, Google's web spam team originally refused to add Froogle to search results because "[o]ur algorithms specifically look for pages like these to either demote or remove from the index."

Google's web spam team also refused to add Google's local property

id="f-effects-of-google-s-serp-changes-on-vertical-rivals">f. Effects of Google's SERP Changes on Vertical Rivals

"Google's prominent placement and display of its Universal Search properties, combined with the demotion of certain vertical competitors in Google's natural search results, has resulted in significant loss of traffic to many competing vertical websites"

"Google's internal data confirms the impact, showing that Google anticipated significant traffic loss to certain categories of vertical websites when it implemented many of the algorithmic changes described above"

"While Google's changes to its SERP led to a significant decrease in traffic for the websites of many vertical competitors, Google's prominent showcasing of its vertical properties led to gains in user share for its own properties"

"For example, Google's inclusion of Google Product Search as a Universal Search result took Google Product Search from a rank of seventh in page views in July 2007 to the number one rank by July 2008. Google product search leadership acknowledged that "[t]he majority of that growth has been driven through product search universal."

"Beyond the direct impact on traffic to Google and its rivals, Google's changes to its SERP have led to reduced investment and innovation in vertical search markets. For example, as a result of the rise of Google Product Search (and simultaneous fall of rival comparison shopping websites), NexTag has taken steps to reduce its investment in this area. Google's more recent launch of its flight search product has also caused NexTag to cease development of an 'innovative and competitive travel service.'"

id="2-google-s-scraping-of-rivals-vertical-content">2. Google's "Scraping" of Rivals' Vertical Content

"Staff has investigated whether Google has "scraped" - or appropriated - the content of rival vertical websites in order to improve its own vertical properties SO as to maintain, preserve, or enhance Google's monopoly power in the markets for search and search advertising. We recommend that the Commission issue a complaint against Google for this conduct."

In addition to developing its own vertical properties, Google scraped content from existing vertical websites (e.g., Yelp, TripAdvisor, Amazon) in order to improve its own vertical listings, "e.g., GOOG-Texas-1380771-73 (2009), at 71-72 (discussing importance of Google Places carrying better review content from Yelp)."

id="a-the-local-story">a. The "Local" Story

"Some local information providers, such as Yelp, TripAdvisor, and CitySearch, disapprove of the ways in which Google has made use of their content"

“Google recognized that review content, in particular, was “critical to winning in local search,” but that Google had an ‘unhealthy dependency’ on Yelp for much of its review content. Google feared that its heavy reliance on Yelp content, along with Yelp’s success in certain categories and geographies, could lead Yelp and other local information websites to siphon users’ local queries away from Google”

“concern that Yelp could become competing local search platforms” (Goog-Texas-0975467-97)

Google Local execs tried to convince Google to acquire Yelp, but failed

Yelp, on finding that Google was going to use reviews on its own property, discontinued its feed and asked for Yelp content to be removed from Google Local

“after offering its own review site for more than two years, Google recognized that it had failed to develop a community of users - and thus, the critical mass of user reviews - that it needed to sustain its local product.”, which led to failed attempt to buy Yelp

To address this problem, Google added Google Places results on SERP: “The listing for each business that came up as a search result linked the user directly to Google’s Places page, with a label indicating that hundreds of reviews for the business were available on the Places page (but with no links to the actual sources of those reviews). On the Places Page itself, Google provided an entire paragraph of each copied review (although not the complete review), followed by a link to the source of the review, such as Yelp (which it crawled for reviews) and TripAdvisor (which was providing a feed).”

Yelp noticed this in July 2010, that Google was featuring Yelp’s content without a license and protested to Google. TripAdvisor chose not to renew license with Google after finding same

Google implemented new policy that would ban properties from Google search if they didn’t allow their content to be used in Google Places

“GOOG-Texas-1041511-12 (2010), at 12 (“remove blacklist of yelp [reviews] from Web-extracted Reviews once provider based UI live”); GOOG-Texas-1417391-403 (2010), at 394 (“stating that Google should wait to publish a blog post on the new UI until the change to “unblacklist Yelp” is “live”).”

Along with this policy, launched new reviews product and seeded it reviews from 3rd party websites without attribution

Yelp, CitySearch, and TripAdvisor all complained and were all told that they could only remove their content if they were fully removed from search results. “This was not technically necessary - it was just a policy decision by Google.”

Yelp sent Google a C&D

Google claimed it was technically infeasible to remove Yelp content from Google Places without also banning Yelp from search result

Google later did this, making it clear that the claim that it was technically infeasible was false

Google still maintained that it would be technically infeasible to remove Yelp from Google Places without removing it from “local merge” interface on SERP. Staff believes this assertion is false as well because Google maintains numerous “blacklists” that prevent content from being shown in specific locations

Mayer later admitted during hearing that the infeasible claim was false and that Google feared consequences of allowing websites to opt out of Google Places while staying in “local merge”

“Yelp contends that Google’s continued refusal to link to Yelp on Google’s ‘local merge’ interface on the main SERP is simply retaliation for Yelp seeking removal from Google Places.”

“Publicly, Google framed its changes to Google Local as a redesign to move toward the provision of more original content, and thereby, to remove all third-party content and review counts from Google Local, as well as from the prominent “local merge” Universal Search interface on the main SERP. But the more likely explanation is that, by July 2011, Google had already collected sufficient reviews by bootstrapping its review collection on the display of other websites’ reviews. It no longer needed to display third-party reviews, particularly while under investigation for this precise conduct.”

id=“b-the-shopping-story”>b. The “Shopping” Story

[full notes omitted; story is similar to above, but with Amazon; similar claims of impossibility of removing from some places and not others; Amazon wanted Google to stop using Amazon star ratings, which Google claimed was impossible without blacklisting Amazon from all of web search, etc.; there’s also a parallel story about Froogle’s failure and Google’s actions after that]

id=“c-effects-of-google-s-scraping-on-vertical-rivals”>c. Effects of Google’s “Scraping” on Vertical Rivals

“Because Google scraped content from these vertical websites over an extended period of time, it is difficult to point to declines in traffic that are specifically attributable to Google’s conduct. However, the natural and probable effect of Google’s conduct is to diminish the incentives of companies like Yelp, TripAdvisor, CitySearch, and Amazon to invest in, and to develop, new and innovative content, as the companies cannot fully capture the benefits of their innovations”

id=“3-google-s-api-restrictions”>3. Google’s API Restrictions

“Staff has investigated whether Google’s restrictions on the automated cross-management of advertising campaigns has unlawfully contributed to the maintenance, preservation, or enhancement of Google’s monopoly power in the markets for search and search advertising. Microsoft alleges that these restrictions are anticompetitive because they prevent Google’s competitors from achieving efficient scale in search and search advertising. We recommend that the Commission issue a complaint against Google for this conduct.”

id=“a-overview-of-the-adwords-platform”>a. Overview of the AdWords Platform

To set up AdWords, advertisers prepare bids. Can have thousands or hundreds of thousands of keywords.

E.g., DirectTV might bid on “television”, “TV”, and “satellite” plus specific TV show names, such as “Friday Night Lights”, as well as misspellings

Bids can be calibrated by time and location

Advertisers then prepare ads (called “creatives”) and match with various groups of keywords

Advertisers get data from AdWords, can evaluate effectiveness and modify bids, add/drop keywords, modify creative

This is called “optimization” when done manually; expensive and time-intensive

Initially two ways to access AdWords system, AdWords Front End and AdWords Editor

Editor is a program. Allows advertisers to download campaign information from Google, make bulk changes offline, then upload changes back to AdWords

Advertisers would make so many changes that system’s capacity would be exceeded, causing outages

In 2004, Google added AdWords API to address problems

[description of what an API is omitted]

id=“b-the-restrictive-conditions”>b. The Restrictive Conditions

AdWords API terms and conditions non-negotiable, apply to all users

One restriction prevents advertisers from using 3rd party tool or have 3rd party use a tool to copy data from AdWords API into ad campaign on another search network

Another, can’t use 3rd party tool or have 3rd party use a tool to combine AdWords campaign data with data from another search engine

The two conditions above will be referred to as “the restrictive conditions”

“These restrictions essentially prevent any third-party tool developer or advertising agency from creating a tool that provides a single user interface for multiple advertising campaigns. Such tools would facilitate cross-platform advertising.”

“However, the restrictions do not apply to advertisers themselves, which means that very large advertisers, such as Amazon and eBay, can develop - and have developed - their own multi-homing tools that simultaneously manage campaigns across platforms”

“The advertisers affected are those whose campaign volumes are large enough to benefit from using the AdWords API, but too small to justify devoting the necessary resources to develop in-house the software and expertise to manage multiple search network ad campaigns.”

id=“c-effects-of-the-restrictive-conditions”>c. Effects of the Restrictive Conditions

id=“i-effects-on-advertisers-and-search-engine-marketers-sems”>i. Effects on Advertisers and Search Engine Marketers (“SEMs”)

Prevents development of tools that would allow advertisers from managing ad campaigns on multiple search ad networks simultaneously

Google routinely audits API clients for compliance

Google has required SEMs to remove functionality, “e.g., GOOGEC-0180810-14 (2010) (Trada); GOOGEC-0180815-16 (2010) (MediaPlex); GOOGEC-0181055-58 (2010) (CoreMetrics); GOOGEC-0181083-87 (2010) (Keybroker); GOOGEC-0182218-330 (2008) (Marin Software). 251 Acquisition IR (Sep. 12, 2011); Efficient Frontier IR (Mar. 5, 2012)”

Other SEMs have stated they would develop this functionality without restrictions

“Google anticipated that the restrictive conditions would eliminate SEM incentives to innovate.”, “GOOGKAMA-000004815 (2004), at 2.”

“Many advertisers have said they would be interested in buying a tool that had multi-homing functionality. Such functionality would be attractive to advertisers because it would reduce the costs of managing multiple ad campaigns, giving advertisers access to additional advertising opportunities on multiple search advertising networks with minimal additional investment of time. The advertisers who would benefit from such a tool appear to be the medium-sized advertisers, whose advertising budgets are too small to justify hiring a full service agency, but large enough to justify paying for such a tool to help increase their advertising opportunities on multiple search networks.”

id=“ii-effects-on-competitors”>ii. Effects on Competitors

Removing restrictions would increase ad spend on networks that compete with Google

Data on advertiser multi-homing show some effects of restrictive conditions. Nearly all the largest advertisers multi-home, but percentage declines as spend decreases

Advertisers would also multi-home with more intensity

Microsoft claims that multi-homing advertisers optimize their Google campaigns almost-daily, Microsoft campaigns less frequently, weekly or bi-weekly

Without incremental transaction costs, “all rational advertisers would multi-home”

Staff interviewed randomly selected small advertisers. Interviews “strongly supported” thesis that advertisers would multi-home if cross-platform optimization tool were available

Some advertisers don’t advertise on Bing due to lack of tool, the ones that do do less optimization

id=“d-internal-google-discussions-regarding-the-restrictions”>d. Internal Google Discussions Regarding the Restrictions

Internal discussions support the above

PM wrote the following in 2007, endorsed by director of PM Richard Holden:

“If we offer cross-network SEM in [Europe], we will give a significant boost to our competitors. Most advertisers that I have talked to in [Europe] don’t bother running campaigns on [Microsoft] or Yahoo because the additional overhead needed to manage these other networks outweighs the small amount of additional traffic. For this reason, [Microsoft] and Yahoo still have a fraction of the advertisers that we have in [Europe], and they still have lower average CPAs [cost per acquisition]”

“This last point is significant. The success of Google’s AdWords auctions has served to raise the costs of advertising on Google. With more advertisers entering the AdWords auctions, the prices it takes to win those auctions have naturally risen. As a result, the costs per acquisition on Google have risen relative to the costs per acquisition on Bing and Yahoo!. Despite these higher costs, as this document notes, advertisers are not switching to Bing and Yahoo! because, for many of them, the transactional costs are too great.”

Advertisers seek out SEMs and agencies for cross-network management technology and services;

The restrictive conditions make the market more inefficient;

Removing the restrictive conditions would “open up the market” and give Google the opportunity to compete with a best-in-class SEM tool with “a streamlined workflow”;

Removing the restrictive conditions would allow SEMs to improve their tools as well;

While there is a risk of additional spend going to competing search networks, it is unlikely that Google would be seriously harmed because “advertisers are going where the users are,” i.e., to Google

“internally, Google recognized that removing the restrictions would create a more efficient market, but acknowledged a concern that doing so might diminish Google’s grip on advertisers.”

“Nonetheless, following up on that meeting, Google began evaluating ways to improve the DART Search program. DART Search was a cross-network campaign management tool owned by DoubleClick, which Google acquired in 2008. Google engineers were looking at improving the DART Search product, but had to confront limitations imposed by the restrictive conditions. During his investigational hearing, Richard Holden steadfastly denied any linkage between the need to relax the restrictive conditions and the plans to improve DART Search.²⁷⁴ However, a series of documents - documents authored by Holden - explicitly link the two ideas.”

Dec 2008 Holden to SVP of ad products, Susan Wojcicki and others met.

Holden wrote: “[O]ne debate we are having is whether we should eliminate our API T&Cs requirement that AW [AdWords] features not be co-mingled with competitor network features in SEM cross-network tools like DART Search. We are advocating that we eliminate this requirement and that we build a much more streamlined and efficient DART Search offering and let SEM tool provider competitors do the same. There was some debate about this, but we concluded that it is better for customers and the industry as a whole to make things more efficient and we will maximize our opportunity by moving quickly and providing the most robust offering”

Feb 2009, Holden wrote exec summary for DART, suggested Google ““alter the AdWords Ts&Cs to be less restrictive and produce the leading cross-network toolset that increases advertiser/agency efficiency.” to “[r]educe friction in the search ads sales and management process and grow the industry faster”

Larry Page rejected this. Afterwards, Holden wrote “We’ve heard that and we will focus on building the product to be industry-leading and will evaluate it with him when it is done and then discuss co-mingling and enabling all to do it.”

Sep 2009, API PM raised possibility of eliminating restrictive conditions to help DART. Comment from Holden:

“I think the core issue on which I’d like to get Susan’s take is whether she sees a high risk of existing spend being channeled to MS/Yahoo! due to a more lenient official policy on campaign cloning. Then, weigh that risk against the benefits: enabling DART Search to compete better against non-compliant SEM tools, more industry goodwill, easier compliance enforcement. Does that seem like the right high level message?”

“The documents make clear that Google was weighing the efficiency of relaxing the restrictions against the potential cost to Google in market power”

“At a January 2010 meeting, Larry Page decided against removing or relaxing the restrictive conditions. However, there is no record of the rationale for that decision or what weight was given to the concern that relaxing the restrictive conditions might result in spend being channeled to Google’s competitors. Larry Page has not testified. Holden testified that he did not recall the discussion. The participants at the meeting did not take notes “for obvious reasons.” Nonetheless, the documents paint a clear picture: Google rejected relaxing the API restrictions, and at least part of the reason for this was fear of diverting advertising spend to Microsoft.”

Holden to Wojcicki: “We didn’t take notes for obvious reasons (hence why I’m not elaborating too much here in email) but happy to brief you more verbally”.

id=“4-google-s-exclusive-and-restrictive-syndication-agreements”>4. Google’s Exclusive and Restrictive Syndication Agreements

“Staff has investigated whether Google has entered into exclusive or highly restrictive agreements with website publishers that have served to maintain, preserve, or enhance Google’s monopoly power in the markets for search, search advertising, or search and search advertising syndication (or “search intermediation”). We recommend that the Commission issue a complaint against Google for this conduct.”

id=“a-publishers-and-market-structure”>a. Publishers and Market Structure

Buyers of search and search ad syndication are website publishers

Largest sites account for vast majority of syndicated search traffic and volume

Biggest customers are e-commerce retailers (e.g., Amazon and eBay), traditional retailers with websites (e.g., Wal-Mart, Target, Best Buy), and ISPs which operate their own portals

Below this group, companies with significant query volume, including vertical e-commerce sites such as Kayak, smaller retailers and ISPs such as EarthLink; all of these are b. Development of the Market for Search Syndication

First AdSense for Search (AFS) agreements with AOL and EarthLink in 2002

Goal then was to grow nascent industry of syndicated search ads

At the time, Google was bidding against incumbent Overture (later acquired by Yahoo) for exclusive agreements with syndication partners

Google’s early deals favored publishers

To establish a presence, Google offered up-front financial guarantees to publishers

c. Specifics of Google’s Syndication Agreements

“Today, the typical AdSense agreement contains terms and conditions that describe how and when Google will deliver search, search advertising, and other (contextual or domain related) advertising services.”

Two main categories are AFS (search) and AFC (content). Staff investigation focused on AFS

For AFS, two types of agreements. GSAs (Google Service Agreements) negotiated with large partners and standard online contracts, which are non-negotiable and non-exclusive

Bulk of AFS partners are on standard online agreements, but those are a small fraction of revenue

Bulk of revenue comes from GSAs with Google’s 10 largest partners (almost 80% of query volume in 2011). All GSAs have some form of exclusivity or “preferred placement” for Google

“Google’s exclusive AFS agreements effectively prohibit the use of non-Google search and search advertising within the sites and pages designated in the agreement. Some exclusive agreements cover all properties held by a publisher globally; other agreements provide for a property-by-property (or market-by-market) assignment”

By 2008, Google began to migrate away from exclusivity to “preferred placement”. Google must display minimum of 3 ads or number of any competitor (whichever is greater), in an unbroken block, with “preferred placement” (in the most prominent position on publisher’s website)

Google had preferred placement restrictions in GSAs and standard online agreement. Google maintains it was not aware of this provision in standard online agreement until investigational hearing of Google VP for search services, Joan Braddi, where staff questioned Braddi

See Letter from Scott Sher, Wilson Sonsini, to Barbara Blank (May 25, 2012) (explaining that, as of the date of the letter, Google was removing the preferred placement clause from

the Online Terms and Conditions, and offering no further explanation of this decision)

id="d-effects-of-exclusivity-and-preferred-placement">d. Effects of Exclusivity and Preferred Placement

Staff interviewed large and small customers for search and search advertising syndication. Key findings:

id="i-common-publisher-responses">i. Common Publisher Responses

Universal agreement that Bing's search and search advertising markedly inferior, not competitive across-the-board

Amazon reports that Bing monetizes at half the rate of Google

business.com told staff that Google would have to cut revenue share from 64.5% to 30% and Microsoft would have to provide 90% share because Microsoft's platform has such low monetization

Customers "generally confirmed" Microsoft's claim that Bing's search syndication is inferior in part because Microsoft's network is smaller than Google's

With a larger ad base, Google more likely to have relevant, high-quality, ad for any given query, which improves monetization rate

A small publisher said, essentially, the only publishers exclusively using Bing are ones who've been banned from Google's service

We know from other interviews this is an exaggeration, but it captures the general tenor of comments about Microsoft

Publishers reported Microsoft not aggressively trying to win their business

Microsoft exec acknowledge that Bing needs a larger portfolio of advertisers, has been focused there over winning new syndication business

Common theme from many publishers is that search is a relatively minor part of their business and not a strategic focus. For example, Wal-Mart operates website as extension to retail and Best Buy's main goal of website is to provide presale info

Most publishers hadn't seriously considered Bing due to poor monetization

Amazon, which does use Bing and Google ads, uses a single syndication provider on a page to avoid showing the user the same ad multiple times on the same page; mixing and matching arrangement generally considered difficult by publishers

Starting in 2008, Google systematically tried to lower revenue share for AdSense partners

E.g., "Our general philosophy with renewals has been to reduce TAC across the board", "2009 Traffic Acquisition Cost (TAC) was down 3 percentage points from 2008 attributable to the application of standardized revenue share guidelines for renewals and new partnerships...", etc.

Google reduced payments (TAC) to AFS partners from 80.4% to 74% between Q1 2009 and Q1 2010

No publisher viewed reduction as large enough to justify shifting to Bing or serving more display ads instead of search ads

id="ii-publishers-views-of-exclusivity-provisions">ii. Publishers' Views of Exclusivity Provisions

Some large publishers reported exclusive contracts and some didn't

Most publishers with exclusivity provisions didn't complain about them

A small number of technically sophisticated publishers were deeply concerned by exclusivity

These customers viewed search and search advertising as a significant part of business, have the sophistication to integrate multiple suppliers into on-line properties

eBay: largest search and search ads partner, 27% of U. S. syndicated search queries in 2011

Contract requires preferential treatment for AdSense ads, which eBay characterizes as equivalent to exclusivity

eBay wanted this removed in last negotiation, but assented to not removing it in return for not having revenue share cut while most other publishers had revenue share cut

eBay's testing indicates that Bing is competitive in some sectors, e.g., tech ads; they believe they could make more money with multiple search providers

NexTag: In 2015, Google's 15th largest AFS customer

Had exclusivity, was able to remove it in 2010, but NexTag considers restrictions "essentially the same thing as exclusivity"; "NexTag reports that moving away from explicit exclusivity even to this kind of de facto exclusivity required substantial, difficult negotiations with Google"

Has had discussions with Yahoo and Bing about using their products "on a filler basis", but unable to do so due to Google contract restrictions

business.com: B2B lead generation / vertical site; much smaller than above. Barely in top 60 of AdSense query volume

Exclusive agreement with Google

Would test Bing and Yahoo without exclusive agreement

Agreement also restricts how business.com can design pages

Loosening exclusivity would improve business.com revenue and allow for new features that make the site more accessible and user-friendly

Amazon: 2nd largest AFS customer after eBay; \$175M from search syndication, \$169M from Google AdSense

Amazon uses other providers despite their poor monetization due to concerns about having a single supplier; because Amazon operates on thin margins, \$175M is a material source of profit

Amazon concerned it will be forced to sign an exclusive agreement in next negotiation

During last negotiation, Amazon wanted 5-year deal, Google would only give 1-year extension unless Amazon agreed to send Google 90% of search queries (Amazon refused to agree to this formally, although they do this)

IAC: umbrella company operating ask.com, Newsweek, CityGrid, Urbanspoon, and other websites

Agreement is exclusive on a per-property basis

IAC concerned about exclusivity. CityGrid wanted mix-and-match options, but couldn't compete with Google's syndication network, forced to opt into IAC's exclusive agreement; CityGrid wants to use other networks (including its own), but can't under agreement with Google

IAC concerned about lack of competition in search and search advertising syndication

Execute who expressed above concerns left, new executive didn't see a possibility of splitting or moving traffic

"The departure of the key executive with the closest knowledge of the issues and the most detailed concerns suggests we may have significant issues obtaining clear, unambiguous testimony from IAC that reflects their earlier expressed concerns."

id="iii-effects-on-competitors">iii. Effects on Competitors

Microsoft asserts even 5%-10% increase in query volume "very meaningful" and Google's exclusive and restrictive agreements deny Microsoft incremental scale to be more efficient competitor

Speciality search ad platforms also impacted; IAC sought to build platform for local search advertising, but Google's exclusivity provisions "make it less likely that small local competitors like IAC's nascent offering can viably emerge."

id="iii-legal-analysis">III. LEGAL ANALYSIS

"A monopolization claim under Section 2 of the Sherman Act, 15 U. S. C. § 2, has two elements: (i) the 'possession of

monopoly power in the relevant market' and (ii) the 'willful acquisition or maintenance of that power as distinguished from growth or development as a consequence of a superior product, business acumen, or historic accident.'"

"An attempted monopolization claim requires a showing that (i) 'the defendant has engaged in predatory or anticompetitive conduct' with (ii) 'a specific intent to monopolize' and (iii) a dangerous probability of achieving or maintaining monopoly power."

id="a-google-has-monopoly-power-in-relevant-markets">A. GOOGLE HAS MONOPOLY POWER IN RELEVANT MARKETS

"A firm is a monopolist if it can profitably raise prices substantially above the competitive level. [M]onopoly power may be inferred from a firm's possession of a dominant share of a relevant market that is protected by entry barriers.' Google has monopoly power in one or more properly defined markets."

id="1-relevant-markets-and-market-shares">1. Relevant Markets and Market Shares

"A properly defined antitrust market consists of 'any grouping of sales whose sellers, if unified by a hypothetical cartel or merger, could profitably raise prices significantly above the competitive level.'"

"Typically, a court examines 'such practical indicia as industry or public recognition of the submarket as a separate economic entity, the product's peculiar characteristics and uses, unique production facilities, distinct customers, distinct prices, sensitivity to price changes, and specialized vendors.'"

"Staff has identified three relevant antitrust markets."

id="a-horizontal-search">a. Horizontal Search

Vertical search engines not a viable substitute to horizontal search; formidable barriers to expanding into horizontal search

Vertical search properties could pick up query volume in response to SSNIP (small, but significant non-transitory increase in price) in horizontal search, potentially displacing horizontal search providers

Google views these with concern, has aggressively moved to build its own vertical offerings

No mechanism for vertical search properties to broadly discipline a monopolist in horizontal search

Web search queries monetized through search ads, ads sold by keyword which have independent demand functions. So, at best, monopolist might be inhibited from SSNIP on a narrow set of keywords with strong vertical competition.

But for billions of queries with no strong vertical, nothing constrains monopolist from SSNIP

Where vertical websites exist, still hard to compete; comprehensive coverage of all areas seems to be important driver of demand, even to websites focusing on specific topics. Eric Schmidt noted this:

“So if you, for example, are an academic researcher and you use Google 30 times for your academics, then perhaps you’ll want to buy a camera... So long as the product is very, very, very good, people will keep coming back... The general product then creates the brand, creates demand and so forth. Then occasionally, these ads get clicked on”

Schmidt’s testimony corroborated by several vertical search firms, who note that they’re dependent on horizontal search providers for traffic because vertical search users often start with Google, Bing, or Yahoo

When asked about competitors in search, Eric Schmidt mentioned zero vertical properties

Google internal documents monitor Bing and Yahoo and compare quality. Sergei Brin testified that he wasn’t aware of any such regular comparison against vertical competitors

Relevant geo for web search limited to U. S. here; search engines return results relevant to users in country they’re serving, so U. S. users unlikely to view foreign-specialized search engines as viable substitute

Although Google has managed to cross borders, other major international search engines (Baidu, Yandex) have filed to do this

Google dominant for “general search” in U. S.; 66.7% share according to ComScore, and also provides results to ask.com and AOL, another 4.6%

Yahoo 15%, Bing 14%

Google’s market share above generally accepted floor for monopolization; defendants with share in this range have been found to have monopoly power

id=“b-search-advertising”>b. Search Advertising

Search ads likely a properly defined market

Search ads distinguishable from other online ads, such as, display ads, contextual ads, behavioral ads, social media ads due to “inherent scale, targetability, and control”

Google: “[t]hey are such different products that you do not measure them against one another and the technology behind the products is different”

Evidence suggests search and display ads are complements, not substitutes

“Google has observed steep click declines when advertisers have attempted to shift budget to display advertising”

Chevrolet suspended search ads for 2 weeks and relied on display ads alone; lost 30% of clicks

New ad offerings don’t fit into traditional search or display categories: contextual, re-targeted display (or behavioral), social media

Only search ads allow advertisers to show ad based on when user is expressing an interest in the moment the ad is shown; numerous advertisers confirmed this point

Search ads convert at much higher rate due to this advantage

Numerous advertisers report they wouldn’t shift ad spend away from search ads if prices increased more than SSNIP. Living Social would need 100% price increase before shifting ads (a minority of advertisers reported they would move ad dollars from search in response to SSNIP)

Google internal documents and testimony confirm lack of viable substitute for search. AdWords VP Nick Fox and chief economist Hal Varian have stated that search ad spend doesn’t come at expense of other ad dollars, Eric Schmidt has testified multiple times that search ads are the most effective ad tool, has best ROI

Google, through AdWords, has 76% to 80% of the market according to industry-wide trackers (rival Bing-Yahoo has 12% to 16%)

[It doesn’t seem wrong to say that search ads are a market and that Google dominates that market, but the primacy of search ads seems overstated here? Social media ads, just becoming important at the time, ended up becoming very important, and of course video as well]

id=“c-syndicated-search-and-search-advertising-search-intermediation”>c. Syndicated Search and Search Advertising (“Search Intermediation”)

Syndicated search and search advertising (“search intermediation”) are likely a properly defined product market

Horizontal search providers sell (“syndicate”) services to other websites

Search engine can also return search ads to the website; search engine and website share revenue

Consumers are websites that want search; sellers are horizontal search providers, Google, Bing, Yahoo

Publishers of various sizes consistent on cross-elasticity of demand; report that search ad syndication monetizes better than display advertising or other content

No publisher told us that modest (5% to 10%) increase in price for search and search ad syndication would favor other forms of advertising or web content

Google's successful efforts to systematically reduce TAC support this, are a natural experiment to determine likely response to SSNIP

Google, via AdSense, is dominant provider of search and search ad syndication; 75% of market according to Com-Score (Microsoft and Yahoo combine for 22%)

id="2-substantial-barriers-to-entry-exist">2. Substantial Barriers to Entry Exist

"Developing and maintaining a competitively viable search or search ad platform requires substantial investment in specialized knowledge, technology, infrastructure, and time. These markets are also characterized by significant scale effects"

id="a-technology-and-specialization">a. Technology and Specialization

[no notes, extremely obvious to anyone technical who's familiar with the area]

id="b-substantial-upfront-investment">b. Substantial Upfront Investment

Enormous investments required. For example in 2011, Google spent \$5B on R&D. And in 2010, MS spent more than \$4.5B developing algorithms and building physical capacity for Bing

id="c-scale-effects">c. Scale Effects

More usage leads to better algorithms and greater accuracy w.r.t. what consumers want

Also leads to greater number of advertisers

Greater number of advertisers and consumers leads to better ad serving accuracy, better monetization of ads, leads to better monetization for search engine, advertisers, and syndication partners

Cyclical effect, "virtuous cycle"

According to Microsoft, greatest barrier is obtaining sufficient scale. Losing \$2B/yr trying to compete with Google, and Bing is only competing horizontal search platform to Google

id="d-reputation-brand-loyalty-and-the-halo-effect">d. Reputation, Brand Loyalty, and the "Halo Effect"

[no notes]

id="e-exclusive-and-restrictive-agreements">>e. Exclusive and Restrictive Agreements -

"Google's exclusive and restrictive agreements pose yet another barrier to entry, as many potential syndication partners with a high volume of customers are locked into agreements with Google."

id="b-google-has-engaged-in-exclusionary-conduct">B. GOOGLE HAS ENGAGED IN EXCLUSIONARY CONDUCT

"Conduct may be judged exclusionary when it tends to exclude competitors 'on some basis other than efficiency,' i.e., when it 'tends to impair the opportunities of rivals' but 'either does not further competition on the merits or does SO in an unnecessarily restrictive way.' In order for conduct to be condemned as 'exclusionary,' Staff must show that Google's conduct likely impairs the ability of its rivals to compete effectively, and thus to constrain Google's exercise of monopoly power"

id="1-google-s-preferencing-of-google-vertical-properties-within-its-serp">1. Google's Preferencing of Google Vertical Properties Within Its SERP

"Although we believe that this is a close question, we conclude that Google's preferencing conduct does not violate Section 2."

id="a-google-s-product-design-impedes-vertical-competitors">a. Google's Product Design Impedes Vertical Competitors

"As a general rule, courts are properly very skeptical about claims that competition has been harmed by a dominant firm's product design changes. Judicial deference to product innovation, however, does not mean that a monopolist's product design decisions are per se lawful", United States v. Microsoft

We evaluate, through Microsoft lens of monopoly maintenance, whether Google took these actions to impede a nascent threat to Google's monopoly power

"Google's internal documents explicitly reflect - and testimony from Google executives confirms - a concern that Google was at risk of losing, in particular, highly profitable queries to vertical websites"

VP of product management Nicholas Fox:

"[Google's] inability to serve this segment [of vertical lead generation] well today is negatively impacting our business. Query growth among high monetizing queries (>\$120 RPM) has declined to 0% in the UK. US isn't far behind (6%). There's evidence (e.g., UK Finance) that we're losing share to aggregators"

Threat to Google isn't vertical websites, displacing Google, but that they'll undercut Google's power over the most lucrative segments of search and search ads portfolio

Additionally, vertical websites could help erode barriers to growth for general search competitors

id="b-google-s-serp-changes-have-resulted-in-anticompetitive-effects">b. Google's SERP Changes Have Resulted In Anticompetitive Effects

Google expanding its own offerings while demoting rival offerings caused significant drops in traffic to rivals, confirmed by Google's internal data

Google's prominent placement of its own Universal Search properties led to gains in share of its own properties

"For example, Google's inclusion of Google Product Search as a Universal Search result turned a property that the Google product team could not even get indexed by Google's web search results into the number one viewed comparison shopping website on Google"

id="c-google-s-justifications-for-the-conduct">c. Google's Justifications for the Conduct

"Product design change is an area of conduct where courts do not tend to strictly scrutinize asserted procompetitive justifications. In any event, Google's procompetitive justifications are compelling."

Google argues design changes to SERP have improved product, provide consumers with "better" results

Google notes that path toward Universal Search and OneBox predates concern about vertical threat

Google justifies preferential treatment of Universal Search by asserting "apples and oranges" problem prevents Google from doing head-to-head comparison of its property vs. competing verticals, verticals and web results ranked with different criteria. This seems to be correct.

Microsoft says Bing uses a single signal, click-through-rate, that can be compared across Universal Search content and web search results

Google claims that its Universal Search results are more helpful than than "blue links" to other comparison shopping websites

Google claims that showing 3rd party data would create technical and latency issues

"The evidence shows that it would be technologically feasible to serve up third-party results in Google's Universal Search results. Indeed, Bing does this today with its flight vertical, serving up Kayak results and Google itself originally considered third-party OneBoxes"

Google defends "demotion" of competing vertical content, "arguing that Google's algorithms are designed solely with the goal of improving a user's search experience"

"one aspect of Google's demotions that especially troubles Staff - and is not addressed by the above justification - is the fact that Google routinely, and prominently, displays

its own vertical properties, while simultaneously demoting properties that are identical to its own, but for the fact that the latter are competing vertical websites", See Brin Tr. 79:16-81:24 (acknowledging the similarities between Google Product Search and its competitors); Fox Tr. 204:6-204:20 (acknowledging the similarities between Google Product Search and its competitors).

id="d-google-s-additional-legal-defenses">d. Google's Additional Legal Defenses

"Google has argued - successfully in several litigations - that it owes no duty to assist in the promotion of a rival's website or search platform, and that it owes no duty to promote a rival's product offering over its own product offerings"

"one reading of Trinko and subsequent cases is that Google is privileged in blocking rivals from its search platform unless its conduct falls into in one of several specific exceptions referenced in Trinko"

"Alternatively, one may argue that Trinko should not be read so broadly as to overrule swathes of antitrust doctrine."

"Google has long argued that its general search results are opinions that are protected speech under the First Amendment, and that such speech should not be subject to government regulation"; staff believes this is overbroad

"the evidence paints a complex portrait of a company working toward an overall goal of maintaining its market share by providing the best user experience, while simultaneously engaging in tactics that resulted in harm to many vertical competitors, and likely helped to entrench Google's monopoly power over search and search advertising"

"The determination that Google's conduct is anticompetitive, and deserving of condemnation, would require an extensive balancing of these factors, a task that courts have been unwilling - in similar circumstances - to perform under Section 2. Thus, although it is a close question, Staff does not recommend that the Commission move forward on this cause of action."

id="2-google-s-scraping-of-rivals-vertical-content-1">2. Google's "Scraping" of Rivals' Vertical Content

"We conclude that this conduct violates Section 2 and Section 5."

id="a-google-s-scraping-constitutes-a-conditional-refusal-to-deal-or-unfair-method-of-competition">a. Google's "Scraping" Constitutes a Conditional Refusal to Deal or Unfair Method Of Competition

Scraping and threats of refusal to deal with some competitors can be condemned as conditional refusal to deal under Section 2

Post-Trinko, identification of circumstances ("[u]nder certain circumstances, a refusal to cooperate with rivals can

constitute anticompetitive conduct and violate § 2”) “subject of much debate”

Aspen Skiing Co. v. Aspen Highlands Skiing Corp: defendant (owner of 3 of 4 ski areas in Aspen) canceled all-ski area ticket with plaintiff (owner of 4th ski area in Aspen)

After demand increasing share of profit, defendant canceled ticket and rejected “increasingly desperate measures” to recreate joint ticket, even rejected plaintiff’s offer to buy tickets at retail price

Supreme court upheld jury’s finding of liability; Trinko court: “unilateral termination of a voluntary (and thus presumably profitable) course of dealing suggested a willingness to forsake short-term profits to achieve an anticompetitive end. Similarly, the defendant’s unwillingness to renew the ticket even if compensated at retail price revealed a distinctly anticompetitive bent”

Appellate courts have focused on Trinko’s reference to “unilateral termination of a voluntary course of dealing”, e.g., in American Central Eastern Texas Gas Co.v. Duke Energy Fuels LLC, Fifth Circuit upheld determination that defendant natural gas processor’s refusal to contract with competitor for additional capacity was unlawful

Plaintiff contracted with defendant for processing capacity; after two years, defendant proposed terms it “knew were unrealistic or completely unviable ... in order to exclude [the plaintiff] from competition with [the defendant] in the gas processing market.”

Case here is analogous to Aspen Skiing and Duke Energy [a lot of detail not written down in notes here]

id=“b-google-s-scraping-has-resulted-in-anticompetitive-effects”>b. Google’s “Scraping” Has Resulted In Anticompetitive Effects

Scraping has lessened the incentives of competing websites like Yelp, TripAdvisor, CitySearch, and Amazon to innovate, diminishes incentives of other vertical websites to develop new products

entrepreneurs more reluctant to develop new sites, investors more reluctant to sponsor development when Google can use its monopoly power to appropriate content it deems lucrative

id=“c-google-s-scraping-is-not-justified-by-efficiencies”>c. Google’s “Scraping” Is Not Justified By Efficiencies

“Marissa Mayer and Sameer Samat testified that was extraordinarily difficult for Google, as a technical matter, to remove sites like Yelp from Google Local without also removing them from web search results”

“Google’s almost immediate compliance after Yelp sent a formal ‘cease and desist’ letter to Google, however, suggests that the “technical” hurdles were not a significant factor in

Google’s refusal to comply with repeated requests to remove competitor content from Google Local”

Partners can opt out of inclusion with Google’s vertical news offering, Google News

“Similarly, Google’s almost immediate removal of Amazon product reviews from Google Product Search indicates that technical barriers were quickly surmounted when Google desired to accommodate a partner.”

“In sum, the evidence shows that Google used its monopoly position in search to scrape content from rivals and to improve its own complementary vertical offerings, to the detriment of those rivals, and without a countervailing efficiency justification. Google’s scraping conduct has helped it to maintain, preserve, and enhance Google’s monopoly position in the markets for search and search advertising. Accordingly, we believe that this conduct should be condemned by the Commission.”

id=“3-google-s-api-restrictions-1”>3. Google’s API Restrictions

“We conclude that Google’s API restrictions violate Section 2.”

AdWords API procompetitive development

But restrictive conditions in API usage agreement anticompetitive, without offsetting procompetitive benefits

“Should the restrictive conditions be found to be unreasonable restraints of trade, they could be removed today instantly, with no adverse effect on the functioning of the API. Any additional engineering required to make the advertiser data interoperable with other search networks would be supplied by other market participants. Notably, because Google would not be required to give its competitors access to the AdWords API, there is no concern about whether Google has a duty to deal with its competitors”

id=“a-the-restrictive-conditions-are-unreasonable”>a. The Restrictive Conditions Are Unreasonable

Restrictive conditions limit ability of advertisers to use their own data, prevent the development and sale of 3rd party tools and services that would allow automated campaign management across multiple search networks

“Even Google is constrained by these restrictions, having had to forgo improving its DART Search tool to offer such capabilities, despite internal estimates that such functionality would benefit Google and advertisers alike”

Restrictive conditions have no procompetitive virtues, anticompetitive effects are substantial

id=“b-the-restrictive-conditions-have-resulted-in-anticompetitive-effects”>b. The Restrictive Conditions Have Resulted In Anticompetitive Effects

Restrictive conditions reduce innovation, increase transaction costs, degrade quality of Google's rivals in search and search advertising

Restrictive conditions increase transaction costs for all advertisers other than those large enough to make internal investments to develop their own tools [doesn't it also, in some amortized fashion, increase transaction costs for companies that can build their own tools?]

Result is that advertisers spend less on non-dominant search networks, reducing quality of ads on non-dominant search networks

id="c-the-restrictive-conditions-are-not-justified-by-efficiencies">c. The Restrictive Conditions Are Not Justified By Efficiencies

Concern about "misaligned incentives" is Google's only justification for restrictive conditions; concern is that SEMs and agencies would adopt a "lowest common denominator" approach and degrade AdWords campaign performance

"The evidence shows that this justification is unsubstantiated and is likely a pretext"

"In brief, these third parties incentives are highly aligned with Google's interests, precisely the opposite of what Google contends."

Google unable to identify an examples of ill effects from misaligned incentives

Terms and Conditions already have conditions for minimum functionality that prevents lowest common denominator concern from materializing

Documents suggest restrictive conditions were not about "misaligned incentives":

"Sergey [Brin] and Larry [Page] are big proponents of a protectionist strategy that prevents third party developers from building offerings which promote the consolidated management of [keywords] on Google and Overture (and whomever else)."

In a 2004 doc, API product manager was looking for "specific points on how we can prevent a new entrant (MSN Ad Network) from benefitting from a common 3rd party platform that is cross-network."

In a related presentation, Google's lists as a concern, "other competitors are buoyed by lowered barriers to entry"; options to prevent this were "applications must have Google-centric UI functions and branding" and "disallow cross-network compatible applications from using API"

id="4-google-s-exclusive-and-restrictive-syndication-agreements-1">4. Google's Exclusive and Restrictive Syndication Agreements

"Staff has investigated whether Google has entered into anticompetitive, exclusionary agreements with websites for syndicated search and search advertising services (AdSense agreements) that serve to maintain, preserve, or enhance Google's monopoly power in the markets for search, search advertising, or search and search advertising syndication (search intermediation). We conclude that these agreements violate Section 2."

id="a-google-s-agreements-foreclose-a-substantial-portion-of-the-relevant-market">a. Google's Agreements Foreclose a Substantial Portion of the Relevant Market

"Exclusive deals by a monopolist harm competition by foreclosing rivals from needed relationships with distributors, suppliers, or end users. For example, in Microsoft, then-defendant Microsoft's exclusive agreements with original equipment manufacturers and software vendors were deemed anticompetitive where they were found to prevent third parties from installing rival browser Netscape, thus foreclosing Netscape from the most efficient distribution channel, and helping Microsoft to preserve its operating system monopoly. The fact that an agreement is not explicitly exclusive does not preclude a finding of liability."

[notes on legal background of computing foreclosure percentage omitted]

Staff relied on ComScore dataset to compute foreclosure; Microsoft and Yahoo's syndicated query volume is higher than in ComScore, resulting in lower foreclosure number. "We are trying to get to the bottom of this discrepancy now. However, based on our broader understanding of the market, we believe that the ComScore set more accurately reflects the relative query shares of each party." [I don't see why staff should believe that ComScore is more accurate than Microsoft's numbers — I would guess the opposite]

[more notes on foreclosure percentage omitted]

id="b-google-s-agreements-have-resulted-in-anticompetitive-effects">b. Google's Agreements Have Resulted In Anticompetitive Effects

Once foreclosure is established as above "safe harbor" levels, need a qualitative, rule of reason analysis of market effects

Google's exclusive agreements impact immediate market for search and search syndication advertising and have broader effects in markets for search and search advertising

In search search ad syndication (search intermediation), exclusivity precludes some of the largest and most sophisticated publishers from using competing platforms. Publishers can't credibly threaten to shift some incremental business to other platforms to get price concessions from Google

Google's aggressive reduction of revenue shares to customers without significant resistance => agreements seem to be further entrenching Google's monopoly position

An objection to this could be that Google's business is because its product is superior

This argument rests on fallacious assumption that Bing's average monetization gap is consistent across the board

[section on CityGrid impact omitted; this section speaks to broader market effects]

Google insists that incremental traffic to Microsoft would be trivial; Microsoft indicates it would be "very meaningful"

Not enough evidence for definitive conclusion, but "internal Google documents suggest that Microsoft's view of things may be closer to the truth. — Google's interest in renewing deals in part to prevent Microsoft from gaining scale. Internal Google analysis of 2010 AOL renewal: "AOL holds marginal search share but represents scale gains for a Microsoft + Yahoo! partnership. AOL/Microsoft combination has modest impact on market dynamics, but material increase in scale of Microsoft's search & ads platform"

When informed that "Microsoft [is] aggressively wooing AOL with large guarantees," a Google exec responded with: "I think the worse case scenario here is that AOL users get sent to Bing, so even if we make AOL a bit more competitive relative to Google, that seems preferable to growing Bing."

Google internal documents show they pursued AOL deal aggressively even though AOL represented "[a] low/no profit partnership for Google."

Evidence is that, in near-term, removing exclusivity would not have dramatic impact; largest and most sophisticated publishers would shift modest amounts of traffic to Bing

Most significant competitive benefits realized over longer period of time

"Removing exclusivity may open up additional opportunities for both established and nascent competitors, and those opportunities may spur more significant changes in the market dynamics as publishers have the opportunity to consider - and test - alternatives to Google's AdSense program."

id="c-google-s-agreements-are-not-justified-by-efficiencies">c. Google's Agreements Are Not Justified By Efficiencies

Google has given three business justifications for exclusive and restrictive syndication agreements

Long-standing industry practice of exclusivity, dating from when publishers demanded large, guaranteed, revenue share payments regardless of performance

"guaranteed revenue shares are now virtually non-existent"

"Google is simply engaging in a vigorous competition with Microsoft for exclusive agreements"

"Google may argue that the fact that Microsoft is losing in a competitive bidding process (and indeed, not competing as vigorously as it might otherwise) is not a basis on which to condemn Google. However, Google has effectively created the rules of today's game, and Microsoft's substantial monetization disadvantage puts it in a poor competition position to compete on an all-or-nothing basis."

"user confusion" — "Google claims that it does not want users to confuse a competitor's poor advertisements with its own higher quality advertisements"

"This argument suffers both from the fact that it is highly unlikely that users care about the source of the ad, as well as the fact that, if users did care, less restrictive alternatives are clearly available. Google has not explained why alternatives such as labeling competitor advertisements as originating from the competitor are unavailing here."

"Google's actions demonstrate that "user confusion" is not a significant concern. In 2008 Google attempted to enter into a non-exclusive agreement with Yahoo! to supplement Yahoo!'s search advertising platform. Under the proposed agreement, Yahoo! would return its own search advertising, but supplement its inventory with Google search advertisements when Yahoo! did not have sufficient inventory.⁵⁸ Additionally, Google has recently eliminated its "preferred placement" restriction for its online partners."

Rule of reasons analysis shows strong evidence of market protected by high entry barriers

Despite limitations to evidence, market is inarguably not robustly competitive today

Google has been unilaterally reducing revenue share with apparent impunity

id="iv-potential-remedies">IV. POTENTIAL REMEDIES

id="a-scraping">A. Scraping

At least two possible remedies

Opt-out to remove snippets of content from Google's vertical properties, while retaining web search results and/or in Universal Search results on main SERP

Google could be required to limit use of content it indexes for web search (could only use content in returning the property in its search results, but not for determining its own product or local rankings) unless given explicit permission

id="b-api-restrictions">B. API Restrictions

Require Google to remove problematic contractual restrictions; no technical fixes necessary

SEMs report that technology for cross-compatibility already exists, will quickly flourish if unhindered by Google's contractual constraints

id="c-exclusive-and-restrictive-syndication-agreements">C. Exclusive and Restrictive Syndication Agreements

Most appropriate remedy is to enjoin Google from entering exclusive agreement with search syndication partners, and to require Google to loosen restrictions surrounding AdSense partners' use of rival search ads

id="v-litigation-risks">V. LITIGATION RISKS

Google does not charge customers, and they are not locked into Google

Universal Search has resulted in substantial benefit to users

Google's organization and aggregation of content adds value to product for customers

Largest advertisers advertise on both Google AdWords and Microsoft AdCenter

Most efficient channel through which Bing can gain scale is Bing.com

Microsoft has the resources to purchase distribution where it seems greatest value

Most website publishers appy with AdSense

id="vi-conclusion">VI. CONCLUSION

"Staff concludes that Google's conduct has resulted - and will result - in real harm to consumers and to innovation in the online search and advertising markets. Google has strengthened its monopolies over search and search advertising through anticompetitive means, and has forestalled competitors' and would-be competitors' ability to challenge those monopolies, and this will have lasting negative effects on consumer welfare"

"Google has unlawfully maintained its monopoly over general search and search advertising, in violation of Section 2, or otherwise engaged in unfair methods of competition, in violation of Section 5, by scraping content from rival vertical websites in order to improve its own product offerings."

"Google has unlawfully maintained its monopoly over general search, search advertising, and search syndication, in violation of Section 2, or otherwise engaged in unfair methods of competition, in violation of Section 5, by entering into exclusive and highly restrictive agreements with web publishers that prevent publishers from displaying competing search results or search advertisements."

"Google has unlawfully maintained its monopoly over general search and search advertising, in violation of Section

2, or otherwise engaged in unfair methods of competition, in violation of Section 5, by maintaining contractual restrictions that inhibit the cross-platform management of advertising campaigns."

"For the reasons set forth above, Staff recommends that the Commission issue the attached complaint."

id="ftc-be-staff-memo">FTC BE staff memo

"Bureau of Economics

August 8, 2012

From: Christopher Adams and John Yun, Economists"

id="executive-summary-1">Executive Summary

Anticompetitive investigation started June 2011

Staff presented theories and evidence February 2012

This memo offers our final recommendation

Four theories of harm

preferencing of search results by favoring own web properties over rivals

exclusive agreements with publishers and vendors, deprive rival platforms of users and advertisers

restrictions on porting advertiser data to rival platforms

misappropriating content from Yelp and TripAdvisor

"our guiding approach must be beyond collecting complaints and antidotes [presumably meant to be anecdotes?] from competitors who were negatively impacted from a firm's various business practices."

Market power in search advertising

Google has "significant" share, 65% of paid clicks and 53% of ad impressions among top 5 U. S. search engines

Market power may be mitigated by the fact that 80% use a search engine other than Google

Empirical evidence consistent with search and non-search ads being substitutes, and that Google considers vertical search to be competitors

Preferencing theory

Theory is that Google is blending its proprietary content with customary "blue links" and demoting competing sites

Google has limited ability to impose significant harm on vertical rivals because it accounts for 10% to 20% of traffic to them. Effect is very small and not statistically significant

[Funny that something so obviously wrong at the time and also seemingly wrong in retrospect was apparently taken seriously]

Universal Search was a procompetitive response to pressure from vertical sites and an improvement for users

Exclusive agreements theory

Access to a search engine's site (i.e., not dependent on 3rd party agreement) is most efficient and common distribution channel, which is not impeded by Google. Additionally, strong reasons to doubt that search toolbars and default status on browsers can be viewed as "exclusives" because users can easily switch (on desktop and mobile)

[statement implies another wrong model of what's happening here]

[Specifically on easy switching on mobile, there's Google's actual blocking of changing the default search engine from Google to what the user wants, but we also know that a huge fraction of users basically don't understand what's happening and can't make an informed decision to switch — if this weren't the case, it wouldn't make sense for companies to bid so high for defaults, e.g. supposedly \$26B/yr to obtain default search engine status on iOS; if users simply switch freely with, default status would be worth close to \$0. Since this payment is, at the margin, pure profit and Apple's P/E ratio is 29.53 as of my typing this sentence, a quick and dirty estimate is that \$776B of Apple's market cap is attributable to taking this payment vs. randomly selecting a default]

[In addition to explicit, measurable, coercion like the above, there were also things like Google pressuring Samsung into shutting down their Android Browser effort in 2012; although enforcing a search engine default on Android was probably not the primer driver on that or other similar pressure that Google applied, many of these sorts of things also had the impact of funneling users into Google on mobile; these economists seem like the incentive-based argument that users will use the best product, so the result we see in the market, but if that's the case, why do companies spend so much effort on ecosystem lock-in, including but not limited to supposedly paying \$18B/yr to own the default setting in one browser? I guess the argument here is that companies are behaving completely irrationally in expending so much effort here, but consumers are behaving perfectly rationally and are fully informed and are not influenced by all of this spending at all?]

In search syndication, Microsoft and Yahoo have a combined greater share than Google's

No support for assertion that rivals' access to users has been impaired by Google. MS and Yahoo have had a steady 30% share for year; query volume has grown faster than Google since alliance was announced

[Another odd statement; at the time, observers didn't see Bing staying competitive without heavy subsidies from MS,

and then MS predictably stopped subsidizing Bing as a big bet and its market share declined. Google's search market share is well above 90% and hasn't been below 90% since the BE memo was written; in the U. S., estimates put Google around 90% share, some a bit below and some a bit above, with low estimates at something like 87%. It's odd that someone could look at the situation at the time and not seeing that this was about to happen]

In December 2011, Microsoft had access to query volume equivalent to what Google had 2 years ago, thus difficult to infer that Microsoft is below some threshold of query volume

[this exact argument was addressed in the BC memo; the BE memo does not appear to refute the BC memo's argument]

[As with a number of the above arguments, this is a strange argument if you understand the dynamics of fast-growing tech companies. When you have rapidly growing companies in markets with network effects or scale effects, being the same absolute size as a competitor a number of years ago doesn't mean that you're in an ok position. We've seen this play out in a ton of markets and it's fundamental to why VCs shovel so much money at companies in promising markets — being a couple years behind often means you get crushed or, if you're lucky, end up as an also ran that's fighting an uphill battle against scale effects]

Characteristics of online search market not consistent with Google buying distribution agreements to raise input costs of rivals

Restrictions on porting advertiser data to AdWords API

Theory is that Google's terms and conditions for AdWords API anticompetitively disadvantages Microsoft's adCenter

Introduction of API with co-mingling restriction made users and Google better off and rivals's costs were unaffected. Any objection therefore implies that when Google introduced the API, it had an obligation to allow its rivals to benefit from increased functionality. Significant risks to long-term innovation incentives from imposing such an obligation [Huh, this seems very weird]

Advertisers responsible for overwhelming majority of search ad spend use both Google and Microsoft. Multihoming advertisers of all sizes spend a significant share of budget on Microsoft [this exact objection is addressed in BC memo]

Evidence from SEMs and end-to-end advertisers suggest policy's impact on ad spend on Microsoft's platform is negligible [it's hard to know how seriously to take this considering the comments on Yelp, above — the model of how tech businesses work seems very wrong, which casts doubt on other conclusions that necessarily require having some kind of model of how this stuff works]

Scraping allegation is that Google has misappropriated content from Yelp and TripAdvisor

Have substantive concerns. Solution proposed in Annex 11

To be an antitrust violation, need strong evidence that it increased users on Google at expense of Yelp or TripAdvisor or decreased incentives to innovate. No strong evidence of either [per above comments, this seems wrong]

Recommendation: recommend investigation be closed

id="1-does-google-possess-monopoly-power-in-the-relevant-antitrust-market">1. Does Google possess monopoly power in the relevant antitrust market?

To be in violation of Section 2 of the Sherman Act, Google needs to be a monopoly or have substantial market power in a relevant market

Online search similar to any other advertising

Competition between platforms and advertisers depends on extent to which advertisers consider users on one platform to be substitutes for another

Google's market power depends on share of internet users

If advertisers can access Google's users at other search platforms, such as Yahoo, Bing, and Facebook, "Google's market power is a lot less"

Substantial evidence contradicting proposition that Google has substantial market power in search advertising

Google's share is large. In Feb 2012, 65% of paid search clicks of top 5 general search engines went through Google, up from 55% in Sep 2008; these figures show Google offers advertisers what they want

Advertisers want "eyeballs"

Users multi-home. About 80% of users use a platform other than Google in a given month, so advertisers can get the same eyeballs elsewhere

Advertiser can get in front of a user on a different query on Yahoo or another search engine

[this is also odd reasoning — if a user uses Google for searches by default, but occasionally stumbles across Yahoo or Bing, this doesn't meaningfully move the needle for an advertiser; the evidence here is comScore saying that 20% of users only use Google, 15% never use Google, and 65% use Google + another search engine; but it's generally accepted that comScore numbers are quite off. Shortly after the report was written, I looked at various companies that reported metrics (Alexa, etc.) and found them to be badly wrong; I don't think it would be easy to dig up the exact info I used at the time now, but on searching for "comscore search engine market accuracy", the first hit I got was someone ex-

plaining that while, today, comScore shows that Google has an implausibly low 67% market share, an analysis of traffic to sites this company has access to showed that Google much more plausibly drove 85% of clicks; it seems worth mentioning that comScore is often considered inaccurate]

Firm-level advertising between search ads and display ads is negatively correlated

[this seems plausible? The evidence in the BC memo for these being complements seemed like a stretch; maybe it's true, but the BE memo's position seems much more plausible]

No claim that these are the same market, but can't conclude that they're unrelated

Google competes with specialized search engines, similar to a supermarket competing with a convenience store [details on this analogy elided; this memo relies heavily on analogies that relate tech markets to various non-tech markets, some of which were also elided above]

For advertising on a search term like "Nikon 5100", Amazon may provide a differentiated but competing product

Google is leading seller of search, but this is mitigated by large proportion of users who also use other search engines, by substitution of display and search advertising, by competition in vertical search

id="theory-1-the-preferencing-theory">Theory 1: The preferencing theory

id="2-1-overview">2.1 Overview

Preferencing theory is that Google's blending of content such as shopping comparison results and local business listings with customary blue links disadvantages competing content sites, such as Nextag, eBay, Yelp, and TripAdvisor

id="2-2-analysis">2.2 Analysis

Blend has two effects, negatively impacting traffic to specialized vertical sites by pushing down sites and impacting Google's incentives to show competing vertical sites

Empirical questions

"To what extent does Google account for the traffic to vertical sites?"

"To what extent do blends impact the likelihood of clicks to vertical sites?"

"To what extent do blends improve consumer value from the search results?"

id="2-3-empirical-evidence">2.3 Empirical evidence

Google search responsible for 10% of traffic to shopping comparison sites, 17.5% to local business search sites. “See Annex 4 for a complete discussion of our platform model”

[Annex 4”, doesn’t appear to be included; but, as discussed above, the authors’ model of how traffic works seems to be wrong]

When blends appear, from Google’s internal data, clicks to other shopping comparison sites drop by a large and statistically significant amount. For example, if a site had a pre-blend CTR of 9%, post-blend CTR would be 5.3%, but a blend isn’t always presented

For local, pre-blend CTR of 6% would be reduced to 5.4%; local blends have smaller impact than shopping

“above result for shopping comparison sites is not the same as finding that overall traffic from Google to shopping sites declined due to universal search. As we describe below, if blends represent a quality improvement, this will increase demand and drive greater query volume on Google, which will boost traffic to all sites.”

All links are substitutes, so we can infer that if user user clicks on ads less, they prefer the content and the user is getting more value. Overall results indicate that blends significantly increase consumer value

[this seems obviously wrong unless the blend is presented with the same visual impact, weight, and position, as normal results, which isn’t the case at all — I don’t disagree that the blend is probably better for consumers, but this methodology seems like a classic misuse of data to prove a point]

id=“2-4-documentary-evidence”>2.4 Documentary evidence

Since the 90s, general search engines have incorporated vertical blends

All major search engines use blends

id=“2-5-summary-of-the-preferencing-theory”>2.5 Summary of the preferencing theory

Google not significant enough source of traffic to foreclose its vertical rivals [as discussed above, the model for this statement is wrong]

id=“theory-2-exclusionary-practices-in-search-distribution”>Theory 2: Exclusionary practices in search distribution

id=“3-1-overview”>3.1 Overview

Theory is that Google is engaging in exclusionary practices in order to deprive Microsoft of economies of scale

Foundational issues

Are Google’s distribution agreements substantially impairing opportunity of rivals to compete for users?

What’s the empirical evidence users are being excluded and denied?

What’s the evidence that Microsoft is at a disadvantage in terms of scale?

id=“3-2-are-the-various-google-distribution-agreements-in-fact-exclusionary”>3.2 Are the various Google distribution agreements in fact exclusionary?

“Exclusionary agreements merit scrutiny when they materially reduce consumer choice and substantially impair the opportunities of rivals”

On desktop, users can access search engine directly, via web browser search box, or a search toolbar

73% of desktop search through direct navigation, all search engines have equal access to consumers in terms of direct access; “Consequently, Google has no ability to impair the opportunities of rivals in the most important and efficient desktop distribution channel.”

[once again, this model seems wrong — if it wasn’t wrong, companies wouldn’t pay so much to become a search default, including shady stuff like Google paying shady badware installers to make Chrome / Google default on people’s desktops . Another model is that if a user uses a search engine because it’s a default, this changes a the probability that they’ll use the search engine via “direct access”; compared to the BE staff model, it’s overwhelmingly likely that this model is correct and the BE staff model is wrong]

Microsoft is search default on Internet Explorer and 70% of PCs sold

For syndication agreement, Google has a base template that contains premium placement provision. This is to achieve minimum level of remuneration in return for Google making its search available. Additionally, clause is often subject to negotiation and can be modified

[this negotiation thing is technically correct, but doesn’t address the statement about this brought up in the BC memo; many, perhaps most, of the points in this memo have been refuted by the BC memo, and the strategy here seems to be to ignore the refutations without addressing them]

“By placing its entire site or suite of suites up for bid, publishers are able to bargain more effectively with search engines. This intensifies the ex ante competition for the contract and lowers publishers’ costs. Consequently, eliminating the ability to negotiate a bundled discount, or exclusivity, based on site-wide coverage will result in higher prices to publishers.” [this seems to contradict what we observe in practice?]

“This suggests that to the extent Google is depriving rivals such as Microsoft of scale economies, this is a result of ‘competition on the merits’— much the same way as if Google had caused Microsoft to lose traffic because it developed a better product and offered it at a lower price.”

Have Google’s premium placement requirements effectively denied Microsoft access to publishers?

Can approach this by considering market share. Google 44%, including AOL and Ask. MS 31%, including Yahoo. Yahoo 25%. Combined, Yahoo and MS are at 56%. “Thus, combined, Microsoft and Yahoo’s syndication shares are higher than their combined shares in a general search engine market” [as noted previously, these stats didn’t seem correct at the time and have gotten predictably less directionally correct over time]

What would MS’s volume be without Google’s exclusionary restrictions

At most a 5% change because Google’s product is so superior [this seems to ignore the primary component of this complaint, which is that there’s a positive feedback cycle]

Search syndication agreements

Final major distribution channel is mobile search

U. S. marketshare: Android 47%, iOS 30%, RIM 16%, MS 5%

Android and iOS grew from 30% to 77% from December 2009 to December 2011, primarily due to decline of RIM, MS, and Palm

Mobile search is 8%. Thus, “small percentage of overall queries and even smaller percentage of search ad revenues”

[The implication here appears to be that mobile is small and unimportant, which was obviously untrue at the time to any informed observer — I was at Google shortly after this was written and the change was made to go “mobile first” on basically everything because it was understood that mobile was the future; this involved a number of product changes that significantly degraded the experience on desktop in order to make the mobile experience better; this was generally considered not only a good decision, but the only remotely reasonable decision. Google was not alone in making this shift at the time. How economists studying this market didn’t understand this after interviewing folks at Google and other tech companies is mysterious]

Switching cost on mobile implied to be very low, “a few taps” [as noted previously, the staggering amount of money spent on being a mobile default and Google’s commitment linked above indicate this is not true]

Even if switching costs were significant, there’s no remedy here. “Too many choices lead to consumer confusion”

Repeat of point that barrier to switching is low because it’s “a few taps”

“Google does not require Google to be the default search engine in order to license the Android OS” [seems technically correct, but misleading at best when taken as part of the broader argument here]

OEMs choose Google search as default for market-based reasons and not because their choice is restricted [this doesn’t address the commitment linked above that actually prevents users from switching the default away from Google; I wonder what the rebuttal to that would be, perhaps also that user choice is bad and confusing to users?]

Opportunities available to Microsoft are larger than indicated by marketshare

Marketshare could change quickly; two years ago, Apple and Google only had 30% share

Default of Google search not anticompetitive and mobile a small volume of queries, “although this is changing rapidly”

Basically no barrier to user switching, “a few taps and downloading other search apps can be achieved in a few seconds. These are trivial switching costs” [as noted above, this is obviously incorrect to anyone who understands mobile, especially the part about downloading an app not being a barrier; I continue to find it interesting that the economists used market-based reasoning when it supports the idea that the market is perfectly competitive, with no switching costs, etc., but decline to use market-based reasoning, such as noting the staggeringly high sums paid to set default search, when it supports the idea that the market is not a perfectly competitive market with no switching costs, etc.]

id=“3-3-are-rival-search-engines-being-excluded-from-the-market”>3.3 Are rival search engines being excluded from the market?

Prior section found that Google’s distribution agreements don’t impair opportunity of rivals to reach users. But could it have happened? We’ll look at market shares and growth trends to determine

“We note that the evidence of Microsoft and Yahoo’s share and growth cannot, even in theory, tell us whether Google’s conduct has had a significant impact. Nonetheless, if we find that rival shares have grown or not diminished, this fact can be informative. Additionally, assuming that Microsoft would have grown dramatically in the counterfactual, despite the fact that Google itself is improving its product, requires a level of proof that must move beyond speculation.” [as an extension of the above, the economists are happy to speculate or even ‘move beyond speculation’ when it comes to applying speculative reasoning on user switching costs, but apparently not when it comes to inferences that can be made about marketshare; why the drastic difference in the standard of proof?]

Microsoft and Yahoo's share shows no design of being excluded, steady 30% for 4 years [as noted in a previous section, the writing was on the wall for Bing and Yahoo at this time, but apparently this would "move beyond speculation" and is not noted here]

Since announcement of MS / Yahoo alliance, MS query volume as grown faster than Google [this is based on comScore qSerach data and the more detailed quoted claim is that MS Query volume increased 134% while Google volume increased 54%; as noted above, this seems like an inaccurate metric, so it's not clear why this would be used to support this point, and it's also misleading at best]

MS-Yahoo have the same number of search engine users as Google in a given month [again, as noted above, this appears to come from incorrect data and is also misleading at best because it counts a single use in a month as equivalent to using something many times a day]

id="3-4-does-microsoft-have-sufficient-scale-to-be-competitive">3.4 Does Microsoft have sufficient scale to be competitive?

In a meeting with Susan Athey, Microsoft could not demonstrate that they had data definitively showing how the cost curve changes as click data changes, "thus, there is basis for suggesting Microsoft is below some threshold point" [the use of the phrase "threshold point" demonstrates either a use of sleight of hand or a lack of understanding of how it works; the BE memo seems to prefer the idea that it's about some threshold since this could be supported by the argument that, if such a threshold were to be demonstrated, Microsoft's growth would have or will carry it past the threshold, but it doesn't make any sense that there would a threshold; also, even if this were important, having a single meeting where Microsoft wasn't able to answer this immediately would be weak evidence]

[many more incorrect comments in the same vein as the above omitted for brevity]

"Finally, Microsoft's public statements are not consistent with statements made to antitrust regulators. Microsoft CEO Steve Ballmer stated in a press release announcing the search agreement with Yahoo: "This agreement with Yahoo! will provide the scale we need to deliver even more rapid advances in relevancy and usefulness. Microsoft and Yahoo! know there's so much more that search could be. This agreement gives us the scale and resources to create the future of search."

[it's quite bizarre to use a press release, which are generally understood to be meaningless puff pieces, as evidence that a strongly supported claim isn't true; again, BE staff seem to be extremely selective about what evidence they look at to a degree that is striking; for example from conversations I had with credible, senior, engineers who worked on search at both Google and Bing, engineers who understand the domain would agree that having more search volume and

more data is a major advantage; instead of using evidence like that, BE staff find a press release that, in the tradition of press releases, has some meaningless and incorrect bragging, and bring that in as evidence; why would they do this?]

[more examples of above incorrect reasoning, omitted for brevity]

id="3-5-theory-based-on-raising-rivals-costs">3.5 Theory based on raising rivals' costs

Despite the above, it could be that distribution agreements deny rivals and data enough that "feedback effects" are triggered

Possible feedback effects

Scale effect: cost per unit of quality or ad matching decreases

Indirect network effect: more advertisers increases number of users

Congestion effect

Cash flow effect

Scale effect was determined to not be applicable[as noted there, the argument for this is completely wrong]

Indirect network effect has weak evidence, evidence exists that it doesn't apply, and even if it did apply, low click-through rate of ads shows that most consumers don't like ads anyway [what? This doesn't seem relevant?], and also, having a greater number of advertises leads to congestion and reduction in the value of the platform to advertisers [this is a reach; there is a sense in which this is technically true, but we could see then and now that platforms with few advertisers are extremely undesirable to advertises because advertisers generally don't want to advertise on a platform that full of low quality ads (and this also impacts the desire of users to use the platform)]

Cash flow effect not relevant because Microsoft isn't cash flow constrained, so cost isn't relevant [a funny comment to make because, not too long after this, Microsoft severely cut back investment in Bing because the returns weren't deemed to be worth it; it seems odd for economists to argue that, if you have a lot of money, the cost of things doesn't matter and ROI is irrelevant. Shouldn't they think about marginal cost and marginal revenue?]

[I stopped taking detailed notes at this point because taking notes that are legible to other people (as opposed to just for myself) takes about an order of magnitude longer, and I didn't think that there was much of interest here. I generally find comments of the form "I stopped reading at X" to be quite poor, in that people making such comments generally seem to pick some trivial thing that's unimportant and then declare and entire document to be worthless based on that. This pattern is also common when it comes to engineers,

institutions, sports players, etc. and I generally find it counterproductive in those cases as well. However, in this case, there isn't really a single, non-representative, issue. The majority of the reasoning seems not just wrong, but highly disconnected from the on-the-ground situation. More notes indicating that the authors are making further misleading or incorrect arguments in the same style don't seem very useful. I did read the rest of the document and I also continue to summarize a few bits, below. I don't want to call them "highlights" because that would imply that I pulled out particularly interesting or compelling or incorrect bits and it's more of a smattering of miscellaneous parts with no particular theme]

There's a claim that removing restrictions on API interoperability may not cause short term problems, but may cause long-term harm due to how this shifts incentives and reduces innovation and this needs to be accounted for, not just the short-term benefit [in form, this is analogous to the argument Tyler Cowen recently made that banning non-competes reduces the incentives for firms to innovate and will reduce innovation]

The authors seem to like refer to advertisements and PR that any reasonable engineer (and I would guess reasonable person) would know are not meant to be factual or accurate. Similar to the PR argument above, they argue that advertising for Microsoft adCenter claims that it's easy to import data from AdWords, therefore the data portability issue is incorrect, and they specifically say that these advertising statements are "more credible than" other evidence

They also relied on some kind of SEO blogspam that restates the above as further evidence of this

The authors do not believe that Google Search and Google Local are complements or that taking data from Yelp or TripAdvisor and displaying it above search results has any negative impact on Yelp or TripAdvisor, or at least that "the burden of proof would be extremely difficult"

id="other-memos">Other memos

[for these, I continued writing high-level summaries, not detailed summaries]

After the BE memo, there's a memo from Laura M. Sullivan, Division of Advertising Practices, which makes a fairly narrow case in a few dimensions, including "we continue to believe that Google has not deceived consumers by integrating its own specialized search results into its organic results" and, as a result, they suggest not pursuing further action.

There are some recommendations, such as "based on what we have observed of these new paid search results [referring to Local Search, etc.], we believe Google can strengthen the prominence and clarity of its disclosure" [in practice, the opposite has happened!]

[overall, the specific points presented here seems like ones a reasonable person could agree with, though whether or not these points are strong enough that they should prevent anti-trust action could be debated]

"Updating the 2002 Search Engine Letter is Warranted"

"The concerns we have regarding Google's disclosure of paid search results also apply to other search engines. Studies since the 2002 Search Engine letter was issued indicate that the standard methods search engines, including Google, Bing, and Yahoo!, have used to disclose their paid results may not be noticeable or clear enough for consumers."²¹ For example, many consumers do not recognize the top ads as paid results ... Documents also indicate Google itself believed that many consumers generally do not recognize top ads as paid. For example, in June 2010, a leading team member of Google's in-house research group, commenting on general search research over time, stated: 'I don't think the research is inconclusive at all - there's definitely a (large) group of users who don't distinguish between sponsored and organic results. If we ask these users why they think the top results are sometimes displayed with a different background color, they will come up with an explanation that can range from "because they are more relevant" to "I have no idea" to "because Google is sponsoring them."' [this could've seemed reasonable at the time, but in retrospect we can see that the opposite of this has happened and ads are less distinguishable from search results than they were in 2012, likely meaning that even fewer consumers can distinguish ads from search results]

On the topic of whether or not Google should be liable for fraudulent ads such as ones for fake weight-loss products or fake mortgage relief services, "there is no indication so far that Google has played any role in developing or creating the search ads we are investigating" and Google is expending some effort to prevent these ads and Google can claim CDA immunity, so further investigation here isn't worthwhile

There's another memo from the same author on whether or not using other consumer data in conjunction with its search advertising business is unfair; the case is generally that this is not unfair and consumers should expect that their data is used to improve search queries

There's a memo from Ken Heyer (at the time, a Director of the Agency's Bureau of Economics)

Suggests having a remedy that seems "quite likely to do more good than harm" before "even considering serious filing a Complaint"

Seems to generally be in agreement with BE memo

On distribution, agrees with economist memo on unimportance of mobile and that Microsoft has good distribution on desktop (due to IE being default on 70% of PCs sold)

On API restrictions, mixed opinion

On mobile, mostly agrees with BE memo, but suggests getting an idea of how much Google pays for the right be default “since if default status is not much of an advantage we would not expect to see large payments being made” and also suggests it would be interesting to know how much switching from the default occurs

Further notes that mobile is only 8% of the market, too small to be significant [8% should’ve been factually incorrect. By late 2012, when this was written, mobile should’ve been 20% or more of queries; not sure why the economists are so wrong on so many of the numbers]

On vertical sites, agreement with data analysis from BE memo and generally agrees with BE memo

Another Ken Heyer memo

More strongly recommends no action taken than previous memo, recommends against consent decree as well as litigation

Follow-up memo from BC staff (Barbara R. Blank et al.), recommending that staff negotiate a consent order with Google on mobile

Google has exclusive agreement with the 4 major U. S. wireless carriers and Apple to pre-install Google Search; Apple agreement requires exclusivity

Google default on 86% of devices

BC Staff recommends consent agreement to eliminate these exclusive agreements

According to Google documents mobile was 9.5% of Google queries in 2010, 17.3% in 2011 [note that this strongly contradicts the claim from the BE memo that mobile is only 8% of the market here]

Rapid growth shows that mobile distribution channel is significant, and both Microsoft and Google internal documents recognize that mobile will likely surpass desktop in the near future

In contradiction to their claims, Sprint and T-mobile agreements appear to mandate exclusivity, and AT&T agreement is de facto exclusive due to tiered revenue sharing arrangement; Verizon agreement is exclusive

Google business development manager Chris Barton: “So we know with 100% certainty due to contractual terms that: All Android phones on T-Mobile will come with Google as the only search engine out-of-the-box. All Android phones on Verizon will come with Google as the only search engine out-of-the-box. All Android phones on Sprint will come with Google as the only search engine out-of-the-box. I think this approach is really important otherwise Bing or Yahoo can come and steal away our Android search distribution at any time, thus removing the value of entering into contracts

with them. Our philosophy is that we are paying revenue share”

Andy Rubin laid out a plan to reduce revenue share of partners over time as Google gained search dominance and Google has done this over time

Carriers would not switch even without exclusive agreement due to better monetization and/or bad PR

When wrapping up Verizon deal, Andy Rubin said “[i]f we can pull this off ... we will own the US market”

Memo from Willard K. Tom, General Counsel

“In sum, this may be a good case. But it would be a novel one, and as in all such cases, the Commission should think through carefully what it means.”

Memo from Howard Shelanski, Director in Bureau of Economics

Mostly supports the BE memo and the memo from Ken Heyer, except on scraping, where there’s support for the BC memo

id=“fn: K”>

By analogy to a case that many people in tech are familiar with, consider this exchange between Oracle counsel David Boies and Judge William Alsup on the rangeCheck function, which checks if a range is a valid array access or not given the length of an array and throws an exception if the access is out of range:

Boies : [argument that Google copied the rangeCheck function in order to accelerate development]

Alsup : All right. I have — I was not good — I couldn’t have told you the first thing about Java before this trial. But, I have done and still do a lot of programming myself in other languages. I have written blocks of code like rangeCheck a hundred times or more. I could do it. You could do it. It is so simple. The idea that somebody copied that in order to get to market faster, when it would be just as fast to write it out, it was an accident that that thing got in there. There was no way that you could say that that was speeding them along to the marketplace. That is not a good argument.

Boies : Your Honor

Alsup : [cutting off Boies] You’re one of the best lawyers in America. How can you even make that argument? You know, maybe the answer is because you are so good it sounds legit. But it is not legit. That is not a good argument.

Boies : Your Honor, let me approach it this way, first, okay. I want to come back to rangeCheck. All right.

Alsup : RangeCheck. All it does is it makes sure that the numbers you’re inputting are within a range. And if they’re

not, they give it some kind of exceptional treatment. It is so — that witness, when he said a high school student would do this, is absolutely right.

Boies : He didn't say a high school student would do it in an hour, all right.

Alsup : Less than — in five minutes, Mr. Boies.

Boies previously brought up this function as a non-trivial piece of work and then argues that, in their haste, a Google engineer copied this function from Oracle. As Alsup points out, the function is trivial, so trivial that it wouldn't be worth looking it up to copy and that even a high school student could easily produce the function from scratch. Boies then objects that, sure, maybe a high school student could write the function, but it might take an hour or more and Alsup correctly responds that an hour is implausible and that it might take five minutes.

Although nearly anyone who could pass a high school programming class would find Boies's argument not just wrong but absurd³, more like a joke than something that someone might say seriously, it seems reasonable for Boies to make the argument because people presiding over these decisions in court, in regulatory agencies, and in the legislature, sometimes demonstrate a lack of basic understanding of tech. Since my background is in tech and not law or economics, I have no doubt that this analysis will miss some basics about law and economics in the same way that most analyses I've read seem miss basics about tech, but since there's been extensive commentary on this case from people with strong law and economics backgrounds, I don't see a need to cover those issues in depth here because anyone who's interested can read another analysis instead of or in addition to this one.

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id="fn: B">

Although this document is focused on tech, the lack of hands-on industry-expertise in regulatory bodies, legislation, and the courts, appears to cause problems in other industries as well. An example that's relatively well known due to a NY Times article that was turned into a movie is DuPont's involvement in the popularization of PFAS and, in particular, PFOA. Scientists at 3M and DuPont had evidence of the harms of PFAS going back at least to the 60s, and possibly even as far back as the 50s. Given the severe harms that PFOA caused to people who were exposed to it in significant concentrations, it would've been difficult to set up a production process for PFOA without seeing the harm it caused, but this knowledge, which must've been apparent to senior scientists and decision makers in 3M and DuPont, wasn't understood by regulatory agencies for almost four decades after it was apparent to chemical companies.

By the way, the NY Times article is titled "The Lawyer Who Became DuPont's Worst Nightmare" and it describes how DuPont made \$1B/yr in profit for years while hiding

the harms of PFOA, which was used in the manufacturing process for Teflon. This lawyer brought cases against DuPont that were settled for hundreds of millions of dollars; according to the article and movie, the litigation didn't even cost DuPont a single year's worth of PFOA profit. Also, DuPont manage to drag out the litigation for many years, continuing to reap the profit from PFOA. Now that enough evidence has mounted against PFOA, Teflon is now manufactured using PFO2OA or FRD-903, which are newer and have a less well understood safety profile than PFOA. Perhaps the article could be titled "The Lawyer Who Became DuPont's Largest Mild Annoyance".

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id="fn: J">

In the media, I've sometimes seen this framed as a conflict between tech vs. non-tech folks, but we can see analogous comments from people outside of tech. For example, in a panel discussion with Yale SOM professor Fiona Scott Morton and DoJ Antitrust Principal Deputy AAG Doha Mekki, Scott Morton noted that the judge presiding over the Sprint/T-mobile merger proceedings, a case she was an expert witness for, had comically wrong misunderstandings about the market, and that it's common for decisions to be made which are disconnected from "market realities". Mekki seconded this sentiment, saying "what's so fascinating about some of the bad opinions that Fiona identified, and there are many, there's AT&T Time Warner, Sabre Farelogix, T-mobile Sprint, they're everywhere, there's Amex, you know ..."

If you're seeing this or the other footnote in mouseover text and/or tied to a broken link, this is an issue with Hugo. At this point, I've spent more than an entire blog post's worth of effort working around Hugo breakage and am trying to avoid spending more time working around issues in a tool that makes breaking changes at a high rate. If you have a suggestion to fix this, I'll try it, otherwise I'll try to fix it when I switch away from Hugo.

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FEATURES

Deliveroo's Machine Learning Platform: Powering the Future of ML

Deliveroo Engineering

id="enter-deliveroo's-ml-platform">Enter Deliveroo's ML Platform

For the past three years, we have been building Deliveroo's Machine Learning Platform, or the ML Platform as we like to call it. The ML Platform boosts our model-building and deployment capabilities by standardising ML workflows, streamlining the end-to-end development process and simplifying model deployment. Besides saving software engineering effort through centralising tooling, the ML Platform also reduces the time that our ML engineers spend on infrastructure tasks. As a result, our ML engineers can now iterate their ML models 2-3x faster than before.

id="what-makes-our-ml-platform-tick">What makes our ML Platform tick?

At the core of our ML Platform lies a carefully curated tech stack - an integrated suite of infrastructure tools, services, and libraries. It blends robust open source technologies with purpose-built, in-house components. Key open source tools include Kubernetes, Argo, and Metaflow, all seamlessly connected with leading ML frameworks like TensorFlow and PyTorch. We choose mature, community-driven solutions and actively contribute back where we can. This entire ecosystem is powered by AWS, running on EKS, and anchored by our data warehouse.

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In the sections that follow, we'll take a closer look at the key components that drive our ML Platform.

id="metaflow">Metaflow

To give ML engineers seamless access to scalable compute on Kubernetes, we use Metaflow, a powerful Python library that helps break down complex model-building workflows into smaller, manageable jobs. These jobs are orchestrated by Argo, one of the backbone tools in our infrastructure. One of Metaflow's biggest strengths is its flexibility. It allows engineers to move effortlessly between local development, staging on Kubernetes, and production deployment—helping teams iterate quickly as they experiment and scale. Recently, we added support for GPUs and distributed training, enabling faster training times and the ability to work with much larger datasets.

id="inferoo">Inferoo

For models that require real-time predictions, we built Inferoo, our in-house inference solution. It serves models via both REST and gRPC APIs and currently handles over 1 billion requests per day. The motivation behind Inferoo was simple: to streamline model deployment and offer a consistent, reliable serving interface. Inferoo also integrates tightly with our Experimentation Platform, making it easy to run rapid A/B tests on different model versions. It also

includes observability features for both batch and online models to keep an eye on model staleness, data drift and end-to-end model cost.

id="feature-store">Feature Store

With Feature Store, we standardise how machine learning teams ingest and serve features for real-time inference. It supports batch ingestion from our data warehouse into Redis-backed online stores, with low-latency retrieval via a shared client library. Teams can manage feature definitions, materialisation pipelines, and versioning through a central Git-based registry. The first version of Feature Store focuses on enabling rapid experimentation and consistent deployment for live models. Future plans include real-time feature ingestion, feature sharing across teams, and support for offline training workflows.

id="how-we-work-as-a-team">How we work as a team

But even more important than the tech are the people who built all of it! Our team of software and ML engineers creates documentation, organises upskilling sessions, assists onboarding, maintains and monitors infrastructure, designs policies, selects external tooling and builds in-house apps such as our inference service and feature store. This year, the team is bringing all of our functionality together in an internal ML portal, which will serve as a centralised catalogue for all AI/ML models, offering visibility into model performance, system health, and operational cost metrics. Future iterations of the ML portal will also incorporate platform capabilities, to further simplify infrastructure creation and model deployment.

Three principles that we always keep in mind are automation, cohesion and self-serve:

Automation : Our goal is to develop a platform that automates repetitive tasks, keeping ML engineers focussed on building models and minimising the risk of human error.

Cohesion : We're building a unified platform that emphasises the user journey. This platform needs to seamlessly integrate the entire model lifecycle, minimising the time teams spend connecting different frameworks and onboarding models onto services.

Self-serve : We provide tooling that empower ML engineers to be self-sufficient, pushing the boundaries of technology without requiring the help of a software engineer at each turn. For example, we recently introduced support for GPUs and distributed capabilities for model training workflows. Both of these innovations significantly reduced model training time and enabled the handling of larger datasets.

id="whats-next">What's Next?

Deliveroo's ML teams are constantly pushing the envelope, requesting new functionality for our ML Platform. We are already hosting some of the 'next generation' AI models, as well as supporting integrations between our homegrown

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Book Notes: The Dark Art of Linear Algebra by Seth Braver — Chapter 1 Review

Ruslan Spivak

“Mathematics is the art of reducing any problem to linear algebra.” — William Stein

If you’ve ever looked at a vector and thought, “Just a column of numbers, right?”, this chapter will change that. The Dark Art of Linear Algebra (aka DALA) by Seth Braver opens with one of the clearest intros I’ve read. Not every part clicks on the first pass, but the effort pays off. Paired with the author’s videos, this is a strong starting point whether you’re learning math for the first time or coming back to it with purpose.

As I wrote in Unlocking AI with Math and [Book Notes] Infinitesimals, Derivatives, and Beer – Full Frontal Calculus (Ch. 1) , I’m not learning math to pass a test. I’m learning it to understand the machinery behind AI and robotics, and eventually build machines of my own. (That would be fun, right?)

That goal needs a solid grasp of linear algebra. And it starts with understanding what a vector really is. Not just how to work with vectors algebraically, but how they behave in space and fit into a larger structure.

This chapter helped me sharpen that understanding.

id=“chapter-notes”>Chapter Notes

id=“whats-a-vector”>What’s a Vector?

The book makes it clear that the answer to this question will evolve as you go deeper into linear algebra. But Chapter 1 starts simple: a vector is an arrow. A geometric object. A displacement.

In the video that comes with the chapter, the author even says to forget everything you think you know about vectors. He introduces them geometrically, which makes them feel tangible and helps you see familiar algebraic ideas in a visual, spatial way.

id=“vector-addition”>Vector Addition

The book introduces vector addition visually. Once you see vectors as displacements or moves through space, the addition feels natural. Almost obvious.

Image source: DALA Ch1

The text doesn’t focus on vector subtraction, but there’s an exercise on it. The companion video shows two methods. One of them is subtraction by addition: flip the direction of the vector you want to subtract, then add. It reminded me of that Office scene where Andy says “addition by subtraction,” and Michael asks, “What does that even mean?” In that context, it’s just a throwaway phrase. But in vector math, subtraction by addition is a real method. Flip the

vector, then add. If you’ve done engineering, you’ve likely seen this before.

Vector addition also follows familiar rules like commutativity and associativity . If those sound fuzzy, the book and video prove them using triangles and parallelograms. No heavy algebra, just geometry.

One nice bonus is that the commutative proof gives you another way to add vectors. Place both tails at the same point, draw a parallelogram, and the diagonal gives the sum. It’s clean and easy to visualize:

id=“stretching-vectors”>Stretching Vectors

Scalar multiplication is introduced as a way to stretch, shrink, or flip a vector, not just multiply its components.

The author even explains where the word scalar comes from. Numbers are called scalars because they scale vectors. I liked that he doesn’t assume you already know this.

To stretch a vector, multiply by 3.

To flip it, multiply by -1 .

To collapse it, multiply by 0.

The author even explains where the word scalar comes from.

Ruslan Spivak

It’s easier to remember when you learn it by drawing instead of just computing.

id=“standard-basis-vectors”>Standard Basis Vectors

Only after you’ve built a solid geometric understanding does the author introduce the standard basis vectors: i , j , and k . By then, it’s clear that $2i + 3j + 5k$ is just a weighted sum of familiar directions.

The chapter shows how to express vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ using these basis vectors, and how to rewrite them in column form.

id=“length-of-vectors”>Length of Vectors

Be sure to watch the videos that go with this chapter. They walk you through finding the length of a vector visually.

You’ll start with the Pythagorean theorem to calculate the length of a vector in $\mathbb{R}^3$, then extend the idea to $\mathbb{R}^n$. The chapter also proves the general length formula when a vector is written in Cartesian coordinates. Neat.

id=“the-dot-product”>The Dot Product

The chapter defines the dot product using the same geometric approach as earlier sections, and it makes sense. But for me, it really clicked in the physics example where work is

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How bad are search results? Let's compare Google, Bing, Marginalia, Kagi, Mwmb, and ChatGPT

Dan Luu

In The birth & death of search engine optimization , Xe suggests

Here's a fun experiment to try. Take an open source project such as yt-dlp and try to find it from a very generic term like "youtube downloader". You won't be able to find it because of all of the content farms that try to rank at the top for that term. Even though yt-dlp is probably actually what you want for a tool to download video from YouTube.

More generally, most tech folks I'm connected to seem to think that Google search results are significantly worse than they were ten years ago (Mastodon poll , Twitter poll , Threads poll). However, there's a sizable group of vocal folks who claim that search results are still great. E.g., a bluesky thought leader who gets high engagement says:

i think the rending of garments about how even google search is terrible now is pretty overblown 1

I suspect what's going on here is that some people have gotten so used working around bad software that they don't even know they're doing it, reflexively doing the modern equivalent of hitting ctrl+s all the time in editors, or ctrl+a; ctrl+c when composing anything in a text box . Every adept user of the modern web has a bag of tricks they use to get decent results from queries. From having watched quite a few users interact with computers, that doesn't appear to be normal, even among people who are quite competent in various technical fields, e.g., mechanical engineering 2 . However, it could be that people who are complaining about bad search result quality are just hopping on the "everything sucks" bandwagon and making totally unsubstantiated comments about search quality.

Since it's fairly easy to try out straightforward, naive, queries, let's try some queries. We'll look at three kinds of queries with five search engines plus ChatGPT and we'll turn off our ad blocker to get the non-expert browsing experience . I once had a computer get owned from browsing to a website with a shady ad, so I hope that doesn't happen here (in that case, I was lucky that I could tell that it happened because the malware was doing so much stuff to my computer that it was impossible to not notice).

One kind of query is a selected set of representative queries a friend of mine used to set up her new computer. My friend is a highly competent engineer outside of tech and wanted help learning "how to use computers", so I watched her try to set up a computer and pointed out holes in her mental model of how to interact with websites and software 3 .

The second kind of query is queries for the kinds of things I wanted to know in high school where I couldn't find the answer because everyone I asked (teachers, etc.) gave me obviously incorrect answers and I didn't know how to find

the right answer. I was able to get the right answer from various textbooks once I got to college and had access to university libraries, but the questions are simple enough that there's no particular reason a high school student shouldn't be able to understand the answers; it's just an issue of finding the answer, so we'll take a look at how easy these answers are to find. The third kind of query is a local query for information I happened to want to get as I was writing this post.

In grading the queries, there's going to be some subjectivity here because, for example, it's not objectively clear if it's better to have moderately relevant results with no scams or very relevant results mixed interspersed with scams that try to install badware or trick you into giving up your credit card info to pay for something you shouldn't pay for . For the purposes of this post, I'm considering scams to be fairly bad, so in that specific example, I'd rate the moderately relevant results above the very relevant results that have scams mixed in. As with my other posts that have some kind of subjective ranking, there's both a short summary as well as a detailed description of results, so you can rank services yourself, if you like.

In the table below, each column is a query and each row is a search engine or ChatGPT. Results are rated (from worst to best) Terrible, Very Bad, Bad, Ok, Good, and Great, with worse results being more red and better results being more blue.

The queries are:

download youtube videos

ad blocker

download firefox

Why do wider tires have better grip?

Why do they keep making cpu transistors smaller?

vancouver snow forecast winter 2023

YouTube Adblock Firefox Tire CPU Snow

Marginalia Ok Good Ok Bad Bad Bad

ChatGPT V. Bad Great Good V. Bad V. Bad Bad

Mwmb Bad Bad Bad Bad Bad Bad

Kagi Bad V. Bad Great Terrible Bad Terrible

Google Terrible V. Bad Bad Bad Bad Terrible

Bing Terrible Terrible Great Terrible Ok Terrible

Marginalia does relatively well by sometimes providing decent but not great answers and then providing no answers or very obviously irrelevant answers to the questions it can't

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Randomize HN

Dan Luu

You ever notice that there's this funny threshold for getting to the front page on sites like HN? The exact threshold varies depending on how much traffic there is, but, for articles that aren't wildly popular, there's this moment when the article is at $N-1$ votes. There is, perhaps, a 60% chance that the vote will come and the article will get pushed to the front page, where it will receive a slew of votes. There is, maybe, a 40% chance it will never get the vote that pushes it to the front page, causing it to languish in obscurity forever.

It's non-optimal that an article that will receive 50 votes in expectation has a 60% chance of getting 100+ votes, and a 40% chance of getting 2 votes. Ideally, each article would always get its expected number of votes and stay on the front page for the expected number of time, giving readers exposure to the article in proportion to its popularity. Instead, by random happenstance, plenty of interesting content never makes it the front page, and as a result, the content that does make it gets a higher than optimal level of exposure.

You also see the same problem, with the sign bit flipped, on low traffic sites that push things to the front page the moment they're posted, like lobste.rs and the smaller sub-reddits : they displace links that most people would be interested in by putting links that almost no one cares about on the front page just so that the few things people do care about get enough exposure to be upvoted. On reddit, users "fix" this problem by heavily downvoting most submissions, pushing them off the front page, resulting in a problem that's fundamentally the same as the problem HN has.

Instead of implementing some simple and easy to optimize, sites pile on ad hoc rules. Reddit implemented the rising page , but it fails to solve the problem. On low-traffic sub-reddits, like r/programming the threshold is so high that it's almost always empty. On high-traffic sub-reddits, anything that's upvoted enough to make it to the rising page is already wildly successful, and whether or not an article becomes successful is heavily dependent on whether or not the first couple voters happen to be people who upvote the post instead of downvoting it, i.e., the problem of getting onto the rising page is no different than the problem of getting to the top normally.

HN tries to solve the problem by manually penalizing certain domains and keywords. That doesn't solve the problem for the 95% of posts that aren't penalized. For posts that don't make it to the front page, the obvious workaround is to delete and re-submit your post if it doesn't make the front page the first time around , but that's now a ban worthy offense . Of course, people are working around that, and HN has a workaround for the workaround, and so on. It's endless. That's the problem with "simple" ad hoc solutions.

There's an easy fix, but it's counter-intuitive. By adding a small amount of random noise to the rank of an article, we can smooth out the discontinuity between making it onto

the front page and languishing in obscurity. The math is simple, but the intuition is even simpler 1 . Imagine a vastly oversimplified model where, for each article, every reader upvotes with a fixed probability and the front page gets many more eyeballs than the new page. The result follows. If you like, you can work through the exercise with a more realistic model, but the result is the same 2 .

That doesn't solve the problem for the 95% of posts that aren't penalized.

Dan Luu

Adding noise to smooth out a discontinuity is a common trick when you can settle for an approximate result. I recently employed it to work around the classic floating point problem, where adding a tiny number to a large number results in no change, which is problem when adding many small numbers to some large numbers 3 . For a simple example of applying this, consider keeping a reduced precision counter that uses $\log\log(n)$ bits to store the value. Let $\text{countVal}(x) = 2^x$ and $\text{inc}(x) = \text{if } (\text{rand}(2^x) == 0) \text{ } x++4$. Like understanding when to apply Taylor series, this is a simple trick that people are often impressed by if they haven't seen it before 5 .

Update : HN tried this! Dan Gackle tells me that it didn't work very well (it resulted in a lot of low quality junk briefly hitting the front page and then disappearing. I think that might be fixable by tweaking some parameters, but the solution that HN settled on, having a human (or multiple humans) put submissions that are deemed to be good or interesting into a "second chance queue" that boosts the submission onto the front page, works better than an a simple randomized algorithm with no direct human input could with any amount of parameter tweaking. I think this is also true of moderation, where the "new" dang/sctb moderation regime has resulted in a marked increase in comment quality, probably better than anything that could be done with an automated ML-based solution today — Google and FB have some of the most advanced automated systems in the world, and the quality of the result is much worse than what we see on HN.

Also, at the time this post was written (2013), the threshold to get onto the front page was often 2-3 votes, making the marginal impact of a random passerby who happens to like a submission checking the new page very large. Even during off peak times now (in 2019), the threshold seems to be much higher, reducing the amount of randomness. Additionally, the rise in the popularity of HN increased the sheer volume of low quality content that languishes on the new page, which would reduce the exposure that any particular "good" submission would get if it were among the 30 items on the new page that would randomly get boosted onto the front page. That doesn't mean there aren't still problems with the current system: most people seem to upvote and comment based on the title of the article and not the content (to check this, read the comments of articles that are mistitled before someone calls this out for a particular post

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Docker run all the things with user namespaces

Jessie Frazelle

If you weren't aware user namespace support was added to Docker awhile back in the "Experimental" builds. But with the upcoming release of Docker Engine 1.10.0, Phil Estes is working on moving it into stable . Now this is all super exciting and blah blah blah, but what I am going to talk about today is how I started running all the containers from my Docker Containers on the Desktop with the new user namespace support. The containers/images in that post were already doing some linux-y magic, but with a little more, they are perfect. I'm not going to go through them all but I will go through some interesting ones, including even how to run Docker-in-Docker.

id="chrome">Chrome

This one was shockingly easy. The only things I needed to add to my original command were `-group-add video` and `-group-add audio` . Makes sense right.. we obviously want to be a member of those groups to watch Taylor Swift music videos.

The full command is below. I even made a custom seccomp whitelist for chrome, you can view it in my dotfiles repo: github.com/jessfraz/dotfiles . Seccomp will be shipped in 1.10 as well, along with a default whitelist! (But I degress that is not the point of this blog post.)

```
$ docker run -d \ -memory 3gb \ -v /etc/localtime:/etc/localtime:ro \ -v /tmp/.X11-unix:/tmp/.X11-unix \ -e DISPLAY=unix$DISPLAY \ -v $HOME/Downloads:/root/Downloads \ -v $HOME/chrome:/data \ -v /dev/shm:/dev/shm \ -security-opt seccomp:/etc/docker/seccomp/chrome.json \ -device /dev/snd \ -device /dev/dri \ -device /dev/video0 \ -group-add audio \ -group-add video \ -name chrome \ jess/chrome -user-data-dir=/data
```

id="notify-osd-and-irssi">Notify-osd and Irssi

Now I have always run my notifications daemon in a container, because that stuff is nasty to install, so many dependencies, ewwww. This one was a bit more tricky beacuse it involves dbus but it is a way cleaner solution than the way I was originally running it.

```
$ docker run -d \ -v /etc/localtime:/etc/localtime:ro \ -v /tmp/.X11-unix:/tmp/.X11-unix \ -v /etc \ -v /home/user/.dbus \ -v /home/user/.cache/dconf \ -e DISPLAY=unix$DISPLAY \ -name notify_osd \ jess/notify-osd
```

```
# you can test with $ docker exec -it notify_osd notify-send hello
```

Not to bad right? I am creating those volumes on run so that we can then share them with our irssi container. This way when someone pings me I get a notification, duh!

```
$ docker run -rm -it \ -v /etc/localtime:/etc/localtime:ro \ -v $HOME/.irssi:/home/user/.irssi \ -volumes-from notify_osd
```

```
\ -e DBUS_SESSION_DBUS_ADDRESS="unix:abstract=/home/user/.dbus/session-bus/$(docker exec notify_osd ls /home/user/.dbus/session-bus/)" \ -name irssi \ jess/irssi
```

So this is pretty simple as well even considering the real gross part is trying to get the `DBUS_SESSION_DBUS_ADDRESS` from our `notify_osd` container.

Let's get into the fun part.

id="docker-in-docker">Docker-in-Docker

When running the docker daemon with user namespace support, you cannot use docker run flags like `-privileged` , `-net host` , `-pid host` , etc. These are for pretty obvious reasons so I'm not going to get into it, if you want to know more RTFM.

Okay so we can't use `-privileged` , but but but that's how I run docker-in-docker... ok let's think about it. What is `-privileged` actually doing? Well for starters it's allowing all capabilities, but... do we really need them all? The answer is no.. all we really need is `CAP_SYS_ADMIN` and `CAP_NET_ADMIN` . So that gets us pretty far but we also need to disable the default seccomp profile (because it drops clone args, mount , and a bunch of others). Lastly we need to run with a different apparmor profile so we can have more capabilities as well.

This leaves us with:

Dockerfile \$ docker run -d \ -v /etc/localtime:/etc/localtime:ro \ -v /tmp/.

Jessie Frazelle

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Files are hard

Dan Luu

I haven't used a desktop email client in years. None of them could handle the volume of email I get without at least occasionally corrupting my mailbox. Pine, Eudora, and outlook have all corrupted my inbox, forcing me to restore from backup. How is it that desktop mail clients are less reliable than gmail, even though my gmail account not only handles more email than I ever had on desktop clients, but also allows simultaneous access from multiple locations across the globe? Distributed systems have an unfair advantage, in that they can be robust against total disk failure in a way that desktop clients can't, but none of the file corruption issues I've had have been from total disk failure. Why has my experience with desktop applications been so bad?

Well, what sort of failures can occur? Crash consistency (maintaining consistent state even if there's a crash) is probably the easiest property to consider, since we can assume that everything, from the filesystem to the disk, works correctly; let's consider that first.

id="crash-consistency">Crash Consistency

Pillai et al. had a paper and presentation at OSDI '14 on exactly how hard it is to save data without corruption or data loss.

Let's look at a simple example of what it takes to save data in a way that's robust against a crash. Say we have a file that contains the text a foo and we want to update the file to contain a bar . The pwrite function looks like it's designed for this exact thing. It takes a file descriptor, what we want to write, a length, and an offset. So we might try

```
prwrite([file], "bar", 3, 2) /\ write 3 bytes at offset 2
```

What happens? If nothing goes wrong, the file will contain a bar , but if there's a crash during the write, we could get a boo , a far , or any other combination. Note that you may want to consider this an example over sectors or blocks and not chars/bytes.

If we want atomicity (so we either end up with a foo or a bar but nothing in between) one standard technique is to make a copy of the data we're about to change in an undo log file, modify the "real" file, and then delete the log file. If a crash happens, we can recover from the log. We might write something like

```
creat(/dir/log); write(/dir/log, "2,3,foo", 7); pwrite(/dir/orig, "bar", 3, 2); unlink(/dir/log);
```

This should allow recovery from a crash without data corruption via the undo log, at least if we're using ext3 and we made sure to mount our drive with data=journal . But we're out of luck if, like most people, we're using the default 1 – with the default data=ordered , the write and pwrite syscalls can be reordered, causing the write to orig to happen before

the write to the log, which defeats the purpose of having a log. We can fix that.

```
creat(/dir/log); write(/dir/log, "2, 3, foo"); fsync(/dir/log); /\ don't allow write to be reordered past pwrite pwrite(/dir/orig, 2, "bar"); fsync(/dir/orig); unlink(/dir/log);
```

That should force things to occur in the correct order, at least if we're using ext3 with data=journal or data=ordered . If we're using data=writeback , a crash during the the write or fsync to log can leave log in a state where the filesize has been adjusted for the write of "bar", but the data hasn't been written, which means that the log will contain random garbage. This is because with data=writeback , metadata is journaled , but data operations aren't, which means that data operations (like writing data to a file) aren't ordered with respect to metadata operations (like adjusting the size of a file for a write).

We can fix that by adding a checksum to the log file when creating it. If the contents of log don't contain a valid checksum, then we'll know that we ran into the situation described above.

```
creat(/dir/log); write(/dir/log, "2, 3, [checksum], foo"); /\ add checksum to log file fsync(/dir/log); pwrite(/dir/orig, 2, "bar"); fsync(/dir/orig); unlink(/dir/log);
```

That's safe, at least on current configurations of ext3. But it's legal for a filesystem to end up in a state where the log is never created unless we issue an fsync to the parent directory.

```
creat(/dir/log); write(/dir/log, "2, 3, [checksum], foo"); fsync(/dir/log); fsync(/dir); /\ fsync parent directory of log file pwrite(/dir/orig, 2, "bar"); fsync(/dir/orig); unlink(/dir/log);
```

That should prevent corruption on any Linux filesystem, but if we want to make sure that the file actually contains "bar", we need another fsync at the end.

```
creat(/dir/log); write(/dir/log, "2, 3, [checksum], foo"); fsync(/dir/log); fsync(/dir); pwrite(/dir/orig, 2, "bar"); fsync(/dir/orig); unlink(/dir/log); fsync(/dir);
```

That results in consistent behavior and guarantees that our operation actually modifies the file after it's completed, as long as we assume that fsync actually flushes to disk. OS X and some versions of ext3 have an fsync that doesn't really flush to disk. OS X requires fcntl(F_FULLFSYNC) to flush to disk, and some versions of ext3 only flush to disk if the inode changed (which would only happen at most once a second on writes to the same file, since the inode mtime has one second granularity), as an optimization.

Even if we assume fsync issues a flush command to the disk, some disks ignore flush directives for the same reason fsync is gimped on OS X and some versions of ext3 – to look better in benchmarks. Handling that is beyond the scope of this post, but the Rajimwale et al. DSN '11 paper and related

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Your dev environment matters less than you think

Itamar Turner-Trauring

How do you setup your dev environment? Depending on your language there are many choices of editor, package manager, build tool, linter, on and on. And every article you find will have a different combination of suggested tools, each of which claiming that their list is The Right Way To Do Things.

So which do you choose?

The short answer: it doesn't matter. Your choice of dev environment is meaningless.

The slightly less flippant answer is that, yes, there are some constraints on which tools you should pick, but otherwise you should just pick something and move on.

Let's see why dev environments don't matter that much in the end, and what limited constraints you should apply when choosing your tools.

id="learning-how-to-cook">Learning how to cook

Imagine you're training to become a chef. You will need to learn how to use a knife correctly, to chop and dice safely and quickly.

And yes, you need a sharp knife. But when you're starting out, it doesn't matter which knife you use: just pick something sharp and good enough, and move on. After all, the knife is just a tool.

The people eating the food you cook don't care about which knife you used: they care how the food tastes and looks.

After six months in the kitchen, you'll start understanding how you personally use a knife, what cuisines you want to pursue, what techniques you want to vary. And then you'll have the knowledge to pick a specific knife or knives exactly suited to your needs.

But remember: the people eating your food still won't care which knife you used.

id="choosing-a-dev-environment">Choosing a dev environment

When you use a website, you don't care which build tool the programmer used. When you run an app, you don't care which editor they used. You want the software to work, to do what it says, to be easy to use, to get out of your way—and you don't care how they did it.

And that applies just as much to the users of your code: they don't care which tools you used.

And when you're starting out, whether programming in general or a new language or framework, you don't know how you will like to work. So instead of obsessing over

finding the ideal development environment and toolchain, just pick tools that are good enough:

Popular: So you can easily find help.

Easy to get going: Your goal is ship useful code, and as a beginner time spent fiddling with your dev environment won't help with that.

Once you have enough experience, you will start developing opinions. You might become choosy about which tools you use, or end up customizing them to your needs. You might even write an article about your particular dev environment and favorite tools.

But however strong your preferences are, chances are that given tools you don't quite like as much, you will still do just fine. If you know what you're doing, you can chop vegetables with any sharp knife, even if it's not your favorite.

Tired of scrambling to get your job done?

If you were productive enough, you could take the afternoon off, confident you'd produced high value work. Not to mention having an easier time finding a new job when you need one.

Learn the secret skills of productive programmers .

You want the software to work, to do what it says, to be easy to use, to get out of your way—and you don't care how they did it.

Itamar Turner-Trauring

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Let's Build A Simple Interpreter. Part 19: Nested Procedure Calls

Ruslan Spivak

“What I cannot create, I do not understand.” —
Richard Feynman

As I promised you last time, today we're going to expand on the material covered in the previous article and talk about executing nested procedure calls. Just like last time, we will limit our focus today to procedures that can access their parameters and local variables only. We will cover accessing non-local variables in the next article.

Here is the sample program for today:

```
procedure Alpha ( a : integer ; b : integer ) ; var x : integer ;  
  
procedure Beta ( a : integer ; b : integer ) ; var x : integer ;  
begin x := a * 10 + b * 2 ; end ;  
  
begin x := ( a + b ) * 2 ;  
  
Beta ( 5 , 10 ) ; { procedure call } end ;  
  
begin { Main }  
  
Alpha ( 3 + 5 , 7 ) ; { procedure call }  
  
end . { Main }
```

The nesting relationships diagram for the program looks like this:

Some things to note about the above program:

it has two procedure declarations, Alpha and Beta

Alpha is declared inside the main program (the global scope)

the Beta procedure is declared inside the Alpha procedure

both Alpha and Beta have the same names for their formal parameters: integers a and b

and both Alpha and Beta have the same local variable x

the program has nested calls: the Beta procedure is called from the Alpha procedure, which, in turn, is called from the main program

Now, let's do an experiment. Download the part19.pas file from GitHub and run the interpreter from the previous article with the part19.pas file as its input to see what happens when the interpreter executes nested procedure calls (the main program calling Alpha calling Beta):

ENTER: PROGRAM Main CALL STACK 1 : PROGRAM Main

...

ENTER: PROCEDURE Beta CALL STACK 2 : PROCEDURE
Beta a : 5 b : 10 2 : PROCEDURE Alpha a : 8 b : 7 x : 30 1 :
PROGRAM Main

LEAVE: PROCEDURE Beta CALL STACK 2 : PROCEDURE
Beta a : 5 b : 10 x : 70 2 : PROCEDURE Alpha a : 8 b : 7 x : 30
1 : PROGRAM Main

...

LEAVE: PROGRAM Main CALL STACK 1 : PROGRAM Main

It just works! There are no errors. And if you study the contents of the ARs(activation records), you can see that the values stored in the activation records for the Alpha and Beta procedure calls are correct. So, what's the catch then? There is one small issue. If you take a look at the output where it says 'ENTER : PROCEDURE Beta', you can see that the nesting level for the Beta and Alpha procedure call is the same, it's 2 (two). The nesting level for Alpha should be 2 and the nesting level for Beta should be 3. That's the issue that we need to fix. Right now the nesting_level value in the visit_ProcedureCall method is hardcoded to be 2 (two):

```
ar = ActivationRecord ( name = proc_name , type = ARTYPE .  
PROCEDURE , nesting_level = 2 , )
```

```
proc_symbol = node . proc_symbol ...
```

Let's get rid of the hardcoded value. How do we determine a nesting level for a procedure call? In the method above we have a procedure symbol and it is stored in a scoped symbol table that has the right scope level that we can use as the value of the nesting_level parameter (see Part 14 for more details about scopes and scope levels).

How do we get to the scoped symbol table's scope level through the procedure symbol?

Let's look at the following parts of the ScopedSymbolTable class:

```
def insert ( self , symbol ): self . log ( f 'Insert: { symbol .  
name } ' ) self . _symbols [ symbol . name ] = symbol
```

Looking at the code above, we can see that we could assign the scope level of a scoped symbol table to a symbol when we store the symbol in the scoped symbol table (scope) inside the insert method. This way we will have access to the procedure symbol's scope level in the visit_Procedure method during the interpretation phase. And that's exactly what we need.

Let's make the necessary changes:

First, let's add a scope_level member to the Symbol class and give it a default value of zero:

Next, let's assign the corresponding scope level to a symbol when storing the symbol in a scoped symbol table:

```
symbol . scope_level = self . scope_level
```

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Here and Now: Reusing Code at Feedzai with JupyterLab Snippets

João Palmeiro · Feedzai Tech

Data scientists use different Jupyter notebooks every day — ranging from disposable ones for quick tasks to those shareable with clients. Over time, more and more notebooks accumulate, making it increasingly difficult to reuse them in whole or in part. To mitigate this problem and make the most relevant pieces of code quickly accessible to every data scientist, we developed JupyterLab Snippets at Feedzai — our take on leveraging code snippets directly on JupyterLab.

JupyterLab is a computational notebook platform that enables us to carry on data science work (and beyond) via notebooks. These notebooks, where code, text, and images come together, allow us to complete all kinds of tasks, keeping each input close to each output. At Feedzai, data scientists have access to different JupyterLab environments packed with custom notebooks and Python packages. Here they go from analyzing data to training models, from preparing reports to debugging the system — a “lab of all trades”, we could say.

Given the importance of JupyterLab and notebooks in the daily work of our data scientists, we started a research project that culminated in JupyterLab Snippets. First, we collected statistics from internal notebook repositories across teams and conducted user interviews with junior and senior data scientists. We needed to know more about how data scientists actually use JupyterLab and notebooks, and what their ideas are for a better platform.

A sneak peek of JupyterLab Snippets. From the very start, every data scientist can import snippets immediately or start their collection by creating one from scratch.

After compiling the insights, several aspects became clear:

JupyterLab and notebooks are heavily used for all sorts of data science and ad-hoc tasks (typically one per notebook). There is no single type of task (e.g., exploratory data analysis) that is more commonly addressed with notebooks than others.

While file structures may look similar at first glance (each environment has at least a model training notebook, for example), data scientists have different and personal workflows for coding notebooks.

Duplicating past notebooks available in the environment or getting them from repositories is a very common practice.

Notebooks composed of code snippets is also an identified practice. These notebooks are not intended to be run, and contain code and text for future reference (imagine a notepad in notebook format). This case and the previous one imply removal of unnecessary parts and parameterization.

Data scientists regularly communicate within teams and individually. Sharing notebooks/snippets is a common practice.

No significant issues were identified with the JupyterLab/notebook interface.

It is true that data scientists have their own ways of finding whatever they need for their next notebook, and AI coding assistants are proliferating. Even so, would it be beneficial to improve the experience of reusing code in JupyterLab? After analyzing all the insights, we believe so. We believe in an interface that allows every data scientist to easily manage and use “their snippets” now and in one year, while also facilitating their sharing with other data scientists. JupyterLab Snippets aims to improve the developer experience, focusing on productivity and satisfaction, while having a low maintenance cost. Here is JupyterLab Snippets in its essence.

Introducing JupyterLab Snippets

JupyterLab Snippets is a JupyterLab extension for data scientists to create, manage, and use code snippets directly on JupyterLab, facilitating code reuse in their notebooks. To be more precise, the interface is a JupyterLab extension and the backend, for managing the snippet data, is a Jupyter Server one. Both communicate via a REST API.

Two of the three main parts: Snippet Sidebar and Snippet Editor. The snippets are adapted from the Vega-Altair documentation examples.

The interface, a part of JupyterLab like notebooks and the like, is divided into three main components:

Snippets Sidebar : The central hub for JupyterLab Snippets, available as a new tab in the right sidebar of JupyterLab. It allows the data scientist to manage and utilize the code snippets. Each snippet is available on a Snippet Card.

Snippet Editor : The form-like interface for creating new snippets and editing existing ones. When opened, it is available as a tab in the main work area of JupyterLab.

Shortcuts : Options for creating snippets from notebook cells. These options are available from the context menu after right-clicking on a cell, and the option to start a snippet from a cell is also available from the cell toolbar.

Snippet Cards and Shortcuts. Shortcuts consist of new cell context menu options and a new option in the cell toolbar.

In the Snippets Sidebar, in addition to the Personal collection, data scientists will find a set of pre-made snippets: the Feedzai collection. They can switch between both (or view them all at once) by selecting the desired collection in the toolbar just above the list of available snippets. The Feedzai collection is a set of snippets for common actions performed in notebooks, prepared with the help of our data scientists. This collection is specially designed for newcomers.

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Startup options v. cash

Dan Luu

I often talk to startups that claim that their compensation package has a higher expected value than the equivalent package at a place like Facebook, Google, Twitter, or Snapchat. One thing I don't understand about this claim is, if the claim is true, why shouldn't the startup go to an investor, sell their options for what they claim their options to be worth, and then pay me in cash? The non-obvious value of options combined with their volatility is a barrier for recruiting.

Additionally, given my risk function and the risk function of VCs, this appears to be a better deal for everyone. Like most people, extra income gives me diminishing utility, but VCs have an arguably nearly linear utility in income. Moreover, even if VCs shared my risk function, because VCs hold a diversified portfolio of investments, the same options would be worth more to them than they are to me because they can diversify away downside risk much more effectively than I can. If these startups are making a true claim about the value of their options, there should be a trade here that makes all parties better off.

In a classic series of essays written a decade ago, seemingly aimed at convincing people to either found or join startups, Paul Graham stated "If you wanted to get rich, how would you do it? I think your best bet would be to start or join a startup. That's been a reliable way to get rich for hundreds of years" and "Risk and reward are always proportionate." This risk-reward assertion is used to back the claim that people can make more money, in expectation, by joining startups and taking risky equity packages than they can by taking jobs that pay cash or cash plus public equity. However, the premise — that risk and reward are always proportionate — isn't true in the general case. It's basic finance 101 that only assets whose risk cannot be diversified away carry a risk premium (on average). Since VCs can and do diversify risk away, there's no reason to believe that an individual employee who "invests" in startup options by working at a startup is getting a deal because of the risk involved. And by the way, when you look at historical returns, VC funds don't appear to outperform other investment classes even though they get to buy a kind of startup equity that has less downside risk than the options you get as a normal employee.

So how come startups can't or won't take on more investment and pay their employees in cash? Let's start by looking at some cynical reasons, followed by some less cynical reasons.

id="cynical-reasons">Cynical reasons

One possible answer, perhaps the simplest possible answer, is that options aren't worth what startups claim they're worth and startups prefer options because their lack of value is less obvious than it would be with cash. A simplistic argument that this might be the case is, if you look at the amount investors pay for a fraction of an early-stage or mid-

stage startup and look at the extra cash the company would have been able to raise if they gave their employee option pool to investors, it usually isn't enough to pay employees competitive compensation packages. Given that VCs don't, on average, have outsized returns, this seems to imply that employee options aren't worth as much as startups often claim. Compensation is much cheaper if you can convince people to take an arbitrary number of lottery tickets in a lottery of unknown value instead of cash.

Some common ways that employee options are misrepresented are:

id="strike-price-as-value">Strike price as value

A company that gives you 1M options with a strike price of \$10 might claim that those are "worth" \$10M. However, if the share price stays at \$10 for the lifetime of the option, the options will end up being worth \$0 because an option with a \$10 strike price is an option to buy the stock at \$10, which is not the same as a grant of actual shares worth \$10 a piece.

id="public-valuation-as-value">Public valuation as value

Let's say a company raised \$300M by selling 30% of the company, giving the company an implied valuation of \$1B. The most common misrepresentation I see is that the company will claim that because they're giving an option for, say, 0.1% of the company, your option is worth $\$1B * 0.001 = \$1M$. A related, common, misrepresentation is that the company raised money last year and has increased in value since then, e.g., the company has since doubled in value, so your option is worth \$2M. Even if you assume the strike price was \$0 and go with the last valuation at which the company raised money, the implied value of your option isn't \$1M because investors buy a different class of stock than you get as an employee.

There are a lot of differences between the preferred stock that VCs get and the common stock that employees get; let's look at a couple of concrete scenarios.

Let's say those investors that paid \$300M for 30% of the company have a straight (1x) liquidation preference, and the company sells for \$500M. The 1x liquidation preference means that the investors will get 1x of their investment back before lowly common stock holders get anything, so the investors will get \$300M for their 30% of the company. The other 70% of equity will split \$200M: your 0.1% common stock option with a \$0 strike price is worth \$285k (instead of the \$500k you might expect it to be worth if you multiply \$500M by 0.001).

The preferred stock VCs get usually has at least a 1x liquidation preference. Let's say the investors had a 2x liquidation preference in the above scenario. They would get 2x their investment back before the common stockholders split the rest of the company. Since $2 * \$300M$ is greater than \$500M, the investors would get everything and the remaining equity holders would get \$0.

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Transcript of Elon Musk on stage with Dave Chapelle

Dan Luu

This is a transcription of videos Elon Musk's appearance on stage with Dave Chapelle using OpenAI's Whisper model with some manual error corrections and annotations for crowd noise.

As with the Exhibit H Twitter text message release, there are a lot of articles that quote bits of this, but the articles generally missing a lot of what happened and often paint a misleading picture of happened and the entire thing is short enough that you might as well watch or read it instead of reading someone's misleading summary. In general, the media seems to want to paint a highly unflattering picture of Elon, resulting in articles and virtual tweets that are factually incorrect. For example, it's been widely incorrectly reported that, during the "I'm rich, bitch" part, horns were played to drown out the crowd's booing of Elon, but the horn sounds were played when the previous person said the same thing, which was the most cheered statement that was recorded. The sounds are much weaker when Elon says "I'm rich, bitch" and can't be heard clearly, but it sounds like a mix of booing and cheering. It was probably the most positive crowd response that Elon got from anything and it seems inaccurate in at least two ways to say that horns were played to drown out the booing Elon was receiving. On the other hand, even though the media has tried to paint as negative a picture of Elon as possible, it's done quite a poor job and a boring, accurate, accounting of what happened in many of other sections are much less flattering than the misleading summaries that are being passed around.

Video 1

Dave : Ladies and gentlemen, make some noise for the richest man in the world.

Crowd : [mixed cheering, clapping, and boos; boos drown out cheering and clapping after a couple of seconds and continue into next statements]

Dave : Cheers and boos, I say

Crowd : [brief laugh, boos continue to drown out other crowd noise]

Dave : Elon

Crowd : [booing continues]

Elon : Hey Dave

Crowd : [booing intensifies]

Elon : [unintelligible over booing]

Dave : Controversy, buddy.

Crowd : [booing continues; some cheering can be heard]

Elon : Weren't expecting this, were ya?

Dave : It sounds like some of them people you fired are in the audience.

Crowd : [laughs, some clapping can be heard]

Elon : [laughs]

Crowd : [booing resumes]

Dave : Hey, wait a minute. Those of you booing

Crowd : [booing intensifies]

Dave : Tough [unintelligible due to booing] sounds like

Dave : You know there's one thing. All those people are booing. I'm just. I'm just pointing out the obvious. They have terrible seats. [unintelligible due to crowd noise]

Crowd : [weak laughter]

Dave : All coming from wayyy up there [unintelligible] last minute non-[unintelligible] n****. Boooo. Booooooooo.

Crowd : [quiets down]

Dave : Listen.

Crowd : [booing resumes]

Dave : Whatever. Look motherfuckas. This n**** is not even trying to die on earth

Crowd : [laughter mixed with booing, laughter louder than boos]

Video 2

Dave : His whole business model is fuck earth I'm leaving anyway

Crowd : [weak laughter, weak mixed sounds]

Dave : Do all you want. Take me with you n**** I'm going to Mars

Crowd : [laughter]

Dave : Whatever kind of pussy they got up there, that's what we'll be doin

Crowd : [weak laughter]

Dave : [laughs] Anti-gravity titty bars. Follow your dreams bitch and the money just flow all over the room

Crowd : [weak laughter]

Elon : [laughs]

Crowd : [continued laughter drowned out by resumed boo-

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Personal Infrastructure

Jessie Frazelle

This post is kind of like “part two” on my series on all the weird things I do for my personal infrastructure. If you missed “part one”, you should check out Home Lab is the Dopest Lab .

I run a lot of little things to make my life easier, like a CI, some bots, and a bunch of services just for the lolz. This post will go over all of those. These run scattered across my NUCs and the cloud.

Let’s start with the most useful.

id=“continuous-integration”>Continuous Integration

I host my own continuous integration server. Yes, you guessed it... it’s Jenkins. I use the Jenkins DSL plugin to keep everything in sync. You can find all my DSLs in my repo github.com/jessfraz/jenkins-dsl . This has all the configurations for views, keeps forks up to date, mirrors all my repositories to private git (more on this in [git](#)), builds all Dockerfiles to push to Docker Hub and my private registry (more on this in [private docker registry](#)) and a bunch of maintenance scripts.

The Makefile in this repo calls out to bash scripts which generate new DSLs for any new GitHub repos I create. Yep I even generate the automation...

There’s a bunch of other fun things in there as well that you can discover by poking around yourself.

I host my own postfix server alongside Jenkins. You can find the postfix docker image at r.j3ss.co/postfix or the Dockerfile . It’s super minimal and less gross than literally every other postfix image in existence.

You can run it with:

```
$ docker run --restart always -d \ --name postfix \ --net container:jenkins \ -e “ROOT_ALIAS=root@blah.com” \ -e “RELAY=[smtp-relay.gmail.com]:587” \ -e “TLS=1” \ -e “MY_DESTINATION=..., localhost” \ -e “MAIL-NAME=blah.com” \ r.j3ss.co/postfix
```

id=“private-docker-registry”>Private Docker Registry

I host my own private docker registry with my own notary server and authentication server. Why? Well because about 4 years ago when I started using docker, Docker Hub was super slow and I came to love having my own super fast one.

I still push all the images to both Docker Hub and my registry and both are signed so it’s really like I am using Docker Hub as my backup. Yay, highly available... just kidding.

I made a pretty shitty UI for it. You can play with it at r.j3ss.co . The UI is from my reg project but the server component lives in the server subdirectory .

The really nice thing about both the reg command line and server is that you can get a list of CVEs on an image.

I do this by hosting my own instance of CoreOS’s Clair .

Most of my Dockerfiles live at github.com/jessfraz/dockerfiles if you are curious.

I also went over all of this on my talk on Over Engineering my Laptop / Container Linux on the Desktop . This includes all the reasons why I have continuous integration as well.

I have a script to cleanup the registry of old images clean-registry . This deletes old registry blobs that are not used in the latest version of the tag. I don’t really care about old images and I don’t want to have a huge registry filled with old shit. There is a jenkins DSL to run this.

id=“git-server”>Git Server

I host my own git server. You can find the gitserver docker image at r.j3ss.co/gitserver or the Dockerfile .

You can run it with:

```
$ docker run --restart always -d \ --name gitserver \ -p 127.0.0.1:22:22 \ -e “PUBKEY=$(cat /ssh/authorized_keys)” \ -v “/mnt/disks/gitserver:/home/git” \ r.j3ss.co/gitserver
```

It has it’s own UI that is run with Gitiles. You can find the Gitiles docker image at r.j3ss.co/gitiles or the Dockerfile .

You can run it with:

```
$ docker run --restart always -d \ --name gitiles \ -p 127.0.0.1:8080:8080 \ -e BASE_GIT_URL=“git@git.blah.com” \ -e SITE_TITLE=“git.blah.com” \ -v “/mnt/disks/gitserver:/home/git” \ -w /home/git \ r.j3ss.co/gitiles
```

id=“ghb0t”>ghb0t

This is one of my most useful things. It’s a GitHub Bot to automatically delete your fork’s branches after a pull request has been merged.

I am so OCD about keeping git repos clean and this is my little helper.

Check out the repo: github.com/jessfraz/ghb0t .

dir=“ltr” lang=“en”>I go to fork your thing and there is like 300 branches my face is like pic.twitter.com/JpdpO447KS

— jessie frazelle (@jessfraz) January 23, 2017

id=“irc-bouncer”>IRC Bouncer

I host my own IRC Bouncer with ZNC. You can find the ZNC docker image at r.j3ss.co/znc or the Dockerfile .

You can run it with:

```
$ docker run --restart always -d \ --name znc \ -p 6697:6697 \
```

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Are closed social networks inevitable?

Dan Luu

This is an archive of an old Google Buzz conversation (circa 2010?) on a variety of topics, including whether or not it's inevitable that a closed platform will dominate social.

Piaw : Social networks will be dominated primarily by network effects. That means in the long run, Facebook will dominate all the others.

Rebecca : ... which is also why no one company should dominate it. "The Social Graph" and its associated apps should be like the internet, distributed and not confined to one company's servers. I wish the narrative surrounding this battle was centered around this idea, and not around the whole Silicon Valley "who is the most genius innovator" self-aggrandizing unreality field. Thank god Tim Berners Lee wasn't from Silicon Valley, or we wouldn't have the Internet the way we know it in the first place.

I suppose I shouldn't be being so snarky, revealing how much I hate your narratives sometimes. But I think for once this isn't, as it usually is, merely harmlessly cute and endearing - you all collectively screwing up something actually important, and I'm annoyed.

Piaw : The way network effects work, one company will control it. It's inevitable.

Rebecca : No it is not inevitable! What is inevitable is either that one company controls it or that no company controls it. If you guys had been writing the narrative of the invention of the internet you would have been arguing that it was inevitable that the entire internet live on one companies servers, brokered by hidden proprietary protocols. And obviously that's just nuts.

Piaw : I see, the social graph would be collectively owned. That makes sense, but I don't see why Facebook would have an incentive to make that happen.

Rebecca : Of course not! That's why I'm biting my fingernails hoping for some other company to be the white knight and ride up and save the day, liberating the social graph (or more precisely, the APIs of the apps that live on top of them) from any hope of control by a single company. Of course, there isn't a huge incentive for any other company to do it either — the other companies are likely to just gaze enviously at Facebook and wish they got there first. Tim Berners Lee may have done great stuff for the world, but he didn't get rich or return massive value to shareholders, so the narrative of the value he created isn't included in the standard corporate hype machine or incentives.

Google is the only company with the right position, somewhat appropriate incentives, and possibly the right internal culture to be the "Tim Berners Lee" of the new social internet. That's what I was hoping for, and I'm am more than a bit bummed they don't seem to be stepping up to the plate in

an effective way in this case, especially since they are doing such a fabulous job in an analogous role with Android.

Rebecca : There is a worldview lurking behind these comments, which perhaps I should try to explain. I'm been nervous about this because it contains some strange ideas, but I'm wondering what you think.

Here's a very strange assertion: Mark Zuckerberg is not a capitalist, and therefore should not be judged by capitalist logic. Before you dismiss me as nuts, stop and think for a minute. What is the essential property that makes someone a capitalist?

For instance, when Nike goes to Indonesia and sets up sweatshops there, and if communists, unhappy with low pay & terrible conditions, threaten to rebel, they are told "this is capitalism, and however noxious it is, capitalism will make us rich, so shut up and hold your peace!" What are they really saying? Nike brings poor people sewing machines and distribution networks so they can make and sell things they could not make and sell otherwise, so they become more productive and therefore richer. The productive capacity is scarce, and Nike is bringing to Indonesia a piece of this scarce resource (and taking it away from other people, like American workers.) So Indonesia gets richer, even if sweatshop workers suffer for a while.

So is Mark Zuckerberg bringing to American workers a piece of scarce productive capacity, and therefore making American workers richer? It is true he is creating for people productive capacity they did not have before — the possibility of writing social apps, like social games. This is "innovation" and it does make us richer.

But it is not wealth that follows the rules of capitalist logic. In particular, this kind of wealth of productive capacity, unlike the wealth created by setting up sewing machines, does not have the kind of inherent scarcity that fundamentally drives capitalist logic. Nike can set up its sewing machines for Americans, or Indonesians, but not for everyone at once. But Tim Berners Lee is not forced to make such choices — he can design protocols that allow everyone everywhere to produce new things, and he need not restrict how they choose to do it.

But — here's the key point — though there is no natural scarcity, there may well be "artificial" scarcity. Microsoft can obfuscate Windows API's, and bind IE to Windows. Facebook can close the social graph, and force all apps to live on its servers. "Capitalists" like these can then extract rents from this artificial scarcity. They can use the emotional appeal of capitalist rhetoric to justify their rent-seeking. People are very conditioned to believe that when local companies get rich American workers in general will also get rich — it works for Indonesia so why won't it work for us? And Facebook and Microsoft employees are getting richer. QED.

But they aren't getting richer in the same way that sweatshop employees are getting richer. The sweatshop employ-

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Top lessons to learn from refactoring code with a headless CMS

Gary Butler · Harry's Engineering

The Harry's European website (for Harry's customers in the UK and the rest of Europe), like most websites, is a content-heavy website providing customers with information about our products and services, which means we seek to keep our content management separate from our codebase to enable both to change independently of each other as well as allowing content to be edited without any engineering intervention.

Having worked on a variety of websites using different Content Management Systems (CMS) over the years from WordPress, to Drupal, to SiteCore among others, there is always the challenge of deciding on good abstractions to build between the code I write while not being constrained by what's possible within the CMS. The tricky part is striking the right balance between being fully dynamic and being overly complicated resulting in the maintenance of the codebase becoming a nightmare for the team.

What is a Headless CMS?

What makes a particular CMS “headless” is in regards to how you implement it into your codebase. Unlike traditional Content Management Systems, a headless CMS is agnostic to your codebase, focusing only on the data you want. Returning the data as JSON on a request. This allows more freedom and flexibility for the developer who is not limited by the access library or any templating language.

We're currently using DatoCMS as our headless content management system, which basically means DatoCMS provides a useful dashboard and portal for managing content which enables our Product, Design and Marketing partners to manage content for the website without having to write or know any code. We can then access that content via APIs that DatoCMS provides to pull into our site at build time. This is different to some traditional CMSs because of this decoupling and doesn't prescribe a particular architecture for your site (for example, WordPress or Drupal).

Trial Builder Model within DatoCMS

NOTE: It's worth mentioning at this point that our Harry's European website is currently a statically generated JavaScript-based Single-Page Application (SPA) written in TypeScript with Gatsby and React.

What is the Trial Builder?

One of the most important flows that we have on our site is the 'Trial Builder' which is the name given to the customer journeys where you subscribe for a Harry's trial set (e.g. razor handle, blades and shave gel) and then decide on your subscription cadence for refill blades as well as other products.

See it for yourself, here: <https://www.harrys.com/en/gb/signup/customize?step=1>

This flow references no less than 15 pieces of content in DatoCMS made of approx, 10 images and 25 strings. We also invest a lot of time in experimenting with A/B tests in this customer journey to further optimise it to surface additional features so that customers can get the most benefit from Harry's products. We can then measure the impact on our top ecommerce funnel metrics such as conversion rate (CVR), average order value (AOV) and bill-through rate (BTR).

The Trial Builder is also leveraged by the Subscription Signup flow. They both have a similar customer journey but with an alteration to the steps or content. Though the Trial Builder is the most important of these flows, any changes need to support the other journey to facilitate code reuse. This leads to requirements for the logic to be flexible to handle these variations. With multiple variations, continuous A/B tests and changes based on country, we can quickly start to gain tech debt, spaghetti code and a generally bloated codebase.

What problem are we solving?

As mentioned earlier, the engineering challenge for working with Headless CMSs is to strike the right balance between code maintainability and flexibility (or dynamism). You want to be able to realise the ideas that the Product Managers come up with without having to re-architect your website every time but you also want to make sure you don't over-engineer your code when it's likely you'll never need 100% full flexibility either.

Back in 2023, we got this balance wrong. We attempted to make the most flexible code possible to handle as many possibilities as we could think of but the reality was we never made use of it. We had a clever setup which enabled entire subscription flows to be created from DatoCMS. From how many steps, to what elements would appear on the page. The idea was to remove any engineering effort when setting up changes or new alternative versions of the journey. This did work well at first, the complexity came when we started to make UI changes based on A/B tests by locale. Implementing these changes without breaking the complex dynamic logic became incrementally more difficult over time. Eventually, it started to block requirements.

We decided to make the code interacting with the DatoCMS APIs as flexible as possible but by doing so it caused the codebase to become the opposite.

Though the intention was to enable a customer journey to be completely configurable via the CMS, over time most of the UI changes still needed engineering implementation, but now each change had to work around this complex condition logic setup in the codebase to support our super dynamic relationship with DatoCMS.

Over time this caused each update to take longer and even blocked us from even implementing some features. As the

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HN: the good parts

Dan Luu

HN comments are terrible . On any topic I'm informed about , the vast majority of comments are pretty clearly wrong . Most of the time, there are zero comments from people who know anything about the topic and the top comment is reasonable sounding but totally incorrect. Additionally, many comments are gratuitously mean. You'll often hear mean comments backed up with something like "this is better than the other possibility, where everyone just pats each other on the back with comments like 'this is great'", as if being an asshole is some sort of talisman against empty platitudes. I've seen people push back against that; when pressed, people often say that it's either impossible or inefficient to teach someone without being mean, as if telling someone that they're stupid somehow helps them learn. It's as if people learned how to explain things by watching Simon Cowell and can't comprehend the concept of an explanation that isn't littered with personal insults. Paul Graham has said, " Oh, you should never read Hacker News comments about anything you write ". Most of the negative things you hear about HN comments are true.

And yet, I haven't found a public internet forum with better technical commentary. On topics I'm familiar with, while it's rare that a thread will have even a single comment that's well-informed, when those comments appear, they usually float to the top. On other forums, well-informed comments are either non-existent or get buried by reasonable sounding but totally wrong comments when they appear, and they appear even more rarely than on HN.

By volume, there are probably more interesting technical "posts" in comments than in links. Well, that depends on what you find interesting, but that's true for my interests. If I see a low-level optimization comment from [nkurz](#), a comment on business from [patio11](#), a comment on how companies operate by [nostrademons](#), I almost certainly know that I'm going to read an interesting comment. There are maybe 20 to 30 people I can think of who don't blog much, but write great comments on HN and I doubt I even know of half the people who are writing great comments on HN 1 .

I compiled a very abbreviated list of comments I like because comments seem to get lost. If you write a blog post, people will refer it years later, but comments mostly disappear. I think that's sad – there's a lot of great material on HN (and yes, even more not-so-great material).

`id="what-s-the-deal-with-ms-word-s-file-format-https-news-ycombinator-com-item-id-12472505">` What's the deal with MS Word's file format ?

Basically, the Word file format is a binary dump of memory. I kid you not. They just took whatever was in memory and wrote it out to disk. We can try to reason why (maybe it was faster, maybe it made the code smaller), but I think

the overriding reason is that the original developers didn't know any better.

Later as they tried to add features they had to try to make it backward compatible. This is where a lot of the complexity lies. There are lots of crazy workarounds for things that would be simple if you allowed yourself to redesign the file format. It's pretty clear that this was mandated by management, because no software developer would put themselves through that hell for no reason.

Later they added a fast-save feature (I forget what it is actually called). This appends changes to the file without changing the original file. The way they implemented this was really ingenious, but complicates the file structure a lot.

One thing I feel I must point out (I remember posting a huge thing on slashdot when this article was originally posted) is that 2 way file conversion is next to impossible for word processors. That's because the file formats do not contain enough information to format the document. The most obvious place to see this is pagination. The file format does not say where to paginate a text flow (unless it is explicitly entered by the user). It relies of the formatter to do it. Each word processor formats text completely differently. Word, for example famously paginates footnotes incorrectly. They can't change it, though, because it will break backwards compatibility. This is one of the only reasons that Word Perfect survives today – it is the only word processor that paginates legal documents the way the US Department of Justice requires.

Just considering the pagination issue, you can see what the problem is. When reading a Word document, you have to paginate it like Word – only the file format doesn't tell you what that is. Then if someone modifies the document and you need to resave it, you need to somehow mark that it should be paginated like Word (even though it might now have features that are not in Word). If it was only pagination, you might be able to do it, but practically everything is like that.

I recommend reading (a bit of) the XML Word file format for those who are interested. You will see large numbers of flags for things like "Format like Word 95". The format doesn't say what that is – because it's pretty obvious that the authors of the file format don't know. It's lost in a hopeless mess of legacy code and nobody can figure out what it does now.

`id="fun-with-null-https-news-ycombinator-com-item-id-12007286">` Fun with NULL

Here's another example of this fine feature:

```
#include #include #include #define LENGTH 128
```

```
int main(int argc, char **argv) { char *string = NULL;
int length = 0; if (argc > 1) { string = argv[1]; length =
strlen(string); if (length >= LENGTH) exit(1); }
```

```
char buffer[LENGTH]; memcpy(buffer, string, length);
```

}

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We saw some really bad Intel CPU bugs in 2015 and we should expect to see more in the future

Dan Luu

2015 was a pretty good year for Intel. Their quarterly earnings reports exceeded expectations every quarter. They continue to be the only game in town for the serious server market, which continues to grow exponentially; from the earnings reports of the two largest cloud vendors, we can see that AWS and Azure grew by 80% and 100%, respectively. That growth has effectively offset the damage Intel has seen from the continued decline of the desktop market. For a while, it looked like cloud vendors might be able to avoid the Intel tax by moving their computation onto FPGAs , but Intel bought one of the two serious FPGA vendors and, combined with their fab advantage, they look well positioned to dominate the high-end FPGA market the same way they've been dominating the high-end server CPU market. Also, their fine for anti-competitive practices turned out to be \$1.45B, much less than the benefit they gained from their anti-competitive practices 1 .

Things haven't looked so great on the engineering/bugs side of things, though. We've seen a number of fairly serious CPU bugs and it looks like we should expect more in the future. I don't keep track of Intel bugs unless they're so serious that people I know are scrambling to get a patch in because of the potential impact, and I still heard about two severe bugs this year in the last quarter of the year alone. First, there was the bug found by Ben Serebrin and Jan Beulic , which allowed a guest VM to fault in a way that would cause the CPU to hang in a microcode infinite loop, allowing any VM to DoS its host.

Major cloud vendors were quite lucky that this bug was found by a Google engineer, and that Google decided to share its knowledge of the bug with its competitors before publicly disclosing. Black hats spend a lot of time trying to take down major services. I'm actually really impressed by both the persistence and the cleverness of the people who spend their time attacking the companies I work for. If, buried deep in our infrastructure, we have a bit of code running at DPC that's vulnerable to slowdown because of some kind of hash collision, someone will find and exploit that, even if it takes a long and obscure sequence of events to make it happen. If this CPU microcode hang had been found by one of these black hats, there would have been major carnage for most cloud hosted services at the most inconvenient possible time 2 .

Shortly after the Serebrin/Beulic bug was found, a group of people found that running prime95 , a commonly used tool for benchmarking and burn-in, causes their entire system to lock up . Intel's response to this was :

Intel has identified an issue that potentially affects the 6th Gen Intel® Core™ family of products. This issue only occurs under certain complex workload conditions, like those that may be encountered when running applications like

Prime95. In those cases, the processor may hang or cause unpredictable system behavior.

which reveals almost nothing about what's actually going on. If you look at their errata list, you'll find that this is typical, except that they normally won't even name the application that was used to trigger the bug. For example, one of the current errata lists has entries like

Certain Combinations of AVX Instructions May Cause Unpredictable System Behavior

AVX Gather Instruction That Should Result in #DF May Cause Unexpected System Behavior

Processor May Experience a Spurious LLC-Related Machine Check During Periods of High Activity

Page Fault May Report Incorrect Fault Information

As we've seen, "unexpected system behavior" can mean that we're completely screwed. Machine checks aren't great either – they cause Windows to blue screen and Linux to kernel panic. An incorrect address on a page fault is potentially even worse than a mere crash, and if you dig through the list you can find a lot of other scary sounding bugs.

And keep in mind that the Intel errata list has the following disclaimer:

Errata remain in the specification update throughout the product's lifecycle, or until a particular stepping is no longer commercially available. Under these circumstances, errata removed from the specification update are archived and available upon request.

Once they stop manufacturing a stepping (the hardware equivalent of a point release), they reserve the right to remove the errata and you won't be able to find out what errata your older stepping has unless you're important enough to Intel.

Anyway, back to 2015. We've seen at least two serious bugs in Intel CPUs in the last quarter 3 , and it's almost certain there are more bugs lurking. Back when I worked at a company that produced Intel compatible CPUs, we did a fair amount of testing and characterization of Intel CPUs; as someone fresh out of school who'd previously assumed that CPUs basically worked, I was surprised by how many bugs we were able to find. Even though I never worked on the characterization and competitive analysis side of things, I still personally found multiple Intel CPU bugs just in the normal course of doing my job, poking around to verify things that seemed non-obvious to me. Turns out things that seem non-obvious to me are sometimes also non-obvious to Intel engineers. As more services move to the cloud and the impact of system hang and reset vulnerabilities increases, we'll see more black hats investing time in finding CPU bugs. We should expect to see a lot more of these when people realize that it's much easier than it seems to find these

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Verilog Won & VHDL Lost? — You Be The Judge!

Dan Luu

This is an archived USENET post from John Cooley on a competitive comparison between VHDL and Verilog that was done in 1997.

I knew I hit a nerve. Usually when I publish a candid review of a particular conference or EDA product I typically see around 85 replies in my e-mail “in” box. Buried in my review of the recent Synopsys Users Group meeting, I very tersely reported that 8 out of the 9 Verilog designers managed to complete the conference’s design contest yet none of the 5 VHDL designers could. I apologized for the terseness and promised to do a detailed report on the design contest at a later date. Since publishing this, my e-mail “in” box has become a veritable Verilog/VHDL Beirut filling up with 169 replies! Once word leaked that the detailed contest write-up was going to be published in the DAC issue of “Integrated System Design” (formerly “ASIC & EDA” magazine), I started getting phone calls from the chairman of VHDL International, Mahendra Jain, and from the president of Open Verilog International, Bill Fuchs. A small army of hired gun spin doctors (otherwise know as PR agents) followed with more phone calls. I went ballistic when VHDL columnist Larry Saunders had approached the Editor-in-Chief of ISD for an advanced copy of my design contest report. He felt I was “going to do a hatchet job on VHDL” and wanted to write a rebuttal that would follow my article... and all this was happening before I had even written one damned word of the article!

Because I’m an independent consultant who makes his living training and working both HDL’s, I’d rather not go through a VHDL Salem witch trial where I’m publically accused of being secretly in league with the Devil to promote Verilog, thank you. Instead I’m going present everything that happened at the Design Contest, warts and all, and let you judge! At the end of court evidence, I’ll ask you, the jury, to write an e-mail reply which I can publish in my column in the follow-up “Integrated System Design”.

id=“the-unexpected-results”>The Unexpected Results

Contestants were given 90 minutes using either Verilog or VHDL to create a gate netlist for the fastest fully synchronous loadable 9-bit increment-by-3 decrement-by-5 up/down counter that generated even parity, carry and borrow.

Of the 9 Verilog designers in the contest, only 1 didn’t get to a final gate level netlist because he tried to code a look-ahead parity generator. Of the 8 remaining, 3 had netlists that missed on functional test vectors. The 5 Verilog designers who got fully functional gate-level designs were:

Larry Fiedler NVidea 3.90 nsec 1147 gates Steve Golson Trilobyte Systems 4.30 nsec 1909 gates Howard Landman HaL Computer 5.49 nsec 1495 gates Mark Papamarcos EDA Associates 5.97 nsec 1180 gates Ed Paluch Paluch & Assoc. 7.85 nsec 1514 gates

The surprise was that, during the same time, none of 5 VHDL designers in the contest managed to produce any gate level designs.

id=“not-vhdl-newbies-vs-verilog-pros”>Not VHDL Newbies vs. Verilog Pros

The first reaction I get from the VHDL bigots (who weren’t at the competition) is: “Well, this is obviously a case where Verilog veterans whipped some VHDL newbies. Big deal.” Well, they’re partially right. Many of those Verilog designers are damned good at what they do — but so are the VHDL designers!

I’ve known Prasad Paranjpe of LSI Logic for years. He has taught and still teaches VHDL with synthesis classes at U. C. Santa Cruz University Extension in the heart of Silicon Valley. He was VP of the Silicon Valley VHDL Local Users Group. He’s been a full time ASIC designer since 1987 and has designed real ASIC’s since 1990 using VHDL & Synopsys since rev 1.3c. Prasad’s home e-mail address is “vhdl@ix.netcom.com” and his home phone is (XXX) XXX-VHDL. ASIC designer Jan Decaluwe has a history of contributing insightful VHDL and synthesis posts to ESNUG while at Alcatel and later as a founder of Easics, a European ASIC design house. (Their company motto: “Easics - The VHDL Design Company”.) Another LSI Logic/VHDL contestant, Vikram Shrivastava, has used the VHDL/Synopsys design approach since 1992. These guys aren’t newbies!

id=“creating-the-contest”>Creating The Contest

I followed a double blind approach to putting together this design contest. That is, not only did I have Larry Saunders (a well known VHDL columnist) and Yatin Trivedi (a well known Verilog columnist), both of Seva Technologies comment on the design contest — unknown to them I had Ken Nelsen (a VHDL oriented Methodology Manager from Synopsys) and Jeff Flieder (a Verilog based designer from Ford Microelectronics) also help check the design contest for any conceptual or implementation flaws.

My initial concern in creating the contest was to not have a situation where the Synopsys Design Compiler could quickly complete the design by just placing down a DesignWare part. Yet, I didn’t want to have contestants trying (and failing) to design some fruity, off-the-wall thingy that no one truly understood. Hence, I was restricted to “standard” designs that all engineers knew — but with odd parameters thrown in to keep DesignWare out of the picture. Instead of a simple up/down counter, I asked for an up-by-3 and down-by-5 counter. Instead of 8 bits, everything was 9 bits.

recycled COUNT_OUT [8:0] o----- | up-by-3 |>---carry
--->| D Q |>- CARRY_OUT | [8:0] | down-by-5 |>---borrow
--->| D Q |>- BORROW_OUT | | | | UP --->| logic | | |
| DOWN --->| |o--->---| D[8:0] | | --->| new_count
[8:0] | Q[8:0] |>-o--->---o | | | o--- | o->- COUNT_OUT |
--- [8:0] new_count [8:0] | --->| even | --- o->-| parity
|>-parity->| D Q |>- PARITY_OUT | generator | (1 bit) | |
--->| o->| | --- CLOCK ---o

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Cocktail party ideas

Dan Luu

You don't have to be at a party to see this phenomenon in action, but there's a curious thing I regularly see at parties in social circles where people value intelligence and cleverness without similarly valuing on-the-ground knowledge or intellectual rigor. People often discuss the standard trendy topics (some recent ones I've observed at multiple parties are how to build a competitor to Google search and how to solve the problem of high transit construction costs) and explain why people working in the field today are doing it wrong and then explain how they would do it instead. I occasionally have good conversations that fit that pattern (with people with very deep expertise in the field who've been working on changing the field for years), but the more common pattern is that someone with cocktail-party level knowledge of a field will give their ideas on how the field can be fixed.

Asking people why they think their solutions would solve valuable problems in the field has become a hobby of mine when I'm at parties where this kind of superficial pseudo-technical discussion dominates the party. What I've found when I've asked for details is that, in areas where I have some knowledge, people generally don't know what sub-problems need to be solved to solve the problem they're trying to address, making their solution hopeless. After having done this many times, my opinion is that the root cause of this is generally that many people who have a superficial understanding of topic assume that the topic is as complex as their understanding of the topic instead of realizing that only knowing a bit about a topic means that they're missing an understanding of the full complexity of a topic.

Since I often attend parties with programmers, this means I often hear programmers retelling their cocktail-party level understanding of another field (the search engine example above notwithstanding). If you want a sample of similar comments online, you can often see these when programmers discuss "trad" engineering fields. An example I enjoyed was this Twitter thread where Hillel Wayne discussed how programmers without knowledge of trad engineering often have incorrect ideas about what trad engineering is like, where many of the responses are from programmers with little to no knowledge of trad engineering who then reply to Hillel with their misconceptions. When Hillel completed his crossover project, where he interviewed people who've worked in a trad engineering field as well as in software, he got even more such comments. Even when people are warned that naive conceptions of a field are likely to be incorrect, many can't help themselves and they'll immediately reply with their opinions about a field they know basically nothing about.

Anyway, in the crossover project, Hillel compared the perceptions of people who'd actually worked in multiple fields to pop-programmer perceptions of trad engineering. One of the many examples of this that Hillel gives is when people

talk about bridge building, where he notes that programmers say things like

The predictability of a true engineer's world is an enviable thing. But ours is a world always in flux, where the laws of physics change weekly. If we did not quickly adapt to the unforeseen, the only foreseeable event would be our own destruction.

and

No one thinks about moving the starting or ending point of the bridge midway through construction.

But Hillel interviewed a civil engineer who said that they had to move a bridge! Of course, civil engineers don't move bridges as frequently as programmers deal with changes in software but, if you talk to actual, working, civil engineers, many civil engineers frequently deal with changing requirements after a job has started that's not fundamentally different from what programmers have to deal with at their jobs. People who've worked in both fields or at least talk to people in the other field tend to think the concerns faced by engineers in both fields are complex, but people with a cocktail-party level of understanding of the field often claim that the field they're not in is simple, unlike their field.

A line I often hear from programmers is that programming is like "having to build a plane while it's flying", implicitly making the case that programming is harder than designing and building a plane since people who design and build planes can do so before the plane is flying. But, of course, someone who designs airplanes could just as easily say "gosh, my job would be very easy if I could build planes with 4 9s of uptime and my plane were allowed to crash and kill all of the passengers for 1 minute every week". Of course, the constraints on different types of projects and different fields make different things hard, but people often seem to have a hard time seeing constraints other fields have that their field doesn't. One might think that understanding that their own field is more complex than an outsider might naively think would help people understand that other fields may also have hidden complexity, but that doesn't generally seem to be the case.

If we look at the rest of the statement Hillel was quoting (which is from the top & accepted answer to a stack exchange question), the author goes on to say:

It's much easier to make accurate projections when you know in advance exactly what you're being asked to project rather than making guesses and dealing with constant changes.

The vast majority of bridges are using extremely tried and true materials, architectures, and techniques. A Roman engineer could be transported two thousand years into the future and generally recognize what was going on at a modern construction site. There would be differences, of course, but you're still building arches for load balancing, you're still using many of the same materials, etc. Most

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Windows for Linux Nerds

Jessie Frazelle

I recently started a job at Microsoft. In my first week I have already learned so much about Windows, I figured I would try to put it all into writing. This post is coming to you from a Windows Subsystem for Linux console!

`dir="ltr" lang="en">`I'm headed to Seattle because I'M JOINING MICROSOFT, at the airport wearing this awesome shirt from @listonb & @Taylorb_msft 
`pic.twitter.com/8rnAg1dsPd`

— jessie frazelle (@jessfraz) September 4, 2017

New job and I got a Windows computer and a Linux computer! If you are new to my blog, let me tell you: I love setting up a perfect desktop experience. I've written a few posts on it (for Linux), you should check them out. Setting up a Windows computer is something I have not done in quite some time. I will describe a bit how to set up a windows machine in a reproducible way at the end of this post.

I would like to thank Rich Turner , John Starks , Taylor Brown , and Sarah Cooley for taking the time to explain a lot of the following to me. :)

`id="windows-subsystem-for-linux-wsl">`Windows Subsystem for Linux (WSL)

Let's start with Windows Subsystem for Linux, aka WSL. Even @monkchips wrote that since I joined Microsoft "Linux Subsystem for Windows will definitely be getting a workout." I am super excited about Windows Subsystem for Linux. It is one of the coolest pieces of tech I've seen since I started using Docker.

First, a little background on how WSL works...

You can learn a lot more about this from the Windows Subsystem for Linux Overview . I will go over some of the parts I found to be the most interesting.

The Windows NT kernel was designed from the beginning to support running POSIX, OS/2, and other subsystems. In the early days, these were just user-mode programs that would interact with ntdll to perform system calls. Since the Windows NT kernel supported POSIX there was already a fork system call implemented in the kernel. However, the Windows NT call for fork , NtCreateProcess , is not directly compatible with the Linux syscall so it has some special handling you can read about more under System Calls .

There are both user and kernel mode parts to WSL. Below is a diagram showing the basic Windows kernel and user modes alongside the WSL user and kernel modes.

The blue boxes represent kernel components and the green boxes are Pico Processes. The LX Session Manager Service handles the life cycle of Linux instances. LXCore and lxsys,

lxcore.sys and lxss.sys respectively, translate the Linux syscalls into NT APIs.

`id="pico-processes">`Pico Processes

As you can see in the diagram above, init and /bin/bash are Pico processes. Pico processes work by having system calls and user mode exceptions dispatched to a paired driver. Pico processes and drivers allow Windows Subsystem for Linux to load executable ELF binaries into a Pico process' address space and execute them on top of a Linux-compatible layer of system calls.

You can read even more in depth on this from the MSDN Pico Processes post .

`id="system-calls">`System Calls

One of the first things I did in WSL was run a syscall fuzzer. I knew it would break but it was interesting for the purposes of figuring out which syscalls had been implemented without looking at the source. This was how I realized PID and mount namespaces were already implemented into clone and unshare !

The WSL kernel drivers, lxss.sys and lxcore.sys , handle the Linux system call requests and translate them to the Windows NT kernel. None of this code came from the Linux kernel, it was all re-implemented by Windows engineers. This is truly mind blowing.

When a syscall is made from a Linux executable it gets passed to lxcore.sys which will translate it into the equivalent Windows NT call. For example, open to NtOpenFile and kill to NtTerminateProcess . If there is no mapping then the Windows kernel mode driver will handle the request directly. This was the case for fork , which has lxcore.sys prepare the process to be copied and then call the appropriate Windows NT kernel APIs to create and copy the process.

You can learn more from the MSDN System Calls post .

`id="launching-windows-executables">`Launching Windows Executables

Since WSL allows for running Linux binaries natively (without a VM), this allows for some really fun interactions.

You can actually spawn Windows binaries from WSL. Linux ELF binaries get handled by lxcore.sys and lxss.sys as described above and Windows binaries go through the typical Windows userspace.

You can even launch Windows GUI apps as well this way! Imagine a Linux setup where you can launch PowerPoint without a VM.... well this is it!!

`id="launching-x-applications">`Launching X Applications

You can also run X Applications in WSL. You just need an X server. I used vcxsrv to try it out. I run i3 on all my Linux machines and tried it out in WSL like my awesome coworker

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standard-article(title: [You might not need Kubernetes], author: [Jessie Frazelle], source-name: [Jessie Frazelle], images: (), paragraphs: ([I have realized recently that a lot of people think I am just a skill for Kubernetes and I am not. What I have done is write a few blog posts on some interesting problems to be solved in Kubernetes. But I would like to emphasize that those problems are pretty exclusive to the way Kubernetes was designed and you could easily build your own orchestrator without them.], [id="use-containerd">Use Containerd], [If you need an example of a custom, minimal orchestrator with containerd you should checkout stellar .], [Or see my design doc for a multi-tenant orchestrator .], [I'll let you dive into that in your own time though. Let's take a new look at a blog post I wrote about Building images securely on Kubernetes .], [I feel like I should have more clearly stated how this problem is pretty exclusive to Kubernetes. It's also not really a hard problem. The hard problem I was solving in that post was not how to build images on Kubernetes but how to build images as an unprivileged user in Linux. That is a hard problem. And a serious problem for companies who don't allow root on their machines.], [The easier choice if all you need to do is build an image and you are already using containerd, is to run buildkit on the same machine and then you can use the buildkit API library to build your dockerfiles.], [Or just run docker-in-docker, I have done this for years on my CI with absolutely no problems.], [Anyways, the point I am trying to make is you should use whatever is the easiest thing for your use case and not just what is popular on the internet. With complexity comes a steep learning curve and with a massive number of pluggable layers comes yak shaves until the end of time.], [Think for yourselves, don't be sheep.],), insert-map: (:), word-count: 312, edited-for-length: false, debug-mode: false,)

standard-article(title: [The Brutally Honest Guide to Docker Graphdrivers], author: [Jessie Frazelle], source-name: [Jessie Frazelle], images: (), paragraphs: ([Sup, let me give you fair warning here. Everything contained in this post is my opinion so don't go getting your panties all in a knot on Hacker News because you don't agree with me. I could honestly care less, because that's the thing about my opinion , it's mine.], [I am going to give you my honest and dare I say it "blunt" opinion about each of the Docker graphdrivers so you can decide for yourself which one is the best one for you. None are perfect each has it's flaws and I will be laying those out. Let's begin.], [id="overlay">Overlay], [Overlayfs was added in the 3.18 kernel. This is important to note because if you are running overlay on an older kernel than 3.18 you are either:], [Not running the same overlay.], [Running a kernel with overlayfs backported onto it, which is what we call a "frankenkernel". Frankenkernels are not to be trusted. This is not to say it won't work, hey it might work great, but it's not to be trusted.], [Overlay is great but you need a recent kernel. There are also some super obscure kernel bugs with regard to sockets or certain python packages docker/docker#12080 . But I will say personally I use overlay, I have not hit these bugs recently and I have all my 100+ dockerfiles running as continuous builds on my server with overlay and they all work.], [id="aufs">Aufs], [Aufs is another great one. But it is not in the kernel by default which blows. On Ubuntu/Debian distros this is as easy as installing the kernel extras package but other distros it might not be as simple.], [id="btrfs">Btrfs], [Btrfs is great too but you need to partition the disk you will use for /var/lib/docker as btrfs first. This is kinda a hurdle to jump that I don't think a lot of people are willing to do.], [id="zfs">Zfs], [Zfs is another good one, of course, like btrfs it takes some setup and installing the zfs.ko on your system. But this driver might become a whole lot more popular if Ubuntu 16.04 ships with zfs support.], [id="devicemapper">Devicemapper], [Honestly it makes me

brief-group(([

Mike Ash, Mike Ash (Friday Q&A): I'm afraid I once again don't have a Friday Q&A for you today, but I wrote up the best new features in Swift 4 for the Plausible Labs blog, which is almost as good. Check it out over there! (Read More)

], [

Mike Ash, Mike Ash (Friday Q&A): In my last article I discussed the basics of Swift's new Codable protocol, briefly discussed how to implement your own encoder and decoder, and promised another article about a custom binary coder I've been working on. Today, I'm going to present that binary coder. (Read More)

],))

super disappointed to say this, but buyer beware. Hey on the plus side.... it's in the kernel. You must must must have all the devicemapper options set up perfectly or you will find yourself only being able to launch 2 containers.], [Let me tell you a story.], [My mom once asked her friend for her famous chicken enchilada recipe so she could make it herself. The friend gave the recipe but left out one key ingredient so that my mom's never tasted just right. There was always something off about it.], [This is how I think of devicemapper.], [It works on RedHat.], [id="vfs">Vfs], [I sure hope to hell you are just testing something or clinically insane.], [That's about all. Thanks for reading my opinion if you even made it this far.],), insert-map: (:), word-count: 488, edited-for-length: false, debug-mode: false,)

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